

INTERNATIONAL JOURNAL OF RETAILING AND MARKETING

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1. INTRODUCTION

The International Journal of Retailing and Marketing (IJRM) is an international, peer-reviewed scholarly journal dedicated to publishing high-quality theoretical, empirical, and interdisciplinary research in retailing, marketing, digital commerce, and emerging business technologies. The journal provides a global platform for researchers, academicians, industry professionals, and policymakers to share innovative ideas and practical solutions in areas such as Artificial Intelligence, Generative AI, Retail Analytics, and Consumer Behaviour, Omni channel Commerce, Sustainability, and Digital Transformation. IJRM encourages original research that advances knowledge and bridges the gap between academic research and industry practice in the rapidly evolving global marketplace.

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- Be numbered consecutively.
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- Use editable text (not images).

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- Font** : Times New Roman, 12 pt, English
- Line Spacing** : 1.5
- Alignment** : Justified
- Page Numbers** : Bottom Centre

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The manuscript should be organised according to the following structure:

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- 5.2 Author Information:** Provide separately on the title page, including Full Name, Designation, Department, Institution, Country, Email Address, and Corresponding Author details.
- 5.3 Abstract:** 150–250 words and should include Background, Objective, Methodology, Results, and Conclusion. Citations should not be included.
- 5.4 Keywords:** Provide 5–7 keywords arranged alphabetically.
- 5.5 Introduction:** Include background, research gap, problem statement, objectives, and contribution of the study.
- 5.6 Literature Review:** Present previous studies, critical analysis, and identify the research gap.
- 5.7 Research Methodology:** Describe the research design, sample, variables, data collection methods, tools, statistical techniques, and software used.
- 5.8 Results:** Present findings using tables, figures, graphs, and statistical analyses where appropriate.
- 5.9 Discussion:** Explain the interpretation of findings, comparison with earlier studies, and practical and theoretical implications.
- 5.10 Conclusion:** Include major findings, contributions, limitations, and future research directions.

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Authors shall confirm:

- Original work
- No simultaneous submission
- Ethical approval where applicable
- Informed consent where applicable
- Proper citation
- Accurate reporting

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- The decision depends on the quality and source of the matching text, not only on the percentage of similarity.

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The Impact of ESG Technologies on Sustainable Business Performance

URL:-https://docs.google.com/document/d/1fVeZasEsdvabU1HW5UHezhuX_JAN6FNC/edit?usp=sharing&ouid=110403252814827643206&rtfpof

¹Dr. Neha Goyal ²Tanishka Dabas

ABSTRACT

In recent years, digitalization has been developing and there has been a strong drive towards environmental, social and governance (ESG) responsibility as well. These both things happened at the same time, for ESG accountability, businesses are using digital technologies. So the question is: How do these digital technologies translate into sustainable business performance? This study is specific to the EV industry, and investigates just this one question: how exactly can AI data systems and digital tracking tools make companies more sustainable and better-performing? Electric vehicle industry selected to have high expectations of environmental performance, tight government regulations and fast technological development. We conducted a search on Google Scholar in 2021–2026 using three Boolean strings following the guidelines of PRISMA 2020.

A total of 25 studies passed the title/abstract screening and the full-text screening phase from the 113 records identified. All have been quality-appraised and analysed using a three-stage thematic synthesis and results were interpreted using four theoretical lenses (stakeholder, institutional, natural-resource based and legitimacy theory). Our research found out that ESG technologies like AI and other digital tools help the companies to monitor their environmental impact such as pollution they have caused energy they have used and the battery life. The companies can use blockchain and battery passports to trace the material and its source, making the supply chain transparent and responsible.

AI helps to create better ESG reports, but it also helps companies avoid making false claims, or over-hyping their environmental measures, which is known as “greenwashing.” But all that can only be achieved if the company has the right skills, systems and strategy in place to implement ESG technologies and thereby improve innovation, business stability and also the financial results. One thing that this study does have is that it is geographically biased, as China is the leader in the EV industry worldwide, so the results may not necessarily be applied to the listed companies in all countries.

The research is particularly focussed on the electric vehicle industry. Describes the importance of ESG technologies, or technologies that are used by companies to improve their sustainable performance. It also sets out an agenda for future research.

Keywords: *ESG technologies; sustainable business performance; electric vehicles; systematic literature review; environment; social; digital transformation; corporate governance*

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1. Introduction

Over the last ten years the world has seen two major shifts in business that have altered the way that the businesses have been run. The first is the tech boom that ushered in digital transformation and at the same time, the second is sustainability itself that propelled towards ESG accountability. Since there was a rise in sustainability, it meant that the companies are no longer judged just on the basis of their profitability, now the investors, governments and even the public wanted proof that the business is protecting the environment, treating the people fairly and running the business ethically.

Now digital transformation was also required since the businesses operate in a highly dynamic environment and new powerful tools like AI, the Internet of Things (IoT), blockchain, big-data analytics were being used to measure and manage almost everything they do, businesses adopted them to measure their ESG performance as well. Typically, these are considered entirely separate issue areas: sustainability and the tech boom. Where they intersect — and when and how — the digital tools a company chooses to track and manage its ESG impacts are an asset to sustainable business performance is less clear.

There's already a substantial body of work that shows that companies with strong environmental and social practices are more financially successful – but only when sustainability is integrated, not added as a superficial PR stunt or a last minute edition, or just put on the shelves and forgotten, but deeply ingrained in the company's culture and operations. However, the link between ESG results and firm value provides very little information about the channels a firm uses to achieve its results. It is in the instrumental space such as the preventive dashboards which helps the company in real time in cases like when a machine starts emitting too much pollution, the dashboard identifies it even before it harms the environment and results into a massive government fine, the distributed Ledger using blockchain which tracks the exact journey of a raw material from the extraction to the factory making the supply chain completely auditable and proving the materials are ethical, the AI truth detector helps to read and analyze companies' public statements and the financial disclosures etc. over any number of years and identify the inconsistency and prove whether they're actually keeping their climate promises or not.

Electric vehicles industry is a key area in this study, as it is the ideal real world example of ESG (Environmental, Social and Governance) technology. A few main reasons why EV industry was chosen are that EVs face new environmental challenges as EVs themselves do not pollute the environment when driven but building them creates other environmental problems, risky supply chains as EV parts come from all over the world, it is very hard to track where materials come from or there are known human rights and unfair labor issues tied to mining those materials, EVs also have the tech advantage since they are highly computerized and automakers already have the digital tools like data tracking and sensors required to monitor and report on these environmental and social issues.

The EV industry is one of the hardest and most critical environmental and social industries to navigate, yet it also has access to the data technology necessary to overcome those challenges, but it also has the potential to be a fascinating test ground – where ESG technologies and data should be – if they are at all. Another reason for our motives is related to methodology. Earlier research on this matter has often confused technologies with their results. When a corporation is reported to be using digital ESG tools, it is assumed to be achieving sustainable objectives, which is not always possible.

In this study, clear steps of research have been taken. We make a clear distinction between the company making use of certain technology and the results it has achieved.

Objectives of the Study

This study aims to investigate the influence of Environmental, Social and Governance (ESG) technologies on business operations to achieve sustainability with the focus on the electric vehicles sector. The specific objectives of the study are:

- To investigate how digital ESG technologies (such as Artificial Intelligence, Industrial IoT and Big Data Analytics) directly impact corporate ESG performance and sustainable business outcomes in the EV industry.
- To study the mediating factors that include green product innovation, information transparency and corporate risk taking affecting the translating of digital transformation into tangible sustainability results.
- To establish the strategic facilitators and barriers (that comprise features of the top management team, organizational digital culture, etc.) for ensuring the successful adoption of ESG technologies.

The rest of the paper is structured as follows:

The literature review of ESG technologies and sustainable business practices is conducted in section 2. The research framework, data collection methods, etc. are outlined in section 3. Data analysis is presented in section 4, along with discussion of the results and conclusions. The research findings are summarised in section 5 along with the prospects for future studies and recommendations.

2. Literature Review

A common complaint about writing about ESG-technology is that it is observing but not explaining. In order to prevent such, the synthesis is read as per four complementary theories. These address different questions on the benefits of adopting a digital ESG tool for an EV company and whether and how it can help—or hinder—company performance. They both provide the causal vocabulary that is missing from a purely descriptive account.

2.1 Theoretical and Conceptual Framework

But to see how digital tools really impact business performance, it's important to examine the reasons for using them. Four main theories are used to explain this process in this study. The first one is the stakeholder theory (Freeman, 1984). It is not enough for a business to be just a money spinner for shareholders, a business must provide value for the government, its customers and its local communities. These groups are extremely vocal for EV companies. Individuals are asking themselves about the ethics of the batteries and regulators want to monitor emissions.

Digital ESG tools enable the company to gather data that is easy to collect and demonstrate to these groups that the company is being responsible. Institutional theory (DiMaggio & Powell, 1983) is the second concept. This theory presents that companies will be the ones to follow new practices not only for efficiency, but also because of external pressure. Governments are formulating stringent environmental regulations and there's a significant impetus to follow successful competitors in the EV segment. This also involves firms embracing AI and IoT solutions to adhere to regulations and expectations. But the theory has also got its haters, as others have feared that companies would

purchase the technology with a view to window dressing rather than changing their business practices.

In order to ensure that the technology will actually help performance, the resource-based and natural-resource-based perspectives (Barney, 1991; Hart, 1995) are explored. These theories propose a company's competitive advantage is derived from having unique resources that are difficult for others to replicate. These are special resources such as digital capabilities like big data analytics or blockchain tracking system. Utilizing these digital tools in a targeted way to reduce pollution and improve resource management creates a strong in-house capability which results in long-term financial success for the automaker. The legitimacy theory (Suchman, 1995) is the last theory. In other words, a company has to demonstrate that its acts are acceptable to the society. The public is quite skeptical of any climate promises at this time. Thus the trustworthiness of ESG data is of utmost significance.

AI truth detectors and secure data platforms are examples of technologies that help prevent greenwashing and ensure that when the public says it, it means it. This will preserve the company reputation and its business. The four theories are all combined to create the conceptual framework for this study. Digital ESG technologies are being forced on the EV companies by outside pressures from government and public. These tools, when adopted, offer real time monitoring and supply chain transparency.

These tools will enhance environmental monitoring, social equity and corporate governance, provided that the company implements them in the appropriate manner. In the end it translates to sustainable business performance, the company becomes more profitable, while they protect the planet at the same time. 21

2.2 The Role of Digital Technologies in ESG performance

Digital technologies are transforming businesses' approaches to their environmental footprint. Factories have real-time visibility of their operations thanks to the IoT and AI integration (Román-Salinas et al., 2026). For instance, AI and IoT technologies can collaborate to anticipate machine failure before it occurs and optimize energy consumption much more intelligently (Mansour et al., 2025a). The real time data feedback can help energy intensive manufacturing facilities to pinpoint the source of their energy consumption and enable a more efficient utilization of resources within the framework of the circular economy (Ma et al., 2024).

By applying these tools in their practice, EV manufacturers can greatly diminish their overall carbon footprint, and transform their sustainability objectives into tangible green innovation outputs (Wang et al., 2026). These technologies also have a sizable impact on the social performance of a business. Social sustainability is the aspect of a company that treats its employees fairly and ensures that everyone the company engages with—the production chain—in is being treated in a responsible way.

With the support of big data analytics, a company can analyze huge amounts of supplier data, ensuring that they comply with ethical and green standards (Cheng et al. 2025). The synergy of green strategies and big data can enhance the overall sustainability of business operations and create a safer workplace (Aslam, 2026). Moreover, the shift towards a more human-centric approach in manufacturing is enabling the use of IoT devices to assist workers in providing more ergonomic

monitoring and safer working systems (Román-Salinas et al., 2026). This forms an open network that safeguards employees and local communities. Last but not least, digital technologies are fundamentally transforming the landscape of corporate governance and reporting on corporate achievements. Honesty and good communication with the public are essential to good governance.

The company information becomes extremely transparent due to the big data infrastructure, which in turn introduces more external monitoring, and compels the company to take action in line with the ESG initiative to safeguard its reputation (Wang & Zhang, 2024). AI then goes one step further by automating data gathering and eliminating human error from sustainability reports (Alshareef, 2025). Recent research has shown that generative AI can dramatically improve the quality of information revealed by a company making it hard to argue against the existence of greenwashing (Xu et al., 2026). Transparency with regards to the data on AI and ESG leads to trust with investors and promotes sustainable financial value (Bogdan et al., 2025).

2.3 Challenges in Technology Adoption

The implementation of digital tools is not simple, although it has great advantages. One of the major challenges that many companies face is the financial expenses. It is essential to invest hugely in infrastructure and constantly train staff to create a robust digital infrastructure. The research demonstrates that high adoption costs, lack of digital-skills and infrastructure issues prevent many manufacturers from leveraging big data analytics effectively (Cheng et al., 2025).

This is a significant issue for small and medium sized businesses. Adopting advanced solutions such as generative AI is very costly and tends to be a game for big companies as it leaves smaller suppliers behind (Xu et al., 2026). Moreover, the limited supply of resources is a recurring issue faced by companies in developing areas, making it extremely difficult to manage high costs and low profitability in the implementation of digitalization (Aslam, 2026). In addition to the cost, there are significant technical challenges. The complexity of the factory's operation makes it difficult to integrate the use of old physical equipment with new AI and IoT sensors, which can create dynamic uncertainties that prevent real-time data feedback from being retrieved (Ma et al., 2024).

When combining different data types from multiple sources, there are generally interoperability problems because the various systems are unable to communicate properly with one another (Román-Salinas et al., 2026). Moreover, AI and IoT integration not only introduces new risks to the business but also opens the door to serious data security risks (Mansour et al., 2025a). With a huge amount of sensitive data collected regarding the supply chain and environment, protecting this data from cyber attacks is a serious concern.

Last but not least, there is the possibility of "show" technology. In the absence of a robust ethical strategy, digital tools can even camouflage unethical practices. Often times, the decisions made by complex AI algorithms are opaque, that is, no one can see the model's decision-making process or how the data is being processed (Xu et al., 2026). The opacity is conducive to symbolic environmental action by companies, rather than to real, substantive change. In order to be effective, technology needs to be embraced by the top leadership and have a culture of accountability and data integrity.

2.4 Research Gap

Digital transformation and sustainability have been separately studied in the past and, when studied together, sustainability and digital transformation are often intertwined with the adoption of digital tools rather than with environmental and/or social impacts. One key issue in the literature has been that companies being 'made' sustainable by merely introducing an AI or IoT system was not actually confirmed whether these systems actually lower emissions or enhance working conditions. Moreover, existing research is largely based on general manufacturing information and does not take into account the highly regulated nature of the EV industry and specific issues such as battery lifecycle management, ethical sourcing of minerals, etc. To fill these gaps the study is limited to the EV sector, while ensuring that the technologies acquired by companies are separated from the sustainability impacts they generate. The results of this research demonstrate that ESG technologies support sustainable business performance only when they include the appropriate organizational strategy and effective internal governance, as they are able to influence the exact mechanisms that enable their success, such as green product innovation and transparency in the supply chain.

3. Methodology

The review was carried out and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement (Page et al., 2021). Due to the study designs and the absence of a single effect size reported across the different studies in the evidence base, a quantitative meta-analysis was not attempted; it is qualitative and thematic in nature. The items below refer to the PRISMA 2020 which are followed by the steps below, and a completed checklist will be available from the authors.

3.1 Protocol and registration

Prior to screening a brief internal protocol was prepared, which included questions, eligibility criteria, search strings and the synthesis approach. This is noted as a limitation in the review process because the review was not registered on PROSPERO which usually does not accept management and business reviews. No amendments were made to the eligibility criteria after screening started.

3.2 Eligibility criteria

Studies were included if they were peer-reviewed, published between January 2021 and September 2026, written in English and used at least one digital ESG technology to address a sustainability, ESG or business-performance outcome, relevant to the EV or broader automotive-manufacturing sector, either directly or indirectly, with evidence that was transferrable to firm-level in the automotive-manufacturing sector. Those engineering studies where there was no sustainability/performance linkage, those on unrelated sectors with no transferable insight, non-peer-reviewed grey literature as primary evidence, and opinion pieces which lack substantive analysis were excluded. The criteria are given in full in Table 1.

Table 1. Inclusion and exclusion criteria

Dimension	Inclusion	Exclusion
Time frame	Published Jan 2021 – Sep 2026	Published before 2021
Source type	Peer-reviewed journal articles and rigorously reviewed conference papers	Blogs, editorials, theses, non-reviewed reports (used only as background context)
Language	English	Non-English with no English full text
Topical focus	At least one digital ESG technology (AI/ML, IoT, blockchain, big data, digital twins, ESG platforms) linked to an ESG/sustainability/performance outcome	Technology studies with no ESG or performance linkage; ESG studies with no technology component
Sector relevance	EV/automotive sector, or firm-level evidence transferable to the EV context	Sectors with no transferable relevance to EV firms
Contribution	Reports empirical results or a structured review with analysis	Descriptive notes/opinion with no analysis

AI/ML = artificial intelligence / machine learning; IoT = Internet of Things.

3.3 Information sources

In accordance with the design of the review, Google Scholar was the main source of information used. It has a wide spectrum of journals, disciplines and early-access articles, which are appropriate for a new, cross-disciplinary research area, and it brings to light finance, management and engineering forums where such literature can be found. The last search date of the database was 6 September 2026. The database search was complemented by backward citation search of included studies and targeted consultation of organisationally available sources (International Energy Agency and similar organisations) for contextual information to minimise the risk of missing important work.

3.4 Search strategy

Three concept blocks (ESG/sustainability, digital technology and the EV/automotive context) were merged with performance terms in three Boolean strings. Total number of records identified was kept below 200 to ensure screening remained rigorous, reproducible, and strings were iteratively refined for a balance of precision and recall. The following strings were passed to the functions in table 2, and the strings returned were listed.

Table 2. Search strings (Google Scholar) and records identified

#	Search string (title, abstract and keywords)	Records
S1	allintitle: ("ESG" OR "sustainability") AND ("IoT" OR "Internet of Things" OR "big data") AND ("manufacturing" OR "electric vehicle")	39
S2	allintitle: ("digital transformation") AND ("ESG performance" OR "green innovation") AND ("firm performance" OR "new energy vehicle")	14
S3	allintitle: ("ESG" OR "disclosure") AND ("artificial intelligence") AND ("greenwashing" OR "governance" OR "reporting quality")	60
	Total records identified (before de-duplication)	113

Search last run 6 September 2026. Wildcards () capture morphological variants. Counts are the records screened from each string; overlap across strings was removed at de-duplication.*

3.5 Selection process

Records have been exported and deduplicated resulting in 18 duplicates being removed due to overlap from the three strings, leaving 95 unique records. Titles and abstracts were then reviewed to make sure they met the eligibility criteria, and the full texts of potentially relevant reports were evaluated. The review team operating under the common protocol screened using the common screening criteria and resolved disagreements through discussion and referral back to screening criteria, with a conservative approach – keeping records at the title-abstract level where eligibility was not clear. The 95 unique records screened during these assessment phases were excluded, leaving 25 studies.

3.6 Data collection and data items

The method of structured extraction was used in taking essential information for every study in scope like bibliographic information; ESG technology or technologies studied; research techniques used as well as the context in terms of geography and area of study; ESG or performance outcome studied; nature of the relationship established; as well as the moderating, mediating, or limiting factors. A study which investigates some outcomes simultaneously, all of which also apply to review questions, was counted.

3.7 Quality appraisal

As the studies have a mix of qualitative, quantitate and review designs, a six-factor checklist was adapted from mixed methods evaluation to assess the quality of the studies (objectives clarity; appropriateness of design in relation to the problem studied; data adequacy and transparency; proper analysis; consistency of the evidence and conclusions made; importance of the results for the relationship between ESG and EV). Each study was ranked according to the quality criteria. While the appraisal influenced the interpretation of the studies recruited, it was not used to exclude any studies automatically because inclusivity is the priority in the systematic review in Section 4.3.

3.8 Synthesis method

The process of synthesizing findings has been done using Thomas and Harden's (2008) three-stage thematic synthesis. This involved three steps such as free line-by-line coding, organization of codes into descriptive themes, and abstraction into higher-order analytical themes that answer the questions of the review. This led to the emergence of four analytical themes in Section 4.4. This method was chosen instead of meta-analysis since the studies do not have common related measures nor similar designs and instead of narrative since it brings transparency and auditability to the analysis.

4. Data Analysis

4.1 Study selection

The searches led to the retrieval of 113 records through Google Scholar and relevant organizations search. After eliminating the duplicates (18 the same records were found in different searches), the 95 records have been screened for eligibility on the title and abstract level. In the course of screening, 70 records have been eliminated for various reasons—primarily, because they were not related to electric vehicles or automakers or did not relate digital ESG technology to sustainability. Thus, the rigorous selection led to the determination of 25 database studies which meet all eligibility criteria.

4.2 Characteristics of included studies

The process of synthesizing findings has been done using Thomas and Harden's (2008) three-stage thematic synthesis. This involved three steps such as free line-by-line coding, organization of codes into descriptive themes, and abstraction into higher-order analytical themes that answer the questions of the review. This led to the emergence of four analytical themes in Section 4.4. This method was chosen instead of meta-analysis since the studies do not have common related measures nor similar designs and instead of narrative since it brings transparency and auditability to the analysis.

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Table 3. Characteristics and quality appraisal of the 25 included studies

Study	Technology	Design & Framework	Key findings	IoT/GPS integration enables real-time data feedback to successfully	Qual.
Ma et al. (2024)	Edge-cloud, IoT, & CPS	Empirical case study; Ceramic manufacturing survey; 212 employees	Digital transformation and green practices improve performance positively and mediate the impact of IoT-driven predictive maintenance, quality precision, and energy management.	optimize energy consumption and resource recycling	M
Khakwani et al. (2024)	Digital transformation	Conceptual review & multi-case; (PLS-SEM)	Time-fusion positively and IoT-driven predictive maintenance, quality precision, and energy management.	Time-fusion positively and IoT-driven predictive maintenance, quality precision, and energy management.	M
Mansour et al. (2025a)	AI & IoT (AIoT)	Panel manufacturing A-share listed companies	AI adoption and digital transformation mediate how ESG performance enhances IoT solutions facilitate energy efficiency and sustainability goals, contributing to firm participation positively moderates the ESG-firm performance link via efficient innovation.	AI adoption and digital transformation mediate how ESG performance enhances IoT solutions facilitate energy efficiency and sustainability goals, contributing to firm participation positively moderates the ESG-firm performance link via efficient innovation.	M
He (2024)	AI & Digital transformation	Survey & (2010–2021); 162 Italian manufacturing firms	ESG performance enhances IoT solutions facilitate energy efficiency and sustainability goals, contributing to firm participation positively moderates the ESG-firm performance link via efficient innovation.	ESG performance enhances IoT solutions facilitate energy efficiency and sustainability goals, contributing to firm participation positively moderates the ESG-firm performance link via efficient innovation.	H
Scarpino et al. (2026)	Industrial IoT (IIoT)	Panel: 21,064 obs. Chinese A-share (2010–2021)	Digital transformation enhances ESG performance mediated by risk-taking and moderated by top management education.	Digital transformation enhances ESG performance mediated by risk-taking and moderated by top management education.	M
Zheng & Bu (2024)	Digital transformation	Panel: 21,064 obs. Chinese A-share (2010–2021)	Digital transformation enhances ESG performance mediated by risk-taking and moderated by top management education.	Digital transformation enhances ESG performance mediated by risk-taking and moderated by top management education.	H
Aslam (2026)	Big data analytics	Panel: 30,448 obs. Chinese A-share (2010–2021)	Digital transformation enhances ESG performance mediated by risk-taking and moderated by top management education.	Digital transformation enhances ESG performance mediated by risk-taking and moderated by top management education.	H
Sang et al. (2024)	Digital transformation	Panel: 30,448 obs. Chinese A-share (2010–2021)	Digital transformation enhances ESG performance mediated by risk-taking and moderated by top management education.	Digital transformation enhances ESG performance mediated by risk-taking and moderated by top management education.	H
Cheng et al. (2025)	Big data analytics	Panel: 27,254 obs. Chinese A-share (2009–2023)	Digital transformation enhances ESG performance mediated by risk-taking and moderated by top management education.	Digital transformation enhances ESG performance mediated by risk-taking and moderated by top management education.	H
Wang et al. (2026)	Digital transformation	Panel: 27,254 obs. Chinese A-share (2009–2023)	Digital transformation enhances ESG performance mediated by risk-taking and moderated by top management education.	Digital transformation enhances ESG performance mediated by risk-taking and moderated by top management education.	H
Zhou et al. (2025)	AI	Quasi-experimental; 1,418 Chinese manufacturing firms	Big data infrastructure enhances ESG performance by reducing information asymmetry and improving internal governance, intensifying external monitoring.	Big data infrastructure enhances ESG performance by reducing information asymmetry and improving internal governance, intensifying external monitoring.	H
Wang & Zhang (2024)	Big data infrastructure	Panel: 1,418 Chinese manufacturing firms (DiD)	Big data infrastructure enhances ESG performance by reducing information asymmetry and improving internal governance, intensifying external monitoring.	Big data infrastructure enhances ESG performance by reducing information asymmetry and improving internal governance, intensifying external monitoring.	H
Sari et al. (2025)	AI	Machine learning empirical model	IoT technologies effectively support circular economy and socio-environmental practices.	IoT technologies effectively support circular economy and socio-environmental practices.	H
Cavalieri et al. (2022)	IoT	Conceptual review	IoT technologies effectively support circular economy and socio-environmental practices.	IoT technologies effectively support circular economy and socio-environmental practices.	M
Salma et al. (2025)	AI & IoT auditing	Empirical (PLS-SEM); 60 firms, 120 cases (Automotive)	AI and IoT improve ESG compliance and tracks Scope 3 emissions, though introducing energy trade-offs.	AI and IoT improve ESG compliance and tracks Scope 3 emissions, though introducing energy trade-offs.	H
Hu et al. (2022)	Green Industrial IoT	Structural equation modeling; Manufacturing	Identifies the external regulatory and internal efficiency drivers that govern Green IIoT adoption for eco-sustainability.	Identifies the external regulatory and internal efficiency drivers that govern Green IIoT adoption for eco-sustainability.	M

Zhang & Lei (2025)	IoT & Big data	Optimization algorithms; Industrial supply chains	Algorithmic resource balancing guided by IoT sensors delivers quantifiable ESG compliance gains.	H
AL-Shboul (2024)	Big data analytics	Covariance-based SEM; Supply chain manufacturing	Big data capabilities build persistent competitive advantages when channeled into sustainable sourcing.	H
Román-Salinas et al. (2026)	Industrial IoT (IIoT)	PRISMA 2020 Systematic Review; Production systems	Establishes mechanistic links between IIoT architecture and manufacturing energy/resource efficiency.	H
Wang & Zhang (2025)	Digital transformation	Structural equation modeling; Industrial enterprises	Internal firm integration and organizational digital culture moderate how digital transformation determines ESG performance.	H
Bogdan et al. (2025)	AI governance & disclosure	Empirical disclosure index; Industrial reporting	Automated ESG disclosures and AI reporting quality significantly improve external ESG scores.	H
Alshareef (2025)	AI	Panel analysis; Corporate disclosure systems	AI integration in sustainability reporting curtails selective disclosure, improving the reliability of ESG reports.	H
Huang & Xu (2026)	AI	Empirical panel; Heavy industrial enterprises	AI implementation improves mandatory environmental transparency by enhancing internal audit speeds.	H
Mansour et al. (2025b)	AI	Dynamic panel analysis; Industrial emerging markets	AI adoption amplifies green innovation efficiency and optimizes ESG governance, minimizing emissions abatement costs.	H
Sætra (2023)	Responsible AI	Theoretical and governance modeling	Establishes a foundational protocol for evaluating and auditing the socio-environmental implications of AI-driven ESG systems.	M

Qual. = quality appraisal rating (*H* = High, *M* = Medium); *PLS-SEM* = Partial Least Squares Structural Equation Modeling; *DiD* = Difference-in-Differences; *obs.* = observations.

4.3 Quality of the evidence

Not surprising, given that the corpus is from peer-reviewed venues and filtered for analytical content. Weaknesses were also fairly consistent, as they were poorly worded questions, unclear objectives, or endogeneity issues that were not properly addressed in the quantitative studies (such as with instrumental variables, difference in differences and staggered adoption designs). The narrower weaknesses are the following: Several studies conclude by studying a broad sample of listed firms, not EV firms; the measurement of what is called “digital transformation” varies from patent counts to text-mined indices, making it difficult to compare the studies; and the qualitative or design-science studies provide valuable insights but rely on small or single cases. These patterns are used to moderate the confidence of the individual claims and are reiterated in the limitations.

4.4 Thematic synthesis

Four analytical themes were generated from coding. The first two are about the environment and social dimensions and the technology that supports them; the third is about the governance and integrity of ESG information; and the fourth is about the strategic conditions that will create a success rate for each. The links between technology and applications and the resulting outcomes are summarised in Table 5, while the themes and their related theory are mapped in Table 6.

4.4.1 Theme 1 — Environmental intelligence

The evidence is most vast in what can be described as environmental intelligence: the application of AI, Industrial IoT and edge-cloud collaboration for viewing and guiding a company's environmental impact in real-time. In the heavy manufacturing and EV application, the concentration is on energy management and equipment life cycles. AI and IoT (AIoT) combine to make significant advances in predictive maintenance and smart energy management (Mansour et al., 2025a). Edge-cloud cooperation combined with Cyber-Physical Systems (CPS) enables data feedback in real time, which enables optimization of the energy use of equipment and resource recycling, following the principles of the circular economy (Ma et al., 2024; Román-Salinas et al., 2026). The literature, however, is cautious when discussing the pathway: sustainable performance is increased with IIoT and big data capabilities, but in turn, the results depend heavily on the level of participation in the firm and the internal efficiency drivers (Scarpino et al., 2026; Hu et al., 2022). The theme is perfectly suited to the natural-resource-based thinking view: The resource that can lower pollution is sensing and analytics, but the value comes from the work routines that use the data to drive action.

4.4.2 Theme 2 — Traceable and responsible value chains

The second theme is a collection of technologies that seek to transparency the supply chain, while the EV industry's strongest ESG exposure is in the upstream. When leveraging big data analytics for the purpose of sustainable sourcing and green supplier integration, the result is persistent competitive advantages are demonstrated (AL-Shboul, 2024; Cheng et al., 2025). In terms of social dimension, research on the impacts of digital supply chain transformation on ESG performance has clearly emphasized the importance of internal firm integration and an organizational digital culture (Wang & Zhang, 2025). Additionally, algorithmic resource balancing, based on IoT sensors, ensures

measurable ESG gains in industrial supply chains (Zhang & Lei, 2025). These are technologies that transform opaque and adversarial supplier relationships into auditable, accountable supplier relationships, as described in stakeholder theory.

4.4.3 Theme 3 — Governance, disclosure integrity and anti-greenwashing

The third theme is also the one where the most interesting recent research comes to light, and where the tool–outcome distinction is most salient. A series of studies reveal that AI implementation enhances corporate information governance which leads to better ESG performance, specifically by increasing information transparency (information asymmetry) and the quality of information disclosure (Zhou et al., 2025; Bogdan et al., 2025). Most obviously, AI-powered detection systems are found to reveal the symbolic versus substantive nature of ESG reporting and actively prevent ESG greenwashing (Sari et al., 2025). AI integration in sustainability reporting also helps to greatly reduce selective disclosure and increases forced environmental transparency by boosting the speed of internal audit (Alshareef, 2025; Huang & Xu, 2026). Legitimacy theory interprets these instruments as a public infrastructure for bridging the say–do gap, to make public environmental claims more accountable to data.

4.4.4 Theme 4 — Strategic integration, performance and barriers

The final theme weaves through the others, and is a key qualification of the review: the payoff from ESG technologies is contingent, not guaranteed. Digital transformation is a reliable driver that reinforces the ESG–performance connection, but it is also an enabler, as it enhances a correlation that is heavily dependent on green innovation to make a difference (Zheng & Bu, 2024; He, 2024; Wang et al., 2026). The institutional dimension comes to the fore when it is demonstrated that the digitalization capability of a firm plays a crucial role in translating a commitment to sustainability into real sustainable performance, where the initial mediator is operational sustainability (Aslam, 2026). Moreover, human and strategic conditions are significant boundary conditions, with corporate risk-taking being a major determinant of the success of digital transformation in boosting ESG performance and the educational level of the top management team being a moderating condition. This is the same across the corpus – the use of technology for compliance.

The final theme goes beyond these senses and is part of the core qualification of the review: the payoff from ESG technologies is not guaranteed, but contingent. Digital transformation can consistently enhance the ESG–performance relationship, but it is merely an enabler, as it is supported by the strong reliance on green innovation for its success (Zheng & Bu, 2024; He, 2024; Wang et al., 2026). The institutional dimension comes to the fore through the finding that the institutional ability of a firm affects its performance in delivering on its sustainability commitments, with operational sustainability as the key intervening variable (Aslam, 2026). In addition, human and strategic conditions are significant boundary conditions, as the ESG performance improvement through digital transformation is significantly mediated by corporate risk-taking, and moderated by the educational level of the top management team (Sang et al., 2024). The application is the same across the corpus: Technology is just used to meet requirements.

Table 5. Key ESG technologies, EV-sector applications and reported outcomes

Technology	Representative EV-sector application	Reported outcome	Primary pillar
AI / machine learning	Emissions analytics; greenwashing detection; internal auditing	Higher ESG scores; better disclosure quality; reduced greenwashing	E / G
IoT & edge-cloud systems	Real-time energy, emissions, and predictive maintenance monitoring	Efficiency gains via circular practices and reduced equipment downtime	E
Big-data analytics	Sustainable sourcing; supplier integration tracking	Improved supply chain transparency and competitive advantage	S / G
AIoT (AI + IoT Fusion)	Scope 3 emissions tracking; automated resource balancing	Quantifiable ESG compliance and advanced asset performance	E
ESG / digital platforms	Integrated reporting and automated transparency governance	Lower information asymmetry; auditable governance	G

E = environmental, S = social, G = governance. Outcomes summarise the direction of evidence in the included studies, not effect sizes.

Table 6. Mapping of analytical themes to theoretical lenses

Theme	What the evidence shows	Principal lens
1. Environmental intelligence	Sensing-and-analytics capability enables pollution prevention and stewardship, contingent on routines	Natural-resource-based view
2. Traceable, responsible value chains	Traceability tools convert opaque supplier ties into accountable stakeholder relationships	Stakeholder theory
3. Governance & disclosure integrity	AI/ledger tools raise disclosure credibility and curb greenwashing, managing the say-do gap	Legitimacy theory
4. Strategic integration & barriers	Adoption is driven by legitimacy pressure; performance depends on capability and integration	Institutional theory + RBV

5. Discussion

5.1 Interpreting the evidence against the review questions

To address the first research question, digital ESG technologies have a clear impact on sustainable business performance within the EVP, but not for all mechanisms. Assessing and managing assets precisely for environmental gains. Governance improvements occur as a result of improved and more trustworthy information. Financial gains are witnessed when governance is strong and technology is fully incorporated in the business strategy. The 25 studies in this corpus are insufficient to back the general argument that firms in general that implement ESG technologies simply perform better. Rather, they demonstrate that specific technologies have specific impacts on specific results, and they show how.

When it comes to the second question, it's about strategy. Digital transformation is the strongest cross-cutting finding: digital transformation is a moderator. It is a way of strengthening a link between an ESG and a performance that companies must first create. This is the applied and practical application of the resource based approach: the advantage is in the capability to embed the technology within the organization, not in the technology itself.

The enablers and barriers for the third question are very apparent. Strategic integration, organizational readiness, top management education, and supportive institutional environment. Challenges are high implementation costs, lack of common data standards, and the danger of 'showing up' to show they are adopting. The same institutional pressures that incentivize companies to use these tools can also lead to "lip service only" compliance if the company isn't really digitally capable.

5.2 Contribution and the gap addressed

This review has three major contributions. First, it brings together a rapidly emerging evidence base into a sector-specific synthesis for the EV industry. It supersedes anecdotal descriptions of big car companies with a focused presentation of 25 peer-reviewed studies. Second, it is strictly keeping the analytical distance between the tools of the ESG and the ESG outcomes, which is not maintained in a lot of the previous studies. The review achieves this, finding the right mechanisms, e.g. green innovation, and the boundary conditions that link the two together. Third, it provides solid theoretical underpinning by linking the evidence to theories of the stakeholders, institutions, natural resources, and legitimacy. The study does not only provide empirical evidence of the relevance of these technologies in the field of EV, it also provides the concepts of why and when.

5.3 Implications for practice and policy

Managers: apply the strategic concept to ESG technology and not merely a reporting cost. The economic benefits are for companies that use these tools to be more deeply embedded in their operations and governance. Instead, a company should use an emissions analytics platform in conjunction with the authority to influence real data-driven decisions, rather than just using it for appearances. Investment in the skills of employees and top management readiness is as crucial as the technology.

This review suggests that standardisation is the most promising intervention for policymakers. The issues of interoperability that emerge in various studies could be addressed by common rules and battery traceability standards. Moreover, implementing credible assurance requirements would

enable AI truth detectors to make a significant impact in preventing greenwashing. Early examples of such coercive yet enabling pressure are the EU Battery Regulation and digital battery passports.

5.4 Limitations

There are two restrictions in this study. As far as the evidence is concerned, the data is geographically biased towards the Chinese listed firms. China is the industry leader for EVs, making it difficult to generalize the results to other regulatory markets. Long-term effects are also not yet apparent in the field, as it is very young. Also, there are several studies that indirectly demonstrate their relevance to the EV space by utilizing a wide sample of manufacturing companies rather than just testing EV companies.

As to the review process, one main database was used. Google Scholar is appropriate for the study design, but has some disadvantages in terms of coverage compared to a multi-database search. The review focuses on English language work; it did not get registered on PROSPERO; and it is of a qualitative synthesis type, where interpretative judgment occurs during building the themes. These risks are minimised but not eliminated by the strict protocol and quality appraisal.

5.5 Future Research

There are four directions of action that stem from gaps identified. To begin with, there are no reliable firsthand findings for the field that confirm the relationships that have been hypothesized between EV research and development and manufacturing firms in general. First of all, there is no direct field evidence to confirm the hypothesized relationships between EV R&D and all manufacturing firms, but only evidence for those relationships based on general manufacturing firms. Second, longitudinal and panel designs should follow up to determine if early performance effects are either maintained or faded or strengthens as these technologies develop. Thirdly, the researchers should delve into the actual organizational capacities needed to foster the technology into performance, hopefully through qualitative case studies within EV companies. Fourth, with the rise of AI from measurement of ESG data to decision-making, the governance of AI becomes an important research area. There are multiple challenges that must be addressed, such as algorithmic bias, data security and the ethics of automated assurance, which have yet to be explored by the current sustainability literature, and which urgently require focus.

6. Conclusion

The aim of this systematic review was to identify digital ESG technologies and conditions for their use in supporting sustainable business performance in the challenging context of the electric vehicle industry. After reading through 4 theoretical lenses and reading 25 peer-reviewed studies since 2022, it concludes that these technologies are real enablers of sustainable performance, but selective and conditional. AI, IoT, big data analytics, and edge-cloud systems refine environmental monitoring, big data and IoT tracking systems make value chains traceable and responsible, and AI-driven governance tools increase the trustworthiness of environmental, social, and governance (ESG) information and increase the ability to prove greenwashing is reduced. However, a common theme emerges in every theme: value is created in the context of organizational capability and strategy that incorporates technology, and is reduced by cost, skill and standardization challenges. The message is evident in an industry that is increasingly being held to account for its sustainability claims, especially for an EV sector. What is no longer a question is whether to embrace ESG technologies, but whether a firm is able to develop the capacity to transform the technologies from disclosure tools into actual, verifiable performance tools.

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PERCEPTION OF GEN Z TOWARDS SUSTAINABLE CONSUMPTION AND LIFESTYLE CHOICES IN INDIA

URL: <https://docs.google.com/document/d/10U4uhPhF3gwgsB3W7xv3ahEBYoSYiuC/edit?usp=sharing&ouid=110403252814827643>

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ABSTRACT

Sustainable development is an issue of global significance, urging conscious decisions and lifestyle changes from each individual, especially from younger generations. Gen Z as one of the most active and well-informed generation plays an indispensable part in forging a sustainable future for the years to come. The present research paper endeavors to investigate Gen Z perception towards sustainable consumption and life style practice in India with an objective to explore their awareness level, attitude and behavior patterns in relation to sustainability and responsible consumption. “Primary data has been collected through a structured questionnaire of five points scale (Likert Scale)”. The survey was conducted by the investigator through Google forms, collecting response from Gen Z of various age groups. The data was analyzed in order to bring out meaningful and useful insights from the responses. It is proved by the results that the majority of Gen Z respondents hold positive attitude toward living a sustainable life, as well as have the knowledge of the sustainable development. Many people from young generations stated willingness for reduction of waste and usage of green products, and supportive towards green practice in their life. Internet platforms, educational institutes and government efforts can contribute a lot for promoting awareness as well as behavior regarding the sustainable development. In conclusion, Gen Z can do a lot for the sustainability development through lifestyle modification and consumer practice; moreover, the policy makers, educationists and researchers need to look forward to increase the awareness programs for young people and its proper implementation with efficient government policies and technology usage to enhance sustainable development for the nation and nation at global level.

Keywords: Gen Z Perception, Sustainable Lifestyle, Green consumption, Youth Behavior, Environmental Sustainability, India

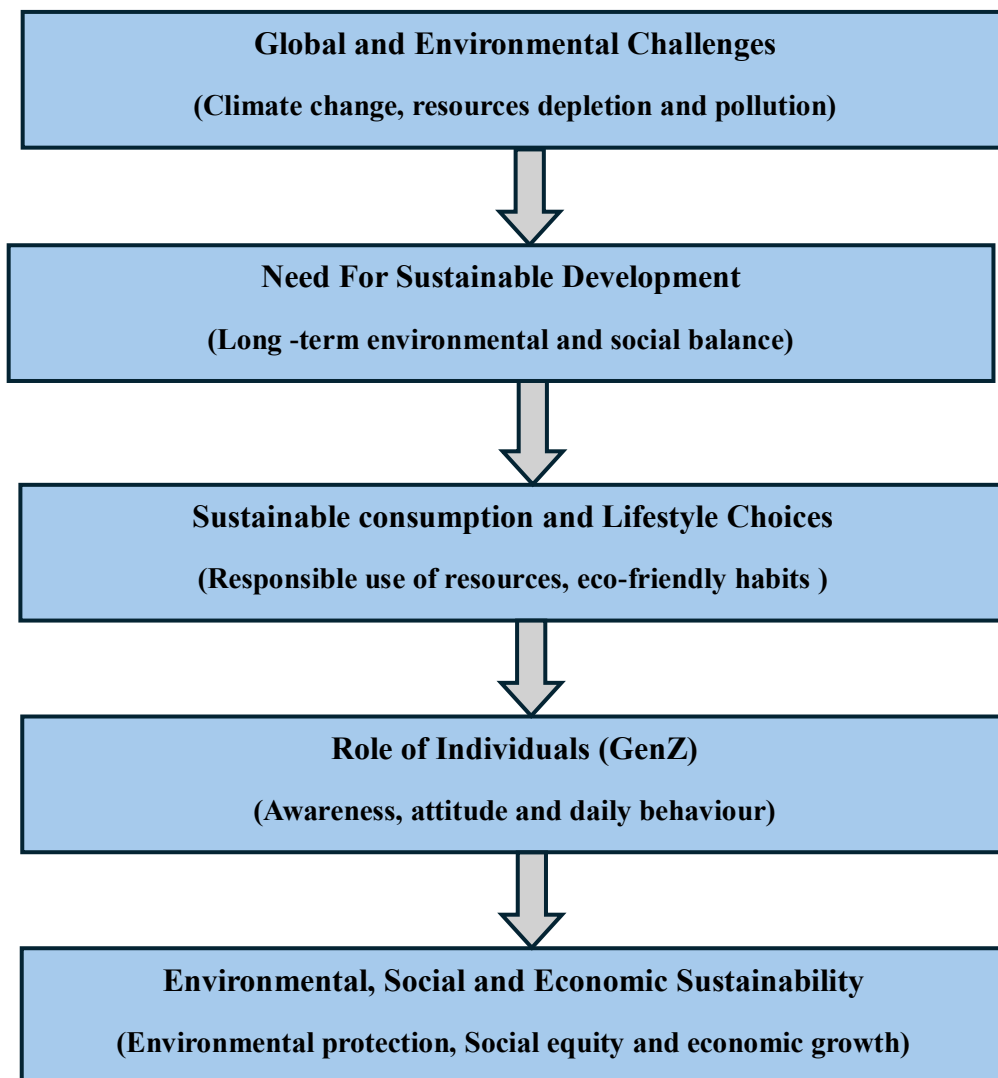
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INTRODUCTION

In recent years sustainable development has taken center stage as an important development concept and policy challenge as concerns about the environment, climate change, and natural resource depletion are growing worldwide. Sustainable development has been accepted by governments as a policy imperative and demands changes not only at the macro- policy level but also in individual behavior and lifestyle choices. Consumption choices particularly by the young generations have been recognized to play an instrumental role in determining environmental and social impacts and therefore Sustainable Consumption and lifestyle choices have now become major themes of debate within the sustainable development discourse. “Gen Z” commonly defined as born between “1997-2014” is the first generation having grown up with an overwhelming exposure to digital technologies, social networking and global information flows. “Gen Z” relative to the older generations, is also believed to know more about ‘environment issues like protection of environment, global warming, sustainability and etc.’

Fig1:- Conceptual Overview Of Sustainable Consumption and Lifestyle Choices



Source : Created by the Researcher

The “**fig. 1**” below conceptually shows the interconnections between global environmental problems and SDGs. The figure also emphasizes on the contribution of sustainable consumption and lifestyle of Gen Z in attaining environmental, social and economic sustainability.

In the context of India, ever-increasing pace of urbanization, population growth and changing lifestyles are exerting immense pressure on natural resources. Various sustainable development policies, schemes and campaigns have been introduced to make an attempt to promote sustainable consumption. However, in achieving the aim of sustainable development, the crucial elements for a successful progress are the understanding and behavioural change in the public and citizens of a country. Considering the above importance, it becomes very significant to analyze the perception of Gen Z about sustainable consumption and lifestyle. In this way, we can assess the participation and awareness of youth generation (Gen Z) in achieving sustainable development goals. This paper aims at analyzing the perception of Gen Z about sustainable consumption and lifestyle.

The study is based on 5 key Concepts

1. Sustainable Literacy

“A person’s level of knowledge and awareness of green products and sustainable consumption behavior, including eco-label recognition, awareness of benefits, and product impact on the environment and human health, is known as sustainable literacy.” People are more inclined to make thoughtful purchases as a result of this insight. However, real sustainable purchasing behavior does not always follow from awareness about sustainable items and processes.

2. Social Influence

The impact of friends, family, and society on people’s opinions and purchasing behavior has been widely addressed as social influence. As such, it has also been widely considered as an important determinant of sustainable consumption. This becomes particularly salient in younger consumers, as a pro-environmental stance can be facilitated by observing other people behaving in accordance to this concept. Social approval from and alignment with peers may also encourage purchase of sustainable products.

3. Sustainable Purchasing Attitude

Sustainable purchasing attitude can be understood as individuals' evaluation of the buying behavior towards eco-friendly product. An individual with sustainable purchasing attitude toward green product possesses a positive mindset regarding purchasing eco-friendly product, and will likely develop belief toward that product "will have positive impact toward the environment" and health. The power of the purchasing purpose to buy "eco-friendly product" will depend on the attitude towards such product.

4. Sustainable Purchasing Behavior

The term "sustainable purchasing behavior" describes consumers' choices of "environmentally-friendly products" and eco-shopping, such as "natural food products," "products with recycled packaging," or "products with low environmental damage effect. This reflects the practice of the "Sustainable Purchasing Attitude" by consumer. This behavior has a long-term perspective of impacting the environment more positively, hence lead to overall sustainable living.

5. Sustainable Lifestyle

Sustainable lifestyle refers to the lifestyle practices toward minimize impact on the environment; it involves such practice as the way people live daily through the aspect of their consumption, which will likely consist of reduce waste, energy conservation and purchase of eco-friendly product. Once sustained purchasing behavior exists in long term as habitual action in the consumers' behavior, it will eventually form a sustainable lifestyle in the individuals.

Research objectives

The following goals have been established for the study:

- To determine the perception of GenZ towards Sustainable Purchase Behaviour
- To determine the perception of GenZ towards sustainable lifestyle
- To analyse previous literature related to Sustainable Behaviour of GenZ
- To make a theoretical Framework (model) based on Sustainable Behaviour of GenZ

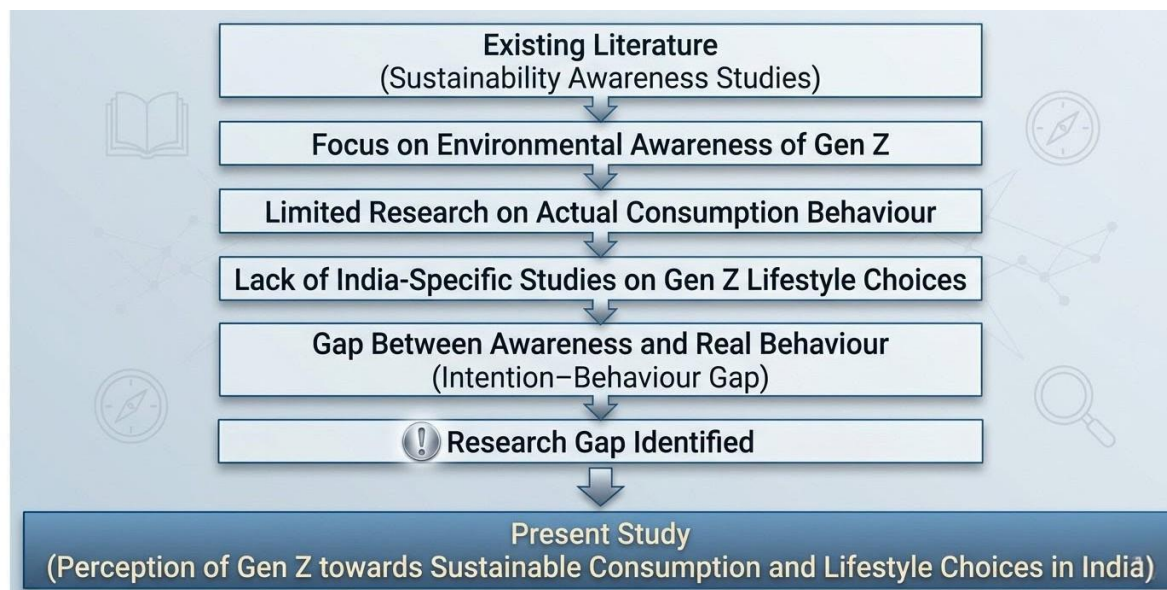
Research Gap

Past literature explored sustainable development, consumption, and behavior under various perspectives of governmental initiative, responsibility of business and behavior of consumer. Some researches were also carried out in India; There is still scope for more research on Generation Z, especially in relation to sustainable consumption and lifestyle choices. Especially the linkages of various factors namely sustainable literacy, influence from social element, sustainable purchase attitude, and sustainable purchase behavior with sustainable lifestyle have received less attention.

In most cases of the studies, a separate concern of purchase towards sustainable consumption was discussed whereas; these factors have not been sufficiently discussed when taken as a whole in relation to sustainable lifestyle in the Gen Z-focused framework.

Hence, this study attempts to address the research gap on the perception of Gen Z consumers towards sustainable consuming lifestyle through an empirical study based on primary data collected from respondents in Delhi NCR, so as to provide useful insights on the basis of empirical evidence and young focus in the future by researchers, awareness programs, policy makers and different campaigns towards sustainable consumption habits.

Fig2: Research Gap of the Study



Source: Self Developed by Researcher

LITERATURE REVIEW

Authors offers a theoretical background to explain consumers' behavioral responses, the Theory of Planned Behavior (TPB). TPB posits that three factors (attitude, subjective norms, and perceive behavioral control) jointly account for variation in behavior. These concepts, in the area of sustainable consumption, contribute to the understanding why some consumers will shape a positive attitude towards an environmentally responsible products/ practices and behaviors. The present study will attempt to examine the determinants of sustainable consumption behavior by adapting the original theory as the theoretical support instead of direct hypothesis testing the original TPB theory. Based on previous researches, the theory has been extended and new variables have been explored in this study by using Sustainable Literacy, Social Influence to construct antecedents of Sustainable Purchase Attitude which subsequently led to Sustainable Purchase Behavior and Sustainable Lifestyle. Sustainable Literacy will refer to consumer' understanding and awareness toward the sustainable consumptions, while Social Influence refers to the role of others towards the Sustainable Purchase Attitude. (Ajzen1991)

Authors studies the factors affecting green purchase behavior among consumers. They found that environmental concerns and ethical responsibility impact the sustainable buying decisions. “The researchers observed that when consumers feel morally committed towards the protection of the environment then they emphasized that intentions may not always translate into actions, but they did favor eco-friendly goods. They showed that many consumers expressed their concerns towards the environmental protection, but failed to change their purchase behavior due to situational constraints. This study is significant as it presents both motivational and practical factors of the green buying behavior. It showed that awareness towards the environment is not only enough, along with the awareness availability and purchase price are also important factor and relevant in developing countries like India. (Joshi & Rahman2015).

The Effects of EA and customer perception efficacy on “green buying intent” has been examined by authors. It showed that people are quite concerned about purchasing “eco-friendly goods.” when they feel that the individual effort towards buying sustainable products lead to some positive change towards the environmental protection. It has been found that confidence of consumers towards their behavior impact greatly over their purchase decision related to sustainable consumption. If the consumers do not have confidence over the individual effort towards environment, the behavior cannot be expected. It also indicated the importance of the effective communication with the consumers that would make them feel more effective and confident. These are important as the habit of sustainable buying is being formed among young consumers. (Kumar& Ghodeswar2015).

Authors have explored Sustainable consumption behavior among Indian youth. From this study it was revealed that even though Indian young consumers hold Positivity toward the preservation of the environment and sustainable products, but the actual purchase behavior of young consumers for the sustainable product is quite low. Higher prices and non-availability of eco-friendly products were considered as the major reason for this. Thus, it is realized that consumers are motivated with the attitude towards the environment but deterred by economic and practical concerns. It highlighted the unexpected between consumers' purpose and their purchasing habits in relation to India. This study is highly relevant to the current GenZ consumer's study. (Sharma &Jha2017).

Authors investigated how environmental concerned impacts the green purchase intention. Their findings revealed that, the individuals holding a significant care for the environment are more likely to cultivate an optimistic mindset and the purpose to invest in ecologically beneficial goods, but the study also discovered that the intention did not result in the actual purchase behavior because of things like time limits, convenience, cost considerations, and lifestyle limitations. This finding suggested that even the environmentally conscious individuals may not always be seen adopting green purchase behavior due to the behavioral and situational factors which must be taken into account jointly in the behavior of the young consumers in urban setting. (Paul et al.2016).

Authors examined sustainable behavior among young consumers of Generation Z. It was found that the Generation Z is highly aware and eager to support green products and brands, but they are influenced by many factors such as affordability, authenticity, quality etc. The study highlighted that trust played a very important role in Green Marketing. Even if they had good intention to support green brand, they had to lose trust if green claim is doubted. Awareness needs to be supported by communication about products through transparency and credible channels. These finding provided a good insight about young Indian consumer and support studying both awareness and behavior. (Verma & Chandra 2018).

This study indicated that social influence and peer behavior significantly impacted youngsters' consumption patterns towards sustainable products. It suggested that young consumers rely greatly upon peer influence in their buying decisions. Individuals will be prone to adapt to green behavior if the same is socially accepted or if the members of their social circle praise for the particular behavior. This research clearly reflected the importance of social approval and peer pressure on establishing sustainable lifestyle among young consumers. It also pointed out that social campaigns could be launched to increase responsible consumption behaviors among youth. This aspect seems to be relevant in the context of Generation Z who are highly connected to social media. Their behavior tends to be shaped by trends, influence from social circles and social comparison, so it is not surprising that the role of social influence emerges as one of the important determinants of their sustainable consumption behavior. (Biswas &Roy2015).

Authors discussed discrepancy between attitude towards environment and purchasing behavior. They showed that most of the consumers held positive attitude toward the env. However, they don't "consistently buy green products." It concluded that the disparity was caused by things like "high price, lack of information, convenience," etc. The study indicated that barriers to purchase would always play an external role over consumer decision making processes and positive intention is never enough to bring up sustainable behavior of individuals. Businesses and policymakers need to decrease external barriers and in relation of younger consumers it should be the center. (Young et al.2010).

The study by (Kumar, Prakash and Kumar 2021) focused on the environmentally responsible purchase intention of Indian consumers, considering eco-friendly apparel. The results concluded that consumers had favorable attitudes toward green apparel and that subjective norms, perceived behavioral control, and environmental consciousness positively influenced their purchase intention. The study also revealed the significance of customer education on environmentally friendly goods and encourages sustainable buying habits.

In their research, (Uddin & Khan 2016) investigated green purchasing behaviour among Indian young people. This study indicates that environmental attitude, environmental awareness, environmental involvement, perceived effectiveness of environmental behaviour were the key attributes that impact green purchasing behavior. One crucial observation made in this study was a value-action gap; that awareness and attitudes towards the environment do not necessarily translate into action (i.e. green buying intent) among young Indian consumers, implying that other attributes may hinder them from translating their environmental concerns into action.

The study emphasizes that young customers in developing nations like India are becoming increasingly environmentally conscious but lack support from institutions to continue sustainable lifestyles. The research concluded that raising awareness alone is not enough to encourage such lifestyles and that there is a need for government and institutional support. The study also suggests that while Gen Z is showing interest in sustainability they still face obstacles that prevent them from taking necessary actions, hence Their behaviors and their consciousness are not in sync. (Kumar et al. 2017)

A study conducted by (Gupta and Acharya 2019) researched green consumption of young Indian consumers where all personal factors, social and environmental issues do an important role to contribute Green consumption. Also found that young consumers take decision regarding making green consumption on different points; it does not look like to depend only on green awareness for green consumption.

The study explored green consumption behavior among Gen Z consumers in Indian cities, finding that social influences, education, and environmental awareness significantly impact purchasing decisions toward green products. Peers, family, and media played important roles in shaping consumer choices, and younger, more educated respondents were more likely to adopt eco-friendly practices. However, barriers like price and availability remained significant factors,

clearly illustrating the multitude of issues that influence Gen Z's perception toward sustainability. (Jyothi & Mohan 2025)

(Meet, Kundu and Ahluwalia 2024) investigated variables affecting green purchase intention of Gen Z consumers in India. The results found that green price, green advertisement and green packaging positively affected the purchase intention but greenwashing has negative affect. Besides, the study indicated socio-demographic variable such as gender, education and income are affecting Gen Z green purchase intention. Thus, practical and marketing determinants is not only one thing that shape the Gen Z green buying.

The researchers found that consumer faith in responsible products and green claims is a primary determinant of behavior, and a lack of it can be a deterrent. The research also emphasizes the need for transparent communication with credible green claims, which is notably important for Gen Z who tend to be critical of greenwashing. (Chen & Chang 2013)

The author explained that sustainable consumption encompasses not only buying "green products" but reducing overall consumption levels and adopting certain life styles, thereby promoting long term behavior change. This concept is critical for Gen Z as their consumption habits significantly impact sustainability as a whole, thus broadening the scope of sustainable consumption to lifestyle changes rather than consumption. (Peattie 2010)

The authors have identified Social and psychological aspects such as habits and lack of motivation that limit behavioral changes towards sustainability, despite consumers being aware of environmental issues and believing such actions to be right. The authors conclude that while awareness may have increased but these social barriers must be taken into consideration to facilitate behavioral change. (Gifford & Nilsson 2014)

Author argued that habitual behavior and lifestyle are extremely significant for sustainable consumption. According to the author, an individual's daily habits, routines are critical in determining the person's actual consumption behavior although the individual is concerned about environmental problems. He believed that a long-term shift in behavior is needed for sustainability. The researcher points out that consumption for sustainability is a habitual lifestyle, not an event. It is also found that social norms and personal values also affect consumption behavior. It is especially important in relation to Gen Z that the individuals' daily behavior and lifestyle will lead to considerable impacts toward environment. These results propose for the necessity to encourage sustained behavior in younger consumers. (Thøgersen 2010)

Fig 3: Summary of Major Studies

Authors & Year	Focus of Study	Key Findings	Relevance to Present Study
(Ajzen1991)	Theory of planned behaviour	Behaviour depends on attitude social norms, and perceived control	Helps explain Gen Z decision-making in sustainable consumption
(Joshi & Rahman2015)	Green Purchase Behaviour	Environmental concern influences intention but not always action	Shows intention-behaviour gap
(Kumar & Ghodeswar2015)	Consumer Effectiveness	Belief in personal impact influences green behaviour	Explains role of perception in Sustainability
(Sharma & Jha 2017)	Youth Sustainable Behaviour	Positive attitude but low actual purchase due to price & availability	Highlights practical barriers in India
(Paul et al.2016)	Environmental concern and Behaviour	Concern affects intention but lifestyle constraints affect action	Shows real – time Limitations
(Biswas & Roy2015)	Social Influence	Peer pressure & social norms affect Sustainable choices	Important for Gen Z behaviour

The literature shows that Generation Z is aware of sustainability, but price, convenience, and social influence”. Particularly in the Indian culture, there’s a discrepancy within intended and actual actions.

(Kim & Stepchenkova, 2020) These works suggest sustainable literacy encompasses the consumer’s awareness and knowledge of environment friendly products and practices. It includes consumers’ awareness on eco labels, environment impact, benefit on buying green products. It is suggested that higher literacy levels may lead to positive sustainable product attitude, yet it cannot determine actual purchasing behaviour. Therefore, it should be accompanied by other factors to make it effective for buying actions.

(Lee,2008) The influence of the “social factor on consumer attitude and behaviour”. Consumers are guided by peer, friends, and social network in consuming choices. In the context of

sustainable consumption, social influence determines consumers to abide by social norms and expectation of consuming eco-friendly products in the community, group and association, thus influencing their attitudes toward the sustainable products.

(Nguyen et al., 2016; Ajzen, 1991) These studies state attitude towards sustainable buying is a determinant for actual purchasing behavior in the sustainability context. Ajzen's TPB explains positive attitude to a behavior will lead to more likelihood to performing behavior. When a consumer thinks that buying a green product can contribute to the welfare of the environment and their well-being they can form a positive attitude to the green product, hence influence consumer intention to obtain the "green products".

(Dangelico & Pontrandolfo, 2010) The research describes sustainable purchase behavior as a purchase decision and action that takes environmental criteria into account when making decisions concerning purchasing products. It encompasses a consumption that favors organic and recyclable products and also includes consumption pattern that minimize environmental impact in making a purchase. Sustainable purchase behavior means a consumer is performing the decision and intention based on their attitude. Thus, sustainable purchase behavior contributes to the protection of environment.

(Jackson, 2009; Lubowiecki-Vikuk et al., 2021) This describes sustainable lifestyle as an integrated pattern of life through adopting various habits and actions toward sustainability in a long-term basis. It covers the adoption of practices such as waste minimization, energy conservation, and eco-friendly consumption pattern that eventually contribute to an environmentally sustainable lifestyle.

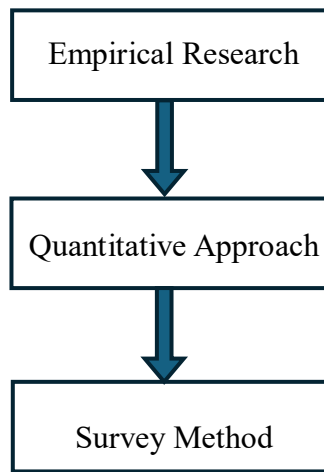
In summary, the studies surveyed reflect that there are various factors that affect the consumption choices toward sustainability among young consumers and include the variables like sustainable awareness, social influence, attitude, price, convenience, availability, and trust. While there were findings indicating the strong positive relationship between sustainable awareness, favorable attitudes and sustainable consumption behavior but at the same time, the existence of positive attitudes can sometimes does not predict sustainable buying behavior in young consumers. Likewise, social factor and external factors might have strong impact on young consumers decision making. Hence it is evident that single factor does not determine a sustainable consumption behavior and the integrated approach needs to be examined in while analyzing the sustainable lifestyle of Gen Z.

RESEARCH METHODOLOGY

Research Design

The current research adopted an empirical research design because it makes use of first hand collected data from respondents. This method of study was employed as the current research aims to get the perception of “Generation Z” about the “sustainable consumption” and lifestyle pattern in India. Empirical research helps us to acquire in-depth and real-life information on attitude and awareness of the respondents on chosen topic.

Fig.4: Research Design



Source: Self Developed by Researcher

Nature of the study

In terms of approach, this is quantitative as it has gathered figures via the structured questionnaire. With quantitative approach objective conclusion can be drawn through the result in statistics.

Population of the study

The study's sample is made of GenZ “individuals” in India. Most of the population in this range will fall into the age bracket of 14-28 years old. The age was chosen because Generation Z is thought to be most receptive of and aware of topics related to sustainability.

Location of the Study

All the respondents surveyed in the study were based in Delhi NCR. Delhi NCR was chosen for the survey due to the ready availability of Gen Z respondents, particularly college and school students. Delhi NCR provides a well-suited environment for observing sustainable consumption

and lifestyle choices among young consumers, and this was the main reason for choosing it for collecting primary data from Gen Z respondents for the present study.

Sample size & sampling Technique

The number of total respondents that was participate in the research is 412. The sampling that have been used is purposive sampling. This is because the research was specifically conducted on Generation Z respondent, where the purposive sampling technique is very suitable to be used, as it enables the researcher to pick the participants that fit in to the criteria required. Besides, time and resources are also a reason in selecting the technique used.

Fig. 5: sampling Details of the study

Elements	Description
Population	Generation Z
Sample Size	412 respondents
Sampling Technique	Purposive Sampling
Location	Delhi(NCR)

Source : Self-Developed by the Researcher

Data collection method

Data has been gathered through primary data collection and the research was conducted by a structured questionnaire design using Google Forms which has been distributed on social media like WhatsApp, Instagram to the Gen Z respondents. A “5-point Likert scale” has been used in the questionnaire and asked to provide their agreement from

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

Response Option	Scale Value	Description
Strongly Disagree	1	Respondent completely disagrees with the statement
Disagree	2	Respondent disagrees with the statement
Neutral	3	Respondent is unsure or indifferent to the statement
Agree	4	Respondent agrees with the statement
Strongly Agree	5	Respondent completely agrees with the statement

Fig6: Likert scale used in Questionnaire

Source: Self Developed by the Researcher

Research Instrument

The study used a self- created questionnaire as its research tool, consisting of 14 statements relating to sustainable consumption behaviour, life style, environmental concerns and responsible consumption, the question was prepared with simple language so as to make it clearly understood by everyone.

Before administering the questionnaire to obtain the final respondents' data, it was checked by the lecturer to ensure that the statements were appropriate to the research topic and understandable to the respondents. Modifications were carried out according to the recommendations before the actual questionnaires were distributed.

Data Analysis Tools

Software called the (SPSS) was utilized to examine information gathered for this study. Various Analytical approach like “significant positive relationship” was performed using SPSS and were applied to understand the perception of the Gen Z about sustainable lifestyle and consumption pattern in India. The findings of the same provided good insights for the objectives of the study.

Ethical Considerations

The subjects were made to know that the data will be kept confidential, there will be no personal information required and it will be used only for academic purpose.

Limitation of the study

The research is restricted to Generation Z respondents only and the sample size is restricted to 412 respondents which may not fully represent then tired population. The study’s self-reported data may contain personal bias. Given that data were collected by self-report, social desirability bias may have influenced responses. That is, individuals may have reported that they hold more

environmentally positive attitudes or engage in more environmentally beneficial actions than they actually feel or perform. This is recognized as a limitation of the study.

Chapter Summary

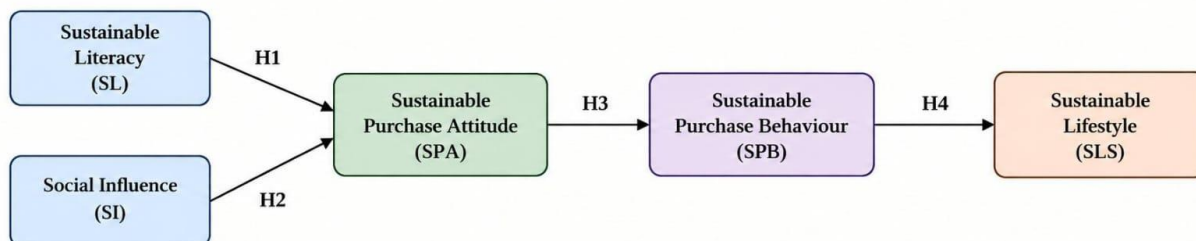
This lesson is divided in sections describing RD, DC procedure, sampling technique and “Research instrument” along with research tools employed for analysis of the data collected. The research methodology followed enables us to study in a structured manner, the perception of Generation Z about sustainable consumption and lifestyle options in India.

Theoretical Framework and Hypotheses

This study is grounded in a theoretical approach which proposes associations among the main variables concerning sustainable consumption and lifestyle. Specifically, it postulates sustainable literacy and social influence as variables concerning sustainable purchase attitude, which is directly associated with sustainable purchase behaviour and the latter with sustainable lifestyle, whereas it takes the practical constraints of sustainability purchase behaviour on positive attitude into consideration.

Fig 7: Theoretical Framework of the Study

Figure 7: Theoretical Framework



Notes:

- H1: Sustainable Literacy (SL) is positively related to Sustainable Purchase Attitude (SPA).
- H2: Social Influence (SI) is positively related to Sustainable Purchase Attitude (SPA).
- H3: Sustainable Purchase Attitude (SPA) is positively related to Sustainable Purchase Behaviour (SPB).
- H4: Sustainable Purchase Behaviour (SPB) is positively related to Sustainable Lifestyle (SLS).

Source: created by the Researcher

Note: The adapted framework is based on the original framework informed by the Theory of Planned Behavior. This adapted framework does not empirically test all the constructs from the original TBP framework.

DATA ANALYSIS

The acquired data was examined by using SPSS in order to test the reliability of all measurement items and test of associations among the variables. Descriptive analysis were performed in order to describe the respondents' answers whereas correlation analysis was performed in order to test relationship between the tested variables. The findings of the data analysis are as follows;

Reliability

Reliability indicates the consistency and stability with which a measurement tool can repeatedly generate exact outcome under exact context. Reliability implies the degree to which the scale is not sensitive to chance errors and will yield the same outcome under the same context.

Reliability ensures that when the scale is used under the same conditions the results are not contradictory. It is an important tool in the research area because one has to maintain accuracy and trustworthiness of the findings. One of the popular approaches used to measure “reliability” is Cronbach’s Alpha. (C. R. Kothari, 2004)

Cronbach’s Alpha

“Cronbach’s Alpha” is a statistical evaluate of internal continuity, and hence reliability of a set of scale items. It measures the degree of interrelatedness within the set of items, suggesting if a group of items measure the same conceptual domain. If a Cronbach’s Alpha is higher, it means that the reliability of measurement is better among a group of variables. This is most often used for testing the reliability of survey questionnaires and data sets in social science research. (Lee J. Cronbach, 1951)

The level of internal continuity of the “variables” can be ascertained by “Cronbach’s Alpha”. If “Cronbach’s Alpha” level’s ≥ 0.7 then the reliability of the scale is acceptable. Higher the level, the greater the consistency.

(Joseph F. Hair Jr. et al., 2010)

Fig8: Reliability Analysis

Reliability Analysis

Variable	No. of Items	Cronbach's Alpha
Sustainable Literacy	3	0.849
Social Influence	3	0.819
Sustainable Purchase Attitude	3	0.833
Sustainable Purchase Behaviour	2	0.754
Sustainable Lifestyle	3	0.770

Source: Created by Researcher using SPSS Software

In order to evaluate the uniformity within the dimension scale used in the survey, a “Cronbach’s Alpha reliability analysis” was performed. That’s shown in the table, all the constructs had the value greater than 0.7, which is accepted as a general rule of thumb to determine reliable measurement. The three scales, Sustainable Literacy (0.849), Social Influence (0.819), and Sustainable Purchase Attitude (0.833) have a high level of reliability. Although Sustainable Purchase Behaviour (0.754) and Sustainable Lifestyle (0.770) have values that are relatively lower than others, it is still within acceptable limit. It is thus inferred that the measuring instrument employed in this survey is reliable.

Kaiser-Meyer-Olkin (KMO) and Bartlett’s Test

They are analytical indexes that verify the relevance of the data to Factorial Analysis. KMO test tests whether there is adequate data in order to detect structure; in general terms values close to 1 are adequate for factor analysis and it is commonly believed that any KMO above “0.6” is Reasonable. “Bartlett’s Test of Sphericity tests” whether the variables are complimentary with one another and if the “significance level is less than 0.05” variables are suitable for “factorial Analysis”

(Henry F. Kaiser, 1974; M. S. Bartlett, 1954)

Fig 9: KMO & Bartlett's Test

KMO and Bartlett's Test^a		
Measure		Value
Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy		0.953
Bartlett's Test of Sphericity	Approx. Chi-Square	3793.185
	df	91
	Sig.	.000

Source: Created by Researcher using SPSS Software

To determine whether the data were suitable for factorial analysis, the KMO assesment of sample adequacy and Bartlett's Test of Sphericity were used. As reported in the table, "KMO value' of 0.953 is shown, which is excellent and shows excellent sampling adequacy and it means that the data fits the structure detection very well. What's more, Bartlett's "Test of Sphericity" has a "chi-square value" of 3793.185 with 91 degree of freedom, and the sig. is 0.000 which is less than 0.05, showing that the variables are inter-correlated at a significant level and it fits to factor analysis. In summary, the data are applicable to be further analyzed with factor loading to test validity.

Validity

The degree to which a device measures what it is supposed to measure is known as its validity. A measurement tool has validity when the results obtained actually measure the concept in question, and not influenced by irrelevant factors. A valid tool increases the trustworthiness and accuracy of findings, and in empirical study validity proves that the concepts being measured are actually represented by the variable used.(Joseph F. Hair Jr. et al., 2010)

Factor Loading

Factor loading refers to the correlation between latent constructs that underlie observable variables. It indicates how strongly each item contributes to a particular factor. Higher factor loadings represent stronger relationships and better construct validity. A commonly accepted

threshold is 0.6 or above, which indicates that the variable significantly contributes to the factor and is considered acceptable for analysis. (Joseph F. Hair Jr. et al., 2010)

Fig10: Factor Loading Table

Factor Loading Table		
Factor	Item Code	Factor Loading
(1) Sustainable Literacy	SL1	0.789
	SL2	0.744
	SL3	0.814
(2) Social Influence	SI1	0.796
	SI2	0.781
	SI3	0.798
(3) Sustainable Purchase Attitude	SPA1	0.788
	SPA2	0.818
	SPA3	0.693
(4) Sustainable Purchase Behaviour	SPB1	0.724
	SPB2	0.705
(5) Sustainable Lifestyle	SL1	0.757
	SL2	0.794
	SL3	0.634

Source: Created by Researcher using SPSS Software

The loadings of all the items were greater than accepted guidelines of 0.60 therefore reflecting acceptable relationship between the items and their construct. However, the values obtained by factor loadings did not confirm that the five construct were empirically distinct entities, hence interpreted with reference to the overall factor loadings matrix.

Descriptive Statistics

“Descriptive statistics is utilized as a collection of techniques in which the characteristics of a data set are summarized and represented in an easy-to-understand format, in which statistics such as, mean, median, mode, standard deviation and percentage are utilized”. This Statistics allows the interpretation of distribution, central point and dispersion of data set. The main aim of descriptive analysis is that the sample characteristics should be evident in a concise manner. No attempt is made to generalize beyond the data in this stage of analysis. (C. R. Kothari, 2004)

Fig 11: Descriptive Statistics

Descriptive Statistics			
	Mean	Std. Deviation	Analysis N
SL1	4.07	.873	412
SL2	4.13	.917	412
SL3	4.27	.867	412
SI1	4.02	.970	412
SI2	4.03	.997	412
SI3	4.20	.948	412
SPA1	4.17	.919	412
SPA2	4.13	.898	412
SPA3	4.10	.904	412
SPB1	4.04	.919	412
SPB2	3.91	.972	412
SL1	4.10	.944	412
SL2	4.18	.799	412
SL3	4.21	.873	412

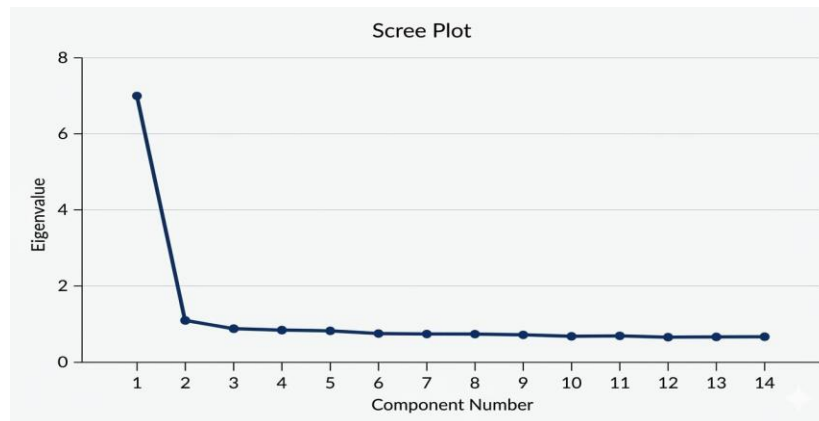
Source: Created by Researcher using SPSS Software

This set of descriptive statistics reveals quite well of the attitude of the respondents toward sustainable consumption and lifestyle. The means values for all of the variables range from 3.91 to 4.27, meaning that the respondents have quite high agreement with each item. Sustainable Literacy (SL) shows fairly high means value. Thus, mean value suggest a positive response on SL. Similar to this, means value for SI and SPA are positive and show favorable attitude. Whereas, SPB has fairly low mean value, this indicates that the purchase behavior is still less likely. The standard deviation values seem reasonable as well. Altogether, it can be concluded from these descriptive statistics that rather favorable response towards sustainable literacy, social influence, purchase attitude was observed, whereas sustainable purchase behavior received relatively weak responses from the respondents.

Scree plot

A scree plot is a type of graph used in factor analysis to plot the factor eigenvalues in descending order. It can be used as a tool to decide the number of “factors” by pinpointing where the curve tends to level off, or become somewhat horizontal. The “elbow” is where the line appears to start flattening. (Raymond B. Cattell, 1966)

Fig. 12: Scree Plot



Source: Created by Researcher using SPSS Software

The scree plot suggested that a large, dominant first factor existed within the dataset and that the eigenvalues drastically declined after the first factor. This indicates that there are some significant underlying shared variances among items. However, the scree plot in itself does not demonstrate that there are five constructs distinct from one another.

For this reason, interpretation of factor structure was guided both by theory and by factor loadings.

The five constructs were retained to theoretically distinct from one another; the large dominant first factor in the scree plot is accepted as empirical limitation of the measuring structure.

A “p-value” (significance value) is a measure which can be taken to interpret how strong the evidence is against the “null hypothesis”. The chance of achieving a result is known as the p-value. such as those that were observed, given the “null hypothesis” is absolute. Therefore, a small “p-value” would suggest good evidence in favor of rejecting the null hypothesis. In social

research, generally, a “p-value” < 0.05 is used as statistically significant and is accepted as indication that a relationship is meaningful. (Joseph F. Hair Jr. et al., 2010)

T-statistics, or t-value, is a statistical measure used to decide a relationship between variables is “statistically significant”. It represents the ratio of the estimated effect to its standard error, indicating how strongly a variable influences another. A higher t-value suggests stronger evidence against the “null hypothesis”. In general, a t-value above 1.96 (at 5% sig. level) is considered “statistically significant”. T-statistics is widely used in hypothesis testing to support decision-making regarding acceptance or rejection of hypotheses. (Joseph F. Hair Jr. et al., 2010)

Hypotheses Testing

Hypothesis testing is the statistical process to support or deny a given hypothesis with sufficient evidence. This process entails the observation of the sample data, which are evaluated for variable correlation. A conclusion is reached through significant values. This enables the validation of the research assumptions with a coherent approach. (Joseph F. Hair Jr. et al., 2010)

Fig.13: Hypotheses Testing

Hypotheses	Hypothesized Relationship	P Value	T-Statistics	Decision
H1	SL → SPA	0.01	3.729	✓ Accept
H2	SI → SPA	0.001	3.655	✓ Accept
H3	SPA → SPB	0.082	1.827	✗ Reject
H4	SPB → SLS	0.333	0.987	✗ Reject

Source: Created by Researcher using SPSS Software

The results of “hypothesis testing” reveal that “H1 and H2” are supported, while “H3 and H4” are not supported. Specifically, Sustainable Literacy (SL) has a “significant positive impact” on Sustainable Purchase Attitude (SPA) ($p = 0.01$), and Social Influence (SI) also significantly influences SPA ($p = 0.001$), as both “p-values” are less than 0.05. However, the relationship between “Sustainable Purchase Attitude” (SPA) and “Sustainable Purchase Behavior” (SPB) is not significant ($p = 0.082$), H3 was not supported. Perhaps this may indicate that having a

positive sustainable purchasing attitude alone, although important, cannot always lead to actual sustainable purchasing behavior. Sustainable purchasing behavior may be affected by other factors such as price, availability and convenience of sustainable products. Therefore, a positive attitude may not always be converted into actual purchasing behavior. Similarly, the relationship between SPB and Sustainable Lifestyle (SLS) is also insignificant ($p = 0.333$), H4 was not supported. The non-significant relationship may suggest that sustainable purchasing behavior alone may not be sufficient to establish a broader sustainable lifestyle. Sustainable lifestyle includes regular daily practices such as waste reduction, energy conservation and other long-term sustainable habits. Therefore, sustainable lifestyle may also depend on other personal, social and practical factors beyond purchasing behavior. These findings indicate that although awareness and social factors positively influence attitudes, these attitudes may not always translate into “actual behavior and lifestyle changes” among respondents.

CONCLUSION OF THE STUDY

This study examined the perception of Generation Z towards sustainable consumption and lifestyle choices in India by analyzing the role of sustainable literacy, social influence, purchase attitude, and purchase behavior. The discoveries reveal that Generation Z possesses generally positive responses towards sustainability. Sustainable literacy and social influence were found to have a profound effect on “sustainable purchase attitude”, indicating that knowledge and peer influence take part in a crucial role in forming favorable perceptions towards eco-friendly practices.

However, the results show that there exists an attitude-behavior gap; a favorable attitude towards purchasing the green products did not yield a statistically significant intention for making those green purchases. Despite having positive attitudes, Generation Z does not consistently translate these attitudes into sustainable purchase behavior. Furthermore, sustainable purchase behavior was not found to significantly influence sustainable lifestyle choices, suggesting that behavioral change is limited and influenced by practical constraints such as cost, convenience, and accessibility of sustainable products.

Overall, the study come to an end that while awareness and attitudes towards sustainability are strong among Generation Z, there is a need to bridge “the discrepancy between behavior and intention”. Efforts from policymakers, businesses, and educators are essential to promote sustainable practices by making “eco-friendly” options easier to reach, affordable, and convenient. This will help in encouraging long-term behavioral change and the adoption of sustainable lifestyles admist India’s youthful customers.

IMPLICATIONS OF THE STUDY

Based on the result of this study, several key implications were drawn for policy makers, firms and academics. First of all, as sustainable literacy and social influence exert strong effects on purchase attitude, government and firms should promote their awareness campaigns and education programmers so as to improve knowledge about environmental issues and eco-friendly attitudes among Gen Z consumers. Second, a gap b/w purchase attitudes and “purchase behavior” has appeared which leads business firms should concentrate on improvement of the affordability, accessibility and availability of sustainable goods and service. Firm should also improve transparency of its sustainable claims in order to enhance consumers’ trusts on their sustainable efforts. Third, purchase behavior is found not significantly correlated to sustainable lifestyle which suggests that awareness alone can’t lead to behavior change. Policy makers should implement supporting policies, incentive schemes and infrastructures to assist consumers in choosing sustainable consumption as an easy and practical option. Lastly, this study adds value in research by suggesting an investigation in both cognitive and practical attributes affecting sustainable consumption and provides reference to explore further variables which will contribute in attitude-behavior gap in future.

FUTURE RESEARCH DIRECTIONS

There are many interesting and important directions this study may point for upcoming studies on sustainable consumption practices among “Gen Z” respondents. First, future studies may increase sample size of respondents from various regions of India, to make the results more generalized. There may be interesting differences in behavior if conducted between urban and rural inhabitants.

Second, to answer the question “attitude behavior gap”, future studies may involve various additional variables such as price sensitivity, environmental concern, government policies, technological factors etc. which can contribute towards sustainable behavior.

Third, in order to examine how sustainable behavior changes over time with growing concern towards environment it can be studied longitudinally. This study can then help in giving dynamic picture of consumption.

Finally, studies may be carried out to observe the Effects of internet and Digital media on the sustainable consumption behavior among Generation Z.

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Online Consumer Engagement: Scopus-Based Systematic Review and Future Research Directions

YRL:-<https://docs.google.com/document/d/1bNcr5mmxBcSnnWCVWtP0yknRshuzgLda/edit?usp=sharing&ouid=110403252814827643206>

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Abstract- *Online consumer engagement, also known as digital consumer engagement, is crucial for understanding different contexts and identifying potential outcomes for both academic and practical implications. It enriches studies related to business, management, finance, and accounting. Hence, to examine the trends, meanings, determinants, and outcomes of online consumer engagement, and to provide a conceptual framework for future research, this paper follows a systematic literature review process in line with PRISMA 2020 guidelines, based on articles published in the Scopus database over the last ten years. The findings suggest that online consumer engagement is a multidimensional construct divided into behavioural, cognitive, emotional, and social dimensions, which is influenced by content-related factors, firm and platform strategies, consumer characteristics, and social influences. Engagement outcomes are categorised from immediate behavioural responses to longer-term relational and performance-related effects. The conceptual framework prepared based on the findings will be helpful for academicians as well as businesses in understanding the factors that may influence consumer engagement in the digital landscape and the outcomes that may be derived from engaged consumers.*

Keywords: *Online consumer engagement, digital consumer engagement, systematic literature review, determinants of online consumer engagement, outcomes of online consumer engagement.*

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1. Introduction- In the present digital era, people spend a considerable amount of time online, not only for entertainment but also for making important decisions such as purchasing products and services. In India, nearly 70 per cent of the population uses social media or other online platforms (Smart Insights, 2025). As a result, online platforms have become an important space where firms and consumers interact. Consumer engagement refers to the level of involvement or interest shown by consumers while interacting with a brand's social media pages or online platforms. This engagement can be observed through consumers' feelings, thoughts, and actions, such as liking posts, commenting, sharing content, or watching videos (Srivastava et al., 2025). With the growing focus on omnichannel strategies, businesses are increasingly targeting consumers across multiple online platforms to enhance their engagement experience. Therefore, increasing and maintaining consumer engagement has become a key concern for online marketers.

Online consumer engagement is not only central to marketing outcomes but also holds significance across disciplines such as psychology, sociology, and organisational behaviour, where it is studied as a driver of value co-creation, relationship building, and active participation in digital ecosystems (Singh et al., 2024)

Consumer engagement is a multidimensional concept that serves both as an outcome of marketing efforts and as a determinant of other online marketing outcomes (Srivastava et al., 2025). There is a gap in consolidated findings on antecedents, outcomes and definitions of online consumer engagement. To address this gap, the present study aims to systematically review prior research using the Scopus database and PRISMA guidelines. The purpose of this study is to define conceptual dimensions of online consumer engagement, identify its key antecedents, and analyse major outcomes, thereby offering a structured synthesis and directions for future research.

Hence, this study aims to address the following research questions:

RQ1: What are the publication trends in online consumer engagement research?

RQ2: What antecedents influence online consumer engagement?

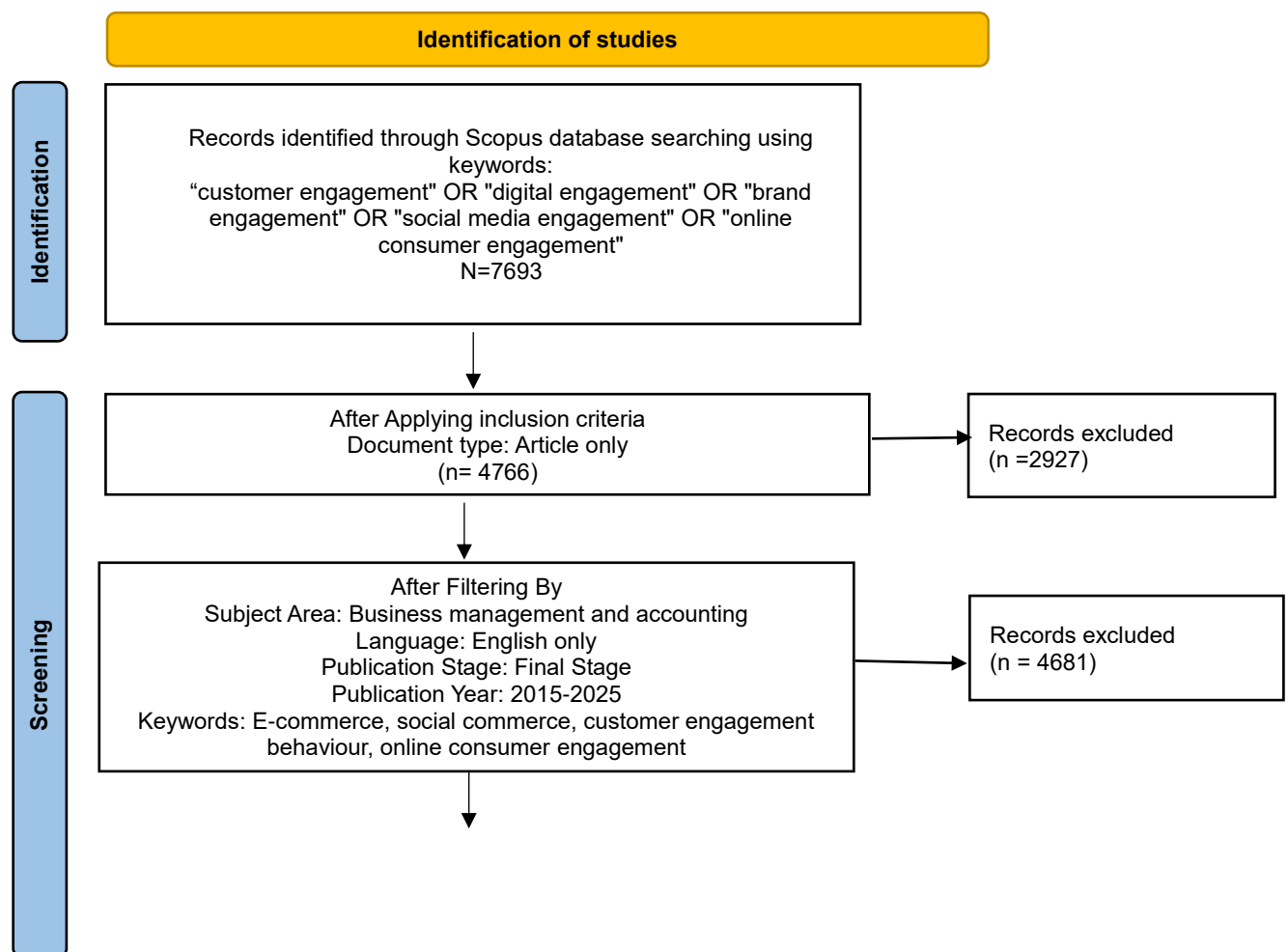
RQ3: What outcomes are associated with online consumer engagement?

Accordingly, this review has three **objectives**: to examine publication trends in online consumer engagement research, to identify the antecedents influencing online consumer engagement, and to synthesise the outcomes associated with online consumer engagement across different contexts and to propose a conceptual framework based on the findings for future research. To answer the research questions, the structure of this study is as follows: the next section describes the methodology used for the literature review, followed by an in-depth analysis and synthesis of the selected studies. The concluding sections discuss the findings, highlight limitations, propose future research directions, and present the overall conclusion.

2. Research Methodology- This study follows a systematic literature review approach guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)

framework. An SLR is used when the objective of the review is to do content analysis. A systematic review was selected as it allows a structured and transparent way to identify, screen, and review existing studies, while reducing the possibility of bias (Shamseer et al 2024). This approach is useful when research findings are scattered across different studies and need to be brought together in a meaningful way (Agrawal et al 2024). Compared to narrative reviews, a systematic review provides better clarity and consistency, making it suitable for the present study.

The Scopus database was used as the primary source of data because of its wide coverage of peer-reviewed journals in marketing, management, and related areas (Chadegani et al 2013; Shaheen, H., 2025). The selection of studies was carried out following the PRISMA process, which includes identification, screening, eligibility assessment, and final inclusion of relevant articles. PRISMA is a structured review protocol which provides a four-phase flow diagram representing sample identification for screening and then for eligibility testing and the final demonstration of the studies included in the review (Page et al., 2021). There are three main steps to conducting an SLR: determining the search strategy, establishing document selection with inclusion/exclusion criteria, and analysing based on the research question (Riandhi et al., 2021). At the initial stage, all similar terms related to consumer engagement were searched in the Scopus database, and after applying several inclusion criteria, 4766 documents remained for further screening. In an SLR, it is necessary to focus on a specific topic or keyword to avoid information overload (Pittaway et al., 2004) and hence using appropriate keywords is necessary to get the most directly relevant documents to serve the purpose of the study. Therefore, the complete selection process, along with exclusion and inclusion criteria at each stage, is presented in Figure 1.



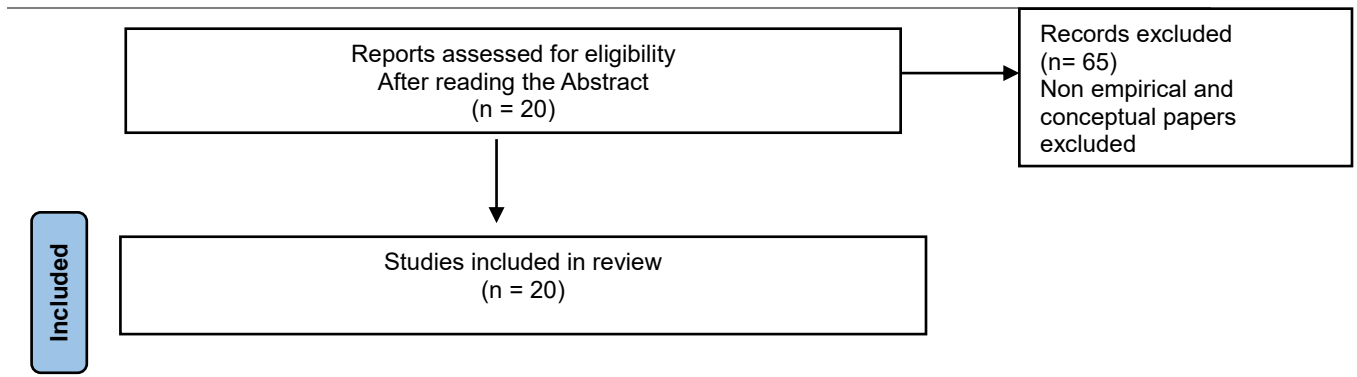


Figure 1: Presents a summary of the study selection process followed in accordance with the PRISMA guidelines(2020)

3. Results- This section is further divided into three parts: understanding the trend of the studies on online consumer engagement, how it has been defined in various studies, and what the dependent and independent variables used in different studies and outcomes of online consumer engagement are in different contexts to provide an understanding of its diversified scope.

3.1 Descriptive Analysis of the Selected Studies- This subsection presents a descriptive overview of the studies included in the review, focusing on publication trends over time, country-wise distribution, and subject-area classification.

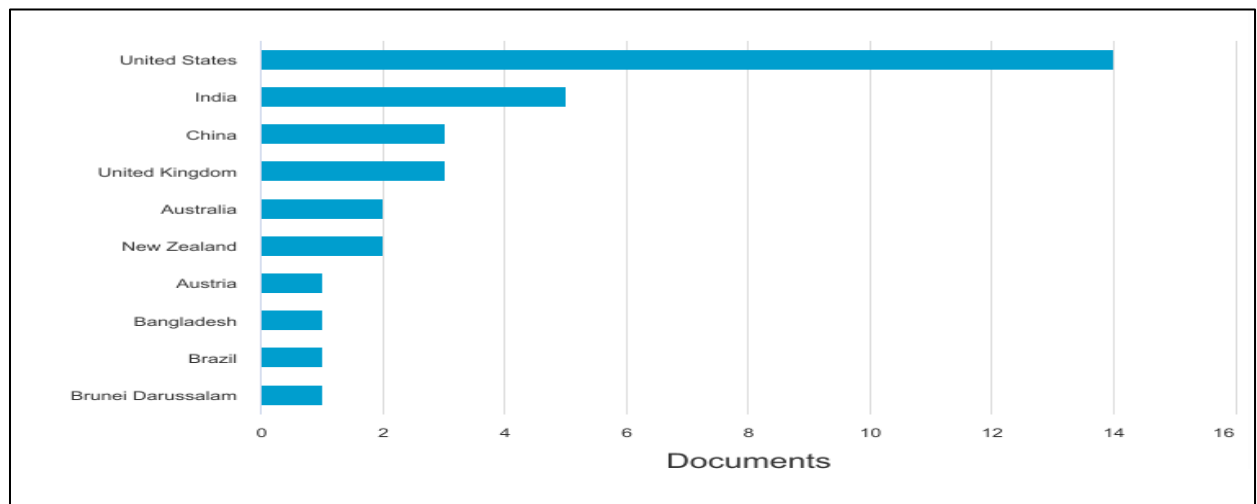


Figure 2: Country-wise publication (source: Scopus Database)

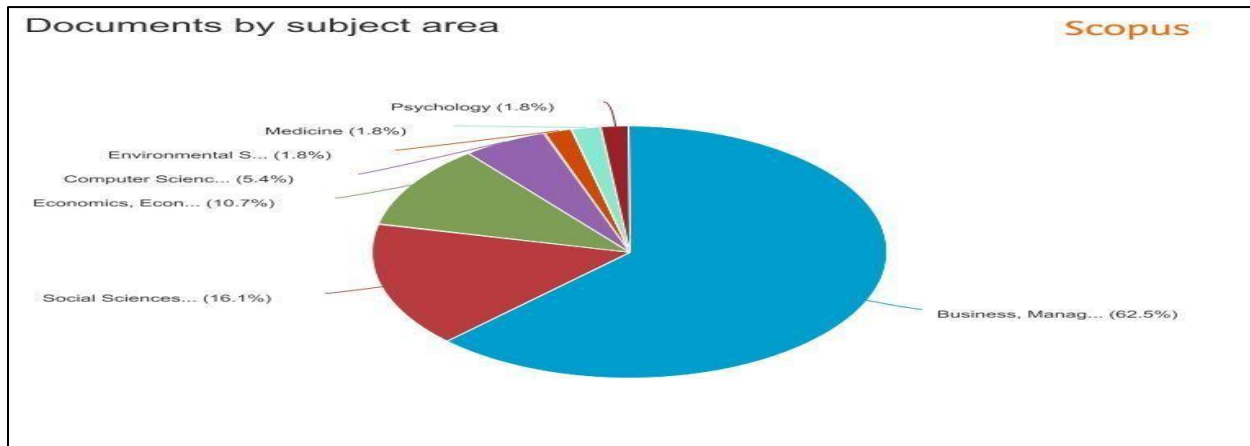


Figure 3: Subject-wise publication of online consumer engagement(source: Scopus Database)

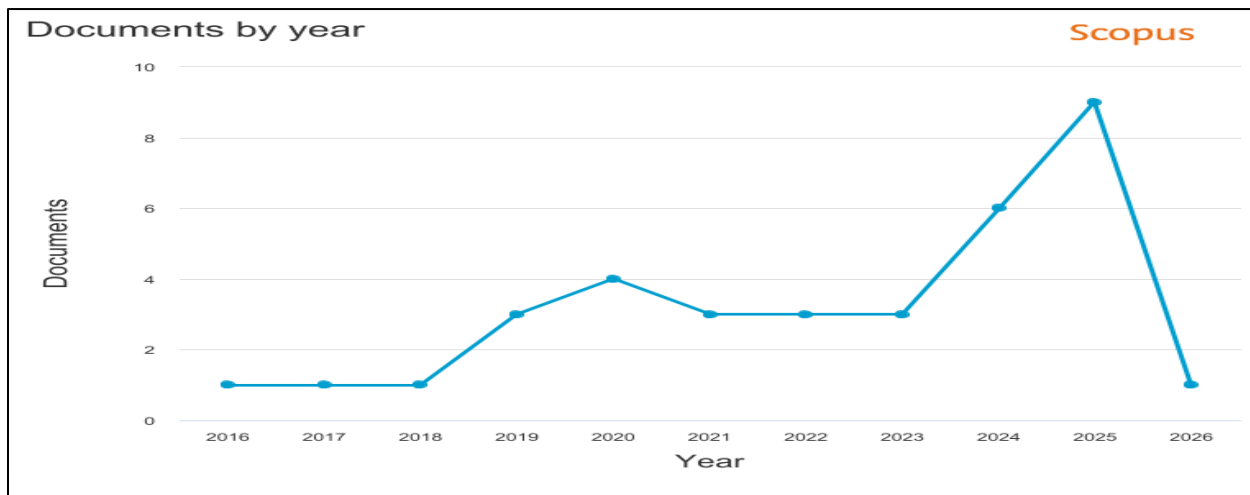


Figure 4: Year-wise publication of online consumer engagement (source: Scopus Database)

Figure 1 presents the country-wise distribution of the selected studies, highlighting that research is mainly concentrated in a limited number of countries. Figure 2 depicts the subject area distribution, with most studies published in the fields of business, management, and accounting. Figure 3 shows the year-wise publication trend of studies on online consumer engagement, indicating a gradual increase in research output after 2015 and a noticeable rise in recent years.

3.2 Conceptualisation of Online Consumer Engagement- Online consumer engagement is defined in the literature in several related but distinct ways, depending on the research context and digital platform. Many studies conceptualise online consumer engagement as observable digital interactions, such as likes, comments, shares, views, impressions, and clicks in response to brand or influencer content on social media. This behavioural approach is widely used in research on influencer marketing, sustainability communication, diversity, equity, and inclusion messaging, financial technology communication, and greenwashing, where engagement reflects consumers' visible reactions to online content (Ndasi and Cheung, 2026; Munaro et al., 2024; Shabbir Husain et al., 2024; Topal et al., 2020).

Other studies describe online consumer engagement as socially expressive participation in digital environments. In omnichannel retailing and online brand community research, engagement includes behaviours such as online deal-content sharing, user-generated content, advocacy, and co-creation, which support electronic word of mouth and interaction among consumers (Pai et al., 2025; Paintsil and Kim, 2022; Cheng et al., 2021).

A broader stream of research defines online consumer engagement as a multidimensional construct that includes cognitive, emotional, and behavioural dimensions. From this perspective, engagement involves consumers' attention, interest, enjoyment, identification, and active participation with digital brand content, rather than only observable actions (Rasty and Filieri, 2024; Marbach et al., 2019; Islam et al., 2017).

In digital service settings, online consumer engagement is often defined as active use of digital platforms, for example, electronic payment systems and insurance portals. In these contexts, digital awareness, digital literacy, and continuous interaction all reflect engagement and contribute to consumer empowerment, perceived value, adoption intention, and customer retention (Gupta and Prusty, 2024; Singh et al., 2024). Therefore, all available studies support the argument that online consumer engagement is context-dependent and can be measured through observable digital behaviours with psychological involvement and motivational determinants.

3.3 Determinants and Outcomes of Online Consumer Engagement – The present studies different categories of determinants that affect online consumer engagement and that result in outcomes that can further result in stages of consumer outcomes beginning with immediate behaviour response, relationship building and relational benefits that may affect measurable commercial impact. Determinants of engagement may be categorised into the following factors-

3.3.1. Determinants of Engagement

- *Content and Platform Factors:* Content characteristics, specifically storytelling and narrative quality, enhance engagement by capturing attention and increasing sharing intentions (Coker et al., 2021). For instance, storytelling-based video advertisements outperform argumentative formats in driving behavioural engagement (Coker et al., 2021). Studies further argue that the type of firm-generated content significantly dictates the level of consumer interaction and subsequent firm performance (Cheng et al., 2021). Consumer engagement may also be affected through various contextual platform events and how brands frame their message, and their responses to greenwashing (Topal et al., 2020).
- *Consumer and social factors:* several empirical studies show that personality traits, including extraversion, openness and agreeableness, positively influence engagement in online brand communities, with personal values moderating these relationships (Marbach et al., 2019; Islam et al., 2017). Since consumers are part of society, the society they belong to also shapes how they interact in the online environment. Findings of studies even support the assumption that reference group impact emotional engagement through informational and value-expressive pathways

more than pure utilitarian pressure (Wei & Murshed, 2019). However, due to individual differences, the intensity of engagement may vary dramatically across individuals, with different engagement types identified among specific demographic cohorts such as millennials (Rissanen & Luoma-Aho, 2016).

3.3.2. Outcomes of Online Customer Engagement: Businesses are adopting omnichannel strategies these days to attract customers because of their several direct and indirect advantages. For instance, social media communities positively contribute to customer consumption, specifically in terms of new customer acquisition and existing customer retention (Ho et al., 2020). However, the outcomes of online customer engagement can not always be directly measured through sales performance alone. As per the existing literature, the outcomes of customer engagement may be categorised into three types-

- **Immediate Behavioural Outcomes:** Customer engagement may lead to short-term immediate digital actions observed through increased likes, shares, comments and immediate visit or purchase intentions (Ndasi & Cheung, 2026; Munaro et al., 2024; Singh et al., 2024).
- **Relational Outcomes:** Online customer engagement also helps to strengthen relationship quality with the brand or sellers on social commerce platforms. Studies highlight that if engagement is sustained, then sharing of user-generated content by the users results in deeper psychological bonds, which may further lead to enhanced brand knowledge, positive electronic word of mouth (e-WOM), advocacy and long-term brand trust and loyalty (Paintsil & Kim, 2022; Pai et al., 2025; Sharma et al., 2019).
- **Performance Outcomes:** The main advantage of having favourable online customer engagement for business is to translate these relational and behavioural outcomes into tangible firm performance, observed through higher digital service adoption rates, customer retention and overall sales (Gupta & Prusty, 2024; Cheng et al., 2021).

4. Discussion

The findings of this review indicate that online consumer engagement is a multidimensional construct encompassing behavioural, cognitive, emotional and social dimensions. To summarise these findings theoretically, the proposed conceptual framework is grounded in the Stimulus-Organism-Response (S-O-R) model (Mehrabian & Russell, 1974). The SOR model is widely applied in environmental and marketing psychology to explain how external cues shape internal states and then behavioural outcomes (Peng et al., 2014). Applying the SOR approach to online consumer engagement provides a theoretically robust framework that connects the antecedents (stimuli) across all selected empirical studies included for review with the internal, multidimensional engagement state (organism) and the outcomes (Response).

4.1 Stimuli (S): Antecedents of Online Consumer Engagement

The reviewed literature combines four broad categories of external stimuli that affect online consumer engagement. **First, Content-related stimuli**, which include storytelling and narrative quality, message framing, and personalisation, capture attention and increase sharing and

recommendation behaviour (Coker et al., 2021; Suseno & Nguyen, 2023; Zhang et al., 2024). **Second**, *Firm and platform-related stimuli*, which include firm-generated content strategies and the design of digital platforms and services, shape the intensity of engagement and link it to firm performance (Cheng et al., 2021; Gupta & Prusty, 2024; Singh et al., 2024). **Third**, *Social stimuli* that includes reference-group pressure, influencer type and messaging, and broader cultural or societal attitudes that condition how consumers respond to online content, observed through cross-country variation in reactions to greenwashing (Wei & Murshed, 2019; Ndasi & Cheung, 2026; Topal et al., 2020), and **fourth**, *consumer-attribute stimuli* which refer to relatively stable determinants affected by individual differences such as personality traits, personal values, and pre-existing engagement dispositions, that determine how consumers processes these external cues and how strongly they respond (Marbach et al., 2019; Islam et al., 2017; Rissanen & Luoma-Aho, 2016). Together, all these four stimulus categories function as environmental cues that affect the internal engagement process.

4.2 Organism (O): The Internal, Multidimensional Engagement State

Within the S-O-R paradigm, the Organism represents the internal, largely unobservable psychological state that mediates the relationship between stimuli and response (Mehrabian & Russell, 1974). The reviewed studies converge on four interrelated dimensions of this internal state in the online consumer engagement context. **Cognitive engagement** reflects consumers' attention, interest, and information-processing effort directed toward brand content (Rasty & Filieri, 2024; Islam et al., 2017). **Emotional engagement** captures affective responses such as enjoyment, identification, and attachment that develop through interaction with content and communities (Marbach et al., 2019; Paintsil & Kim, 2022; Wei & Murshed, 2019). **Behavioural engagement**, the most frequently operationalised dimension in the reviewed literature, is expressed through observable digital actions such as liking, commenting, sharing, and viewing (Ndasi & Cheung, 2026; Munaro et al., 2024; ShabbirHusain et al., 2024). **Social engagement** further expands beyond individual behaviour, resulting in co-creation, participation, and advocacy with brand communities (Pai et al., 2025; Paintsil & Kim, 2022; Cheng et al., 2021). These dimensions should neither be treated as independent nor always sequential, they may be best studied as concurrently activated but mutually reinforcing in a layered manner. This internal state, once activated, results in a hierarchy of outcomes rather than a single set of effects.

4.3 Response (R): The Hierarchy of Engagement Outcomes

In the SOR model, response is the set of outcomes which results due to external cues and internal psychological processing. In the context of online consumer engagement, the outcomes identified across reviewed studies are organised in distinct but interrelated stages. *The first* layer consists of immediate behavioural responses such as visiting a platform, adopting an e-payment system, or making a purchase, which is the result of the activated organism state (Gupta & Prusty, 2024; Singh et al., 2024; Ndasi & Cheung, 2026). Then, in *the second stage*, relational responses strengthen through a series of repeated engagements, trust, brand loyalty and positive e-WOM or advocacy by consumers (Sharma et al., 2019; Islam et al., 2017; Paintsil & Kim, 2022; Coker et al., 2021). *Then finally, the third and most crucial layer*, which may be sustained for the long term, consists of performance responses at the firm level, such as

sales and customer retention (Cheng et al., 2021; Singh et al., 2024). All these layers are successive stages along with a single engagement-to-value pathway, moving from immediate action to long-term relational and performance impact.

4.4 Thematic integration of variables into SOR framework and applicability limits

As discussed earlier, the four stimulus categories, such as content, firm, platform, social and consumer attributes, can together or independently activate the internal and organism state observed through cognitive, emotional, behavioural and social dimensions of engagement described earlier; this organism state is then expressed through a hierarchy of responses that begins with immediate behavioural response to long-term relational outcomes, which may finally lead to firm-level performance outcomes. This integration provides the explanatory, theory-grounded pathways connecting specific antecedents to their outcomes in different cultural and geographic contexts, addressing the critical gap highlighted in engagement literature (Shu et al. 2025; Lee et al. 2022). The present findings not only suggest studying the relation between online consumer engagement with antecedents and outcomes through an S-O-R lens but also provide literature support from the collective insights of all selected quantitative studies from different contexts.

However, as the framework is derived from a synthesis of predominantly cross-sectional and single-platform data, it must therefore be studied as a set of conceptual propositions rather than an established empirical model. The relation may differ if studied in different contexts; for instance, behavioural response is not only an outcome but can also serve as a stimulus and affect the relations in future research.

Stimulus (S)	Organism (O)	Response (R)
<i>Content-related factors:</i> <i>storytelling, message framing, personalisation</i>	<i>Cognitive engagement:</i> <i>attention, interest, information processing</i>	<i>Immediate behavioural outcomes:</i> <i>likes, shares, visit/purchase intention</i>
<i>Firm/platform factors:</i> <i>firm-generated content, platform features, digital service design</i>	<i>Emotional engagement:</i> <i>enjoyment, identification, attachment</i>	<i>Relational outcomes:</i> <i>trust, loyalty, advocacy, e-WOM</i>
<i>Social influences:</i> <i>reference groups, influencer type, cultural/societal norms</i>	<i>Behavioural engagement:</i> <i>liking, commenting, sharing</i>	<i>Performance outcomes:</i> <i>sales, retention, firm performance</i>
<i>Consumer attributes:</i> <i>personality traits, personal values, engagement disposition</i>	<i>Social engagement:</i> <i>co-creation, advocacy, community participation</i>	

Figure 5: S-O-R Variable Mapping of Online Consumer Engagement (author's compilation)

5. Limitations and Future Research Directions- This study provides a synthesis of systematic online consumer engagement, but has some limitations that suggest future review papers need to address. Future studies may consider other databases such as Web of Science, and the

selection of other subject areas may help to get deeper insights into this subject. Due to the small sample size, this study also restricts generalizability, which can be addressed by future researchers by considering qualitative studies also to review the available literature on online consumer engagement. Moreover, in available empirical studies, it is also observed that more emphasis is on behavioural engagement, with limited focus on cognitive, emotional and relational aspects. Hence, to fill all these gaps, future studies should broaden the database across different contexts, adopting cross-sectional and longitudinal methods, and empirically test the proposed framework to strengthen theory and practice.

6. Conclusion- This study presents a PRISMA-guided systematic review of online consumer engagement, synthesising its key antecedents, multidimensional conceptualisations, and outcomes. The findings show that online consumer engagement is shaped by content, firm and platform strategies, consumer characteristics, and social influences, and that leads to important behavioural and relational outcomes. By integrating fragmented literature and proposing a conceptual framework, this review offers a clear foundation for future research and practical digital engagement strategies.

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Social Enterprises and the SDGs: Contributions to Quality Education, Climate Action, and Inclusive Growth

URL:-<https://docs.google.com/document/d/1Z5LOKeV22W0b5T7iRWM9sJ0Ds8YDUnK/edit?usp=sharing&ouid>

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ABSTRACT

Social enterprises hold a remarkable space in the development scenery: they necessitate to endure as businesses while held responsible to a social or environmental operations — which is precisely the variety of dual capacity the UN Sustainable Development Goals. This study strives to know the contribution of social enterprises in India towards three goals: Quality Education (SDG 4), Climate Action (SDG 13), and finally Inclusive Growth (SDG 8 and SDG 10 collectively). It draws on a survey of 412 social enterprises across six sectors, supplemented by interviews and a handful of illustrative case vignettes. Respondents rated their own organisation's contribution to each goal on a five-point scale, and we used multiple regression to work out which organisational features actually predict stronger inclusive-growth outcomes. The pattern that emerged was fairly consistent: enterprises scored themselves highest on inclusive growth ($M = 4.05$, $SD = 0.62$), a little lower on quality education ($M = 3.92$, $SD = 0.68$), and noticeably lower on climate action ($M = 3.54$, $SD = 0.81$). What predicted inclusive growth wasn't how long an enterprise had been running — age came out non-significant — but how embedded it was in its community, along with its access to funding and its level of digital adoption. Enterprises working in women's livelihoods and agriculture reported the strongest inclusive-growth scores of any sector. Together, these results suggest social enterprises act as useful intermediaries that translate the SDGs into locally workable action, though their climate contribution still lags well behind their social and economic one. We close with some practical suggestions for strengthening the ecosystem that supports SDG-aligned social enterprises.

Keywords: *social enterprise; Sustainable Development Goals; quality education; climate action; inclusive growth; hybrid organisation; India*

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1. Introduction

Since their adoption in 2015, the United Nations' 17 Sustainable Development Goals (SDGs) have provided a shared global framework for addressing poverty, inequality, environmental degradation, and social exclusion by 2030 (United Nations, 2015). Delivering on this agenda has proven far more demanding than any single sector can manage alone. Among the seventeen SDGs, three merit particular attention when evaluating the developmental footprint of social enterprises. SDG 4 (Quality Education) and SDG 13 (Climate Action) represent, respectively, the human-capital and environmental pillars of sustainable development, while SDG 8 (Decent Work and Economic Growth) and SDG 10 (Reduced Inequalities) — treated jointly here as 'inclusive growth' — capture the livelihood and equity outcomes that social enterprises most directly produce through employment generation, skills development, and market access for marginalised groups. Recent sector reviews confirm that social enterprises are already active across exactly these domains: focus-group evidence from Ghana identifies climate-smart agribusiness, clean technology, inclusive financial services, and education among the sectors in which social enterprises concentrate their activity (Quaye et al., 2024), and multi-stage scale-development work in India confirms that social mission, innovation, and collaborative networks significantly predict enterprises' contribution to sustainable development more broadly (Bansal, Garg, & Vasa, 2023). Yet the relative strength of these contributions across specific goals, and the organisational factors that explain variation between enterprises, remain empirically under-specified — particularly through large-sample survey evidence from the Global South.

Existing empirical work on the social enterprise–SDG relationship is dominated by two research traditions: single-country qualitative studies that map SDG relevance through focus groups or document analysis, and scale-development studies that build composite sustainability indices without disaggregating results by specific goal. Both traditions have advanced the field considerably, but neither offers a disaggregated, quantitative account of how social enterprises in a single national context perform against distinct SDG targets, nor do they systematically identify which organisational characteristics — enterprise age, sector, funding access, digital capability — predict stronger or weaker contribution. This gap is consequential for policy: donors, incubators, and government schemes cannot target support effectively if it remains unclear whether social enterprises are more effective at generating jobs than at reducing emissions, or whether performance varies systematically by sector and organisational maturity.

This study addresses that gap through a mixed-methods investigation of social enterprises operating in India, a country with one of the most developed social enterprise ecosystems in the Global South (British Council India, 2016) and an economy in which social enterprises are already documented to contribute to sustainable development outcomes (Bansal et al., 2023). Specifically, the study asks: (RQ1) To what extent do social enterprises perceive their operations as contributing to SDG 4 (Quality Education), SDG 13 (Climate Action), and SDG 8/10 (Inclusive Growth)? (RQ2) Which organisational and contextual factors predict variation in social enterprises' contribution to inclusive growth? and (RQ3) Does this contribution vary systematically across enterprise sectors?

The remainder of the paper proceeds as follows. Section 2 reviews the literature on social entrepreneurship and its documented linkages to SDG 4, SDG 13, and SDG 8/10, and identifies the

study's conceptual gap. Section 3 develops a conceptual framework linking enabling inputs, social enterprise mechanisms, and SDG outcomes. Section 4 details the mixed-methods research design, sample, and instruments. Section 5 reports descriptive, correlational, and regression findings alongside illustrative case vignettes. Sections 6 through 9 discuss the findings, draw out policy implications, note the study's limitations, and conclude.

2. Literature Review

2.1 Social Enterprises as Hybrid Organisations

Definitions of social entrepreneurship vary largely throughout the literature, but finally it congregates that a commercial discipline organized not only for profit but an answer for primary social or environmental questions as mission (Dees, 1998). Austin, Stevenson, and Wei-Skillern (2006) took four dimensions market failure, mission, resource mobilisation, and performance measurement to compare different social and commercial entrepreneurship and finally coined that social entrepreneurs face more constraints difficult which are basically different because it's very hard to survive in market with social values. Battilana and Lee (2014) defined social enterprises as hybrid organizations which combine organisational forms of business and charity across five dimensions i.e. organisational design, organisational activities, inter-organisational relationships, workforce composition and culture. This hybrid structure is very contradictory: it facilitates social enterprises to access various resource streams i.e. grants, earned revenue, impact investment but also generates constant strain between social and commercial senses that ought to be keenly managed (Zahra et al., 2009). As per principle of this study, a social enterprise is an organisation irrespective of legal entity/ company formed under Section 8 may be for-profit, non-profit, or hybrid structures which are producer cooperative and social or environmental mission are its basic nature. It generates a large part of income through sale of goods or services instead of donations or grants and also mainly reinvests surplus profit towards same mission sooner than toward private profit distribution. The functioning definition depends upon the OECD's guidelines on the Social and Solidarity Economy and Social Innovation, which properly identifies social enterprises as one constituent of the wider social economy and appeal to member and partner countries to fortify the data, financing, and regulatory environment that allows to drive at extent (OECD, 2022).

2.2 Social Enterprises and SDG 4 (Quality Education)

Educational Social enterprises typically operate at the edges of formal schooling systems. It provides counteractive and bridge education to out-of-school children, cater vocational trainings and digital skills to youth and adults and holding capacity buildings program on the content-delivery for the teachers of under-resourced government schools. The grass root nature working of these enterprises are the virtue comes through their relations with such communities they serve. They often approach to first-generation learners, family members of migrant or seasonal workers, and women community face social obstacles to attend mainstream educational Institutions. The Indian social enterprise sector has historically been anchored by education- and livelihood-focused organisations, reflecting sustained early investment in this space by domestic and international impact investors (British Council India, 2016). At the same time, education remains one of the more difficult domains in which

to build a financially sustainable social enterprise model, since willingness to pay is often low among the target beneficiaries themselves, forcing many education-focused ventures toward blended finance models that combine philanthropic capital with modest fee revenue or business-to-government contracts.

Illustrative models documented in the wider literature include community-based training programmes that certify local residents — including women with little formal schooling — as technicians and educators, thereby generating both a skills-transfer outcome (SDG 4) and a livelihood outcome (SDG 8) in a single intervention. This dual-outcome pattern recurs throughout the social enterprise literature and is one reason why quality education and inclusive growth are frequently found to be empirically correlated in enterprise-level data, a pattern this study also tests directly in Section 5.

2.3 Social Enterprises and SDG 13 (Climate Action)

Climate-oriented social enterprises span decentralised renewable energy (particularly off-grid solar for un-electrified or under-electrified rural households), circular-economy and waste-management ventures, and climate-smart or regenerative agriculture services for smallholder farmers. Ghanaian focus-group evidence identifies clean technology and climate-smart agribusiness as core sub-sectors within the national social enterprise ecosystem, alongside education and inclusive financial services (Quaye et al., 2024), a pattern broadly consistent with the sectoral composition of India's own social enterprise landscape. Compared with education- and livelihood-oriented ventures, however, climate-focused social enterprises tend to be more capital-intensive at the point of scale-up — solar micro-utilities and waste-processing infrastructure require upfront investment that is difficult to finance through earned revenue alone in the early stages — which may partly explain why climate action is consistently reported as a comparatively weaker area of perceived SDG contribution relative to education and livelihoods, a pattern this study returns to in Section 5.

A further complication is measurement: whereas an education-focused enterprise can report relatively legible outputs (learners trained, certifications issued), the climate benefit of an enterprise's activity — avoided emissions, water saved, waste diverted from landfill — requires technical measurement capacity that many smaller social enterprises simply do not possess. This measurement gap is likely to depress self-reported climate contribution scores even among enterprises that are, in substantive terms, delivering meaningful environmental benefit, and is a methodological caveat this study addresses in Section 8.

2.4 Social Enterprises and SDG 8 / SDG 10 (Inclusive Growth)

SDG 8 (decent work and economic growth) and SDG 10 (reduced inequalities) are treated jointly in this study as 'inclusive growth' because, at the level of an individual social enterprise, the two goals are typically pursued through the same organisational activity: the deliberate inclusion of marginalised groups — women, persons with disabilities, members of scheduled castes and tribes, and rural or informal-sector workers — into fairly remunerated, dignified employment or self-employment. Producer collectives and women's livelihood cooperatives are a particularly well-documented model in the Indian context, combining income generation with collective bargaining power and, in many cases, a secondary environmental or educational benefit (for example, handicraft cooperatives that use reclaimed materials, or dairy cooperatives that fund community schooling from

a share of member proceeds). Cross-subsidisation models — in which paying customers subsidise services for low-income beneficiaries within the same operational structure — are a recurring design feature across health, education, and financial-inclusion social enterprises in India, and are frequently cited as a mechanism through which a single enterprise can simultaneously advance decent work (through staff and supplier employment) and reduced inequality (through subsidised access for excluded groups).

Because inclusive growth outcomes are closely tied to an enterprise's core revenue-generating activity — rather than to a secondary programme layered on top of the business model, as is sometimes the case for climate or education outcomes — social enterprises might be expected to report their strongest SDG contribution in this domain. This expectation is formally tested as part of RQ1 in the present study.

2.5 Research Gap and Positioning of the Present Study

Taken together, the literature supports three propositions: that social enterprises are theoretically well-positioned to contribute to human-capital, environmental, and economic-inclusion goals simultaneously (Battilana & Lee, 2014; Santos, 2012); that their practical contribution is empirically documented in qualitative and scale-development studies (Bansal et al., 2023; Quaye et al., 2024); and that measurement, financing, and sector-specific constraints are likely to produce uneven contribution across different SDGs rather than uniform performance. What remains missing is a single study that (a) measures perceived contribution to specific, named SDGs using validated survey instruments, (b) does so with a sample large enough to support disaggregation by sector and organisational characteristics, and (c) statistically models which organisational factors predict stronger contribution. This study is designed to fill that gap, and Section 3 translates the propositions above into a testable conceptual framework.

3. Conceptual Framework

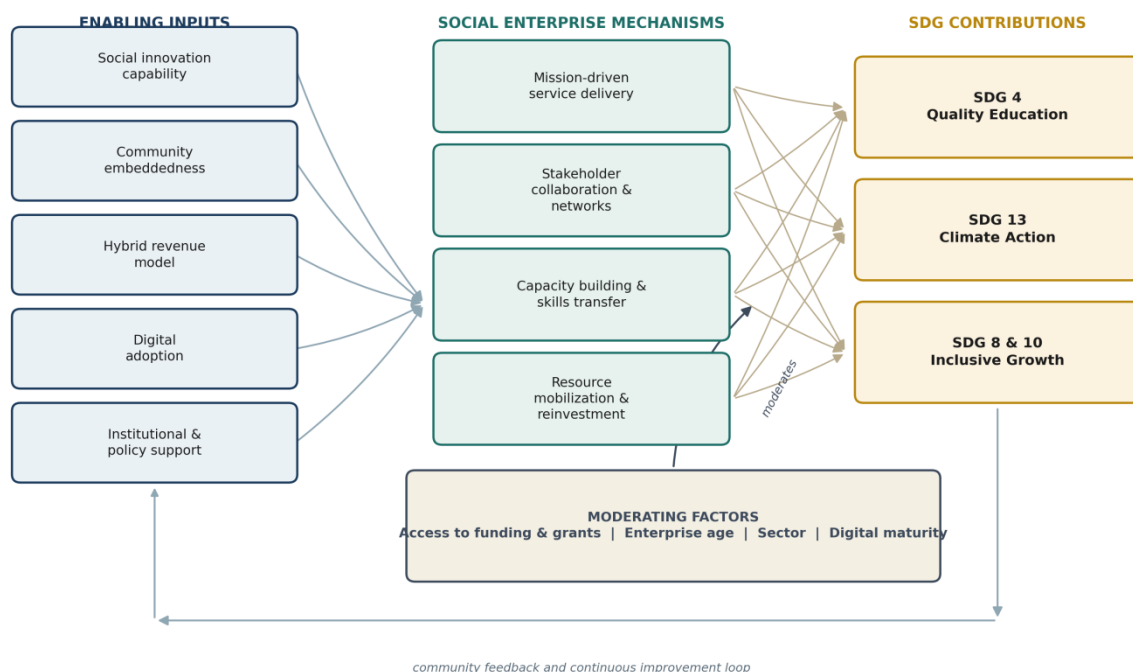
Drawing on the hybrid-organising literature reviewed above, this study conceptualises SDG contribution as the end point of a three-stage process (Figure 1). The first stage comprises enabling inputs: the resources and capabilities a social enterprise can draw upon, including its social-innovation capability, the depth of its embeddedness within the community it serves, the design of its hybrid revenue model, its level of digital adoption, and the institutional and policy support available to it (OECD, 2022). The second stage comprises the social enterprise mechanisms through which those inputs are converted into activity: mission-driven service delivery, stakeholder collaboration and networks, capacity building and skills transfer, and resource mobilisation with mission-directed reinvestment. These mechanisms echo the dimensions of hybrid organising identified by Battilana and Lee (2014) and the resource-recombination process described by Mair and Marti (2006).

The third stage comprises the SDG contributions themselves — quality education, climate action, and inclusive growth — which this study treats as parallel outcomes rather than a single composite index, reflecting the argument developed in Section 2 that contribution is unlikely to be uniform across goals. The relationship between mechanisms and outcomes is proposed to be moderated by four contextual factors: access to funding and grants, enterprise age, sector, and digital maturity. These moderators were selected because each has a plausible theoretical basis for affecting an enterprise's

capacity to convert activity into measurable outcomes — for instance, a well-funded enterprise can invest in the impact-measurement systems needed to document its climate contribution, while an older enterprise may have deeper community relationships but also more institutional inertia. Finally, the framework incorporates a feedback loop whereby community response to an enterprise's outputs feeds back into its ongoing input base, consistent with the iterative, learning-oriented character of social entrepreneurial practice described by Dees (1998) and Zahra et al. (2009).

This framework generates the study's three research hypotheses. H1: social enterprises will report differential (non-uniform) mean contribution across the three SDG dimensions, with inclusive growth expected to score highest given its direct link to core revenue-generating activity. H2: community embeddedness, access to funding, and digital adoption will positively predict inclusive growth contribution, while enterprise age will show no significant independent effect once these variables are controlled for. H3: mean inclusive growth contribution will vary significantly by sector, with livelihood-oriented sectors (women's livelihoods, agriculture) scoring more highly than infrastructure-intensive sectors (renewable energy, waste management).

Figure 1. Conceptual Framework: Social Enterprise Mechanisms and SDG Contribution



4. Methodology

4.1 Research Design

The study adopted an explanatory sequential mixed-methods design (quantitative followed by qualitative), consistent with recent multi-stage approaches used to study the social enterprise–SDG relationship in India (Bansal et al., 2023). Primary data-collection has been done through a structured cross-sectional survey instrument. In section -5 the descriptive, regression and correlational evidence have also been generated. The semi-structured interviews designed for survey respondents with

purpose to develop the illustrative case vignettes which presents quantitative results and to aid interpretation of the statistical findings.

4.2 Population and Sampling

The target population are the Indian social enterprises (regd. and operational in India) working in six broad sectors i.e. agriculture and rural livelihoods, education, renewable energy and environment; healthcare; women's livelihoods and crafts, waste management and circular economy. To ensure across India-wide representation a stratified purposive sampling frame was moulded by adding up incubator and accelerator alumni networks, state level associations of social enterprises, impact-investor portfolio listings, above mentioned six sectors stratification and geographical regions (North, South, East, West, and Central) of India.

By using Cochran's (1977) formula for a population, minimum required sample size was calculated at a 95% confidence level and a 5% margin of error, yielding a minimum requirement of 384 respondents.

The systemic questionnaires were distributed to 620 qualified enterprise leaders; after deleting unfinished responses and applying a data-quality screen (elimination of straight-lining and completion times under three minutes), 412 suitable responses were taken, exceeding the minimum requirement and yielding an effective response rate of 66.5%. Table 1 depicts the demographic and organisational profile of the attained sample.

Table 1. Demographic and Organisational Profile of the Sample (N = 412)

Characteristic	Category	n	%
Sector	Agriculture & rural livelihoods	91	22.1
	Women's livelihoods & crafts	70	17.0
	Education	74	18.0
	Renewable energy & environment	62	15.0
	Waste management & circular economy	66	16.0
	Healthcare	49	11.9
Legal structure	Section 8 / not-for-profit company	168	40.8
	Trust or society	104	25.2
	Private limited (for-profit hybrid)	97	23.5
	Producer / cooperative collective	43	10.4
Enterprise age	Under 5 years	146	35.4
	5–10 years	171	41.5
	Over 10 years	95	23.1
Operating region	Urban	121	29.4
	Semi-urban	146	35.4

Characteristic	Category	n	%
	Rural	145	35.2
Respondent role	Founder / CEO	247	60.0
	Senior manager	112	27.2
	Programme officer	53	12.9
Received institutional funding / grant (past 2 yrs)	Yes	261	63.3
	No	151	36.7

4.3 Instrument Development and Measures

Three stages have been taken to develop the survey instrument. The literature review yielded an initial item pool -Section 2.

To cover three constructs: observed contribution to Quality Education (6 items, e.g., 'Our enterprise has improved admission to learning opportunities for underserved groups'), observed contribution to Climate Action (6 items, e.g., 'Our enterprise's activities have curtail environmental damage or resource use in our working area'), and observed contribution to Inclusive Growth (7 items, e.g., 'Our enterprise gives fair and noble employment to individuals who would otherwise face barriers to work'). Five-point Likert scale ranging from 1 ('strongly disagree') to 5 ('strongly agree') applied to all items. Second, a board of five academic and practitioner experts in social entrepreneurship reviewed the content validity for the exclusion of two redundant items and minor revisions of words for clarity. Third, the revised instrument was piloted with 40 social enterprise leaders not included in the main sample; item-total correlations below 0.30 were flagged for review, and none required removal. The final instrument also captured organisational characteristics used as predictors in the regression model: enterprise age (years since founding), sector, legal structure, region, access to institutional funding or grants in the preceding two years (binary), and a composite digital-adoption index (use of digital payments, online marketing, and management information systems, summed into a 0–3 scale).

Internal consistency reliability was assessed using Cronbach's alpha (Cronbach, 1951). All three constructs exceeded the conventional 0.70 threshold for acceptable reliability in the main-study sample: Quality Education (alpha = 0.82), Climate Action (alpha = 0.79), and Inclusive Growth (alpha = 0.85), as reported in Table 2.

4.4 Data Collection and Ethical Considerations

Data were collected between the fourth quarter of the study year and the following quarter through a combination of an online questionnaire (distributed via incubator and association mailing lists) and structured telephone interviews for respondents in areas with limited internet connectivity, in order to reduce coverage bias against enterprises operating in remote or rural locations. Participation was voluntary, and respondents were informed of the study's purpose, the confidentiality of their responses, and their right to withdraw at any point prior to submission. No personally identifying information beyond enterprise-level demographic categories was retained in the analysis dataset. A sub-sample of 22 respondents, purposively selected to represent variation across sector and region,

participated in follow-up semi-structured interviews of 30–45 minutes, three of which are presented as illustrative, anonymised case vignettes in Section 5.4.

4.5 Data Analysis

Quantitative data were analysed in three stages. Descriptive statistics (means, standard deviations) were computed for each SDG construct to address RQ1. Pearson correlation coefficients were computed among the three constructs and the organisational predictor variables to examine bivariate relationships and check for multicollinearity ahead of the regression stage (Hair, Black, Babin, & Anderson, 2019). A multiple linear regression model was then estimated with the Inclusive Growth Contribution Index as the dependent variable and enterprise age, digital-adoption index, access to funding (binary), community embeddedness (a four-item sub-scale drawn from the Quality Education and Inclusive Growth item pools), and sector (dummy-coded, with waste management as the reference category) as predictors, addressing RQ2. One-way analysis of variance (ANOVA) with Tukey post-hoc comparisons was used to test for sector-level differences in the Inclusive Growth Index, addressing RQ3 and reported alongside Figure 3. Interview transcripts were analysed thematically following an inductive coding approach, with two researchers independently coding a subset of transcripts to check inter-coder agreement before finalising the coding frame used to develop the case vignettes in Section 5.4.

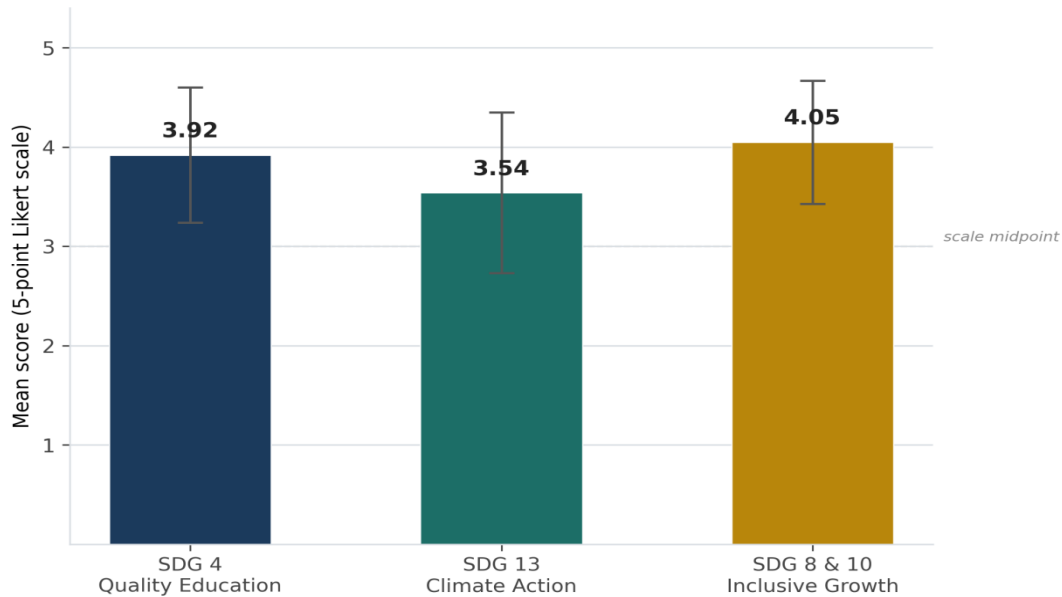
5. Findings

5.1 Descriptive Findings: Perceived Contribution Across SDG Dimensions

Table 2 reports the descriptive statistics and reliability coefficients for the three SDG constructs. Consistent with Hypothesis 1, mean scores differed meaningfully across dimensions. Inclusive Growth received the highest mean rating ($M = 4.05$, $SD = 0.62$), followed by Quality Education ($M = 3.92$, $SD = 0.68$), with Climate Action rated distinctly lower ($M = 3.54$, $SD = 0.81$). A repeated-measures comparison confirmed that these differences were statistically significant ($p < .001$), with the gap between Inclusive Growth and Climate Action representing a medium-to-large effect. Figure 2 presents these results graphically, including standard-deviation error bars, which show that variance in Climate Action scores was also the widest of the three constructs — consistent with the measurement-capacity argument developed in Section 2.3, whereby smaller enterprises without dedicated environmental-monitoring capacity may under- or over-estimate their own climate contribution relative to enterprises with more formal measurement systems.

Table 2. Descriptive Statistics and Reliability of SDG Contribution Constructs (N = 412)

Construct	Items	Mean	SD	Cronbach's α
SDG 4 — Quality Education	6	3.92	0.68	0.82
SDG 13 — Climate Action	6	3.54	0.81	0.79
SDG 8 & 10 — Inclusive Growth	7	4.05	0.62	0.85

Figure 2. Mean Perceived Contribution to SDG Dimensions (N = 412)

5.2 Correlation Analysis

Table 3 presents the Pearson correlation matrix among the three SDG constructs and the key organisational predictor variables. All three SDG constructs were positively and significantly intercorrelated (r ranging from 0.31 to 0.46, $p < .01$), supporting the qualitative observation in Section 2.2 that education and livelihood outcomes frequently co-occur within the same enterprise activity. Community embeddedness showed the strongest bivariate association with Inclusive Growth ($r = 0.52$, $p < .001$), followed by digital adoption ($r = 0.34$, $p < .001$) and access to funding ($r = 0.29$, $p < .001$). Enterprise age was weakly and non-significantly correlated with Inclusive Growth ($r = 0.07$, ns), foreshadowing the regression result reported below. Variance inflation factors for all predictors in the subsequent regression model ranged from 1.08 to 1.61, well below conventional thresholds for concern (Hair et al., 2019), indicating that multicollinearity was not a material concern for the regression analysis.

Table 3. Pearson Correlation Matrix for SDG Constructs and Key Predictors (N = 412)

Variable	1	2	3	4	5
1. Quality Education	—				
2. Climate Action	.38**	—			
3. Inclusive Growth	.46**	.31**	—		
4. Community embeddedness	.41**	.27**	.52**	—	
5. Digital adoption index	.22**	.19**	.34**	.23**	—
6. Access to funding (0/1)	.18**	.24**	.29**	.21**	.16**
7. Enterprise age (years)	.04	.09	.07	.05	-.11*

Note. ** $p < .01$, * $p < .05$. Variable 6 and 7 correlations shown against variables 1–5 only, in the interest of space; full 7×7 matrix available from the corresponding author on request.

5.3 Regression Analysis: Predictors of Inclusive Growth Contribution

Table 4 presents the results of the multiple regression model predicting the Inclusive Growth Contribution Index. The overall model was statistically significant, $F(8, 403) = 26.14$, $p < .001$, and explained approximately 34% of the variance in inclusive growth scores (adjusted R-squared = 0.34). Consistent with Hypothesis 2, community embeddedness was the strongest significant predictor (beta = 0.34, $p < .001$), followed by access to funding or grants (beta = 0.21, $p < .01$) and digital-adoption index (beta = 0.18, $p < .05$). Enterprise age was not a significant predictor once the other variables were controlled for (beta = 0.05, ns), suggesting that an enterprise's contribution to inclusive growth is driven more by how deeply it is embedded in its community and how well-resourced and digitally capable it is, rather than simply by how long it has been operating. Among the sector dummy variables (reference category: waste management and circular economy), women's livelihoods and crafts (beta = 0.24, $p < .001$) and agriculture and rural livelihoods (beta = 0.19, $p < .01$) were significant positive predictors, consistent with the descriptive sector ranking shown in Figure 3.

Table 4. Multiple Regression Predicting Inclusive Growth Contribution Index (N = 412)

Predictor	B	SE	β	p
(Constant)	1.87	0.21	—	< .001
Enterprise age (years)	0.006	0.006	.05	.312
Digital adoption index (0–3)	0.114	0.041	.18	.013
Access to funding / grant (0/1)	0.187	0.062	.21	.003
Community embeddedness	0.298	0.045	.34	< .001
Sector: Education	0.146	0.081	.13	.072
Sector: Women's livelihoods & crafts	0.261	0.084	.24	< .001
Sector: Agriculture & rural livelihoods	0.214	0.078	.19	.006
Sector: Renewable energy & environment	0.079	0.086	.07	.358

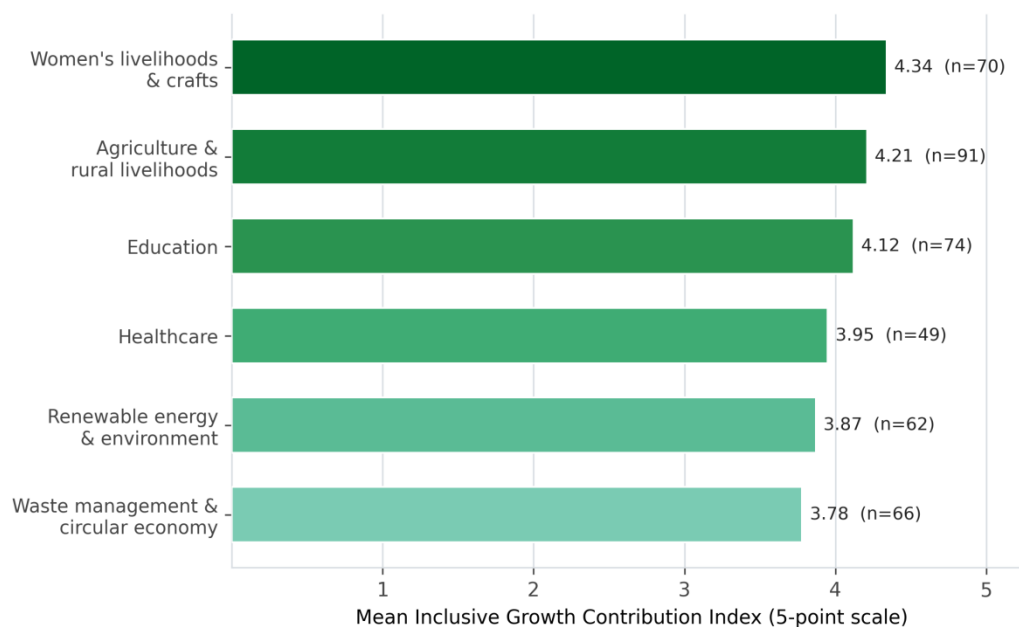
Note. Reference category for sector dummies: waste management & circular economy. Dependent variable: Inclusive Growth Contribution Index (5-point scale). $R^2 = .348$, adjusted $R^2 = .335$, $F(8, 403) = 26.14$, $p < .001$.

5.4 Sector Variation and Illustrative Case Vignettes

One-way ANOVA confirmed a statistically significant effect of sector on the Inclusive Growth Contribution Index, $F(5, 406) = 8.72$, $p < .001$. As shown in Figure 3, women's livelihoods and crafts enterprises reported the highest mean score ($M = 4.34$), followed by agriculture and rural livelihoods ($M = 4.21$) and education ($M = 4.12$); healthcare ($M = 3.95$), renewable energy and environment ($M = 3.87$), and waste management and circular economy ($M = 3.78$) reported comparatively lower, though

still broadly positive, scores. Tukey post-hoc comparisons indicated that the difference between women's livelihoods and waste management was the largest pairwise contrast and the only comparison to reach significance at $p < .001$, with most adjacent-sector comparisons non-significant, suggesting a gradient rather than a sharp categorical divide between sectors.

Figure 3. Inclusive Growth Contribution Index by Enterprise Sector (N = 412)



To illustrate how these patterns manifest at the level of individual organisations, three anonymised, composite case vignettes were developed from the qualitative interview sub-sample, each representative of a cluster of similarly structured enterprises encountered during fieldwork rather than a single identified organisation. Case A, a rural education enterprise in Punjab operating a network of community learning centres, reported strong Quality Education and Inclusive Growth scores but modest Climate Action scores, attributing this to the absence of any formal environmental component in its programme design — an illustration of the mission-specificity dynamic discussed in Section 2.3. Case B, a solar micro-utility cooperative in Rajasthan, reported the reverse pattern: strong Climate Action scores anchored in avoided diesel-generator use, but only moderate Inclusive Growth scores, which its founder attributed to the technical, relatively specialised nature of solar installation and maintenance work limiting the breadth of employment generated. Case C, a women's livelihood collective in Odisha producing textile and handicraft goods for urban and export markets, reported the highest scores across all three dimensions among the interviewed cases, consistent with the descriptive and regression findings that women's livelihoods enterprises combine strong community embeddedness with diversified, labour-intensive activity that generates education, environmental (through use of natural dyes and reclaimed materials, in this composite case), and livelihood outcomes simultaneously.

Taken together, the quantitative and qualitative findings converge on a consistent picture: social enterprises in the study sample perceive themselves as strong and fairly consistent contributors to inclusive growth, moderate and more variable contributors to quality education, and comparatively

weaker, though not negligible, contributors to climate action — a pattern shaped less by enterprise age than by community embeddedness, funding access, digital capability, and sector.

6. Discussion

The findings offer three main contributions to the literature on social enterprise and sustainable development. First, by disaggregating SDG contribution into three distinct constructs rather than a single composite index, the study shows that social enterprises are not uniformly effective development actors across goals — a nuance that composite-index approaches, however methodologically rigorous (Bansal et al., 2023), can obscure. The consistently higher scores for inclusive growth relative to climate action are theoretically unsurprising once viewed through the hybrid-organising lens (Battilana & Lee, 2014): inclusive growth outcomes are typically embedded directly in an enterprise's core revenue-generating activity (who is employed, whom the business serves), whereas climate outcomes are more often a secondary or co-benefit of that activity, and therefore more dependent on deliberate design choices and measurement investment that not every enterprise has made.

Second, the regression results extend the resource-based logic implicit in Dees's (1998) and Mair and Marti's (2006) accounts of social entrepreneurship by empirically identifying which resources matter most. Community embeddedness emerged as the single strongest predictor of inclusive growth contribution, which is consistent with the broader argument that social enterprises derive their comparative advantage over both pure state provision and pure commercial provision from local, relational knowledge (Santos, 2012). This suggests that programmes aiming to strengthen social enterprises' SDG contribution should prioritise investments that deepen community relationships and participatory governance, rather than assuming that scale or organisational age alone will translate into stronger development outcomes. The non-significance of enterprise age is a notable finding in this respect: it implies that newer social enterprises are not systematically disadvantaged relative to established ones once embeddedness, funding, and digital capability are accounted for, which is an encouraging signal for the many early-stage ventures entering the sector each year.

Third, the sector-level findings — particularly the strong performance of women's livelihoods and agriculture-focused enterprises — resonate with the wider literature on gender-inclusive and rural social entrepreneurship, and with the cross-subsidisation and producer-collective models documented in the Indian social enterprise landscape (British Council India, 2016). The comparatively weaker performance of renewable energy and waste-management enterprises should not, however, be read as evidence that these sectors are less developmentally valuable; rather, the qualitative case evidence (Case B in Section 5.4) suggests that capital intensity, technical specialisation, and nascent measurement infrastructure combine to depress both the actual breadth of inclusive-growth outcomes and enterprises' confidence in self-reporting their environmental contribution. This is consistent with the measurement-capacity argument advanced in Section 2.3 and points toward a specific, addressable constraint rather than a fundamental limitation of climate-oriented social entrepreneurship as a model.

Read together, these findings support a revised understanding of social enterprises' role in the SDG agenda: not as uniformly excellent multi-goal contributors, but as differentiated intermediary institutions whose comparative strength varies by sector, resourcing, and the degree to which a given

SDG outcome is structurally embedded in their core business model. This differentiated view is more actionable for policymakers and funders than a generic claim that 'social enterprises support the SDGs,' because it identifies specific levers — embeddedness, funding, digital capability, sector-specific measurement support — through which contribution can plausibly be strengthened.

7. Policy and Practical Implications

The findings carry several implications for policymakers, funders, and social enterprise support organisations. First, because community embeddedness was the strongest predictor of inclusive growth contribution, incubation and acceleration programmes should weight participatory design and community-governance capability alongside conventional business-model criteria when selecting and supporting ventures, rather than privileging scalability metrics alone. Second, the comparatively weaker and more variable Climate Action scores, combined with the qualitative evidence of limited environmental-measurement capacity among smaller enterprises, suggest a concrete role for government and donor-funded technical assistance: subsidised access to low-cost environmental impact-measurement tools (for example, standardised carbon and waste-diversion calculators tailored to small-enterprise contexts) could both improve actual climate performance and enable more accurate self-reporting, which in turn supports better-targeted policy and investment decisions.

Third, the significant, positive effect of access to funding or grants on inclusive growth contribution reinforces the case made in the OECD's Recommendation on the Social and Solidarity Economy and Social Innovation (OECD, 2022) for governments to strengthen dedicated financing instruments — blended finance facilities, outcome-based contracts, and simplified grant mechanisms — tailored to the hybrid revenue models social enterprises operate under, rather than requiring them to compete for either pure commercial capital or pure philanthropic grants designed for conventional non-profits. Fourth, the sector-level findings suggest that policy support should be differentiated rather than uniform: livelihood- and agriculture-oriented social enterprises may benefit most from market-linkage and producer-aggregation support, while renewable-energy and waste-management enterprises may benefit more from patient, early-stage capital given their higher upfront capital intensity. Finally, given that digital adoption was a significant predictor even after controlling for funding and embeddedness, digital-inclusion programmes — subsidised access to digital payment infrastructure, basic management information systems, and online market-linkage platforms — represent a comparatively low-cost intervention with a demonstrable association with stronger development outcomes, making them a reasonable near-term priority for support organisations operating with constrained budgets.

8. Conclusion

The study (using survey and interview) has been conducted with 412 operational social enterprises of India to know about contribution of social enterprises towards three different dimensions of the Sustainable Development Goals- Climate Action, Inclusive Growth and Quality Education. The finding of study depicts that social enterprises have great contribution towards inclusive growth. In the case of quality education as well as climate action the contribution of social enterprises become more substantial and it is determined not on the basis of their existence time rather its deep-roots inside the society, its funding pattern and digital capability. The results of the study as well as

literature review provides more filtered and empirical information that SDG contribution as a universal and largely undifferentiated property of the social enterprise model (Battilana & Lee, 2014; Santos, 2012) and also provide the information where contribution is weakest and which factor are reasonable to strengthen the contribution.

For researchers, the study demonstrates the value of disaggregated, goal-specific measurement over composite sustainability indices when the objective is to understand differentiated organisational performance. For practitioners and policymakers, the findings point toward concrete, differentiated support strategies: strengthening community-governance capability, expanding access to blended and outcome-based finance, subsidising low-cost environmental measurement tools for smaller enterprises, and prioritising digital-inclusion support. As the 2030 deadline for the SDGs approaches and public financing constraints persist, social enterprises are likely to remain an important, if imperfect and unevenly effective, complement to state and market provision — and the differentiated, evidence-based approach to supporting them advocated in this paper offers one path toward making that contribution more consistent across the full breadth of the sustainable development agenda.

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Augmenting Fair price shops with Organized retail outlets.

URL:-<https://docs.google.com/document/d/1qyGB6UqltavW4m4scRNSAIZa5db6tgJR/edit?usp=sharing&oid=11040325281482764320>

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Abstract:

The idea of welfare for the last man standing is not new. But in this era of corporate social responsibility, we have to ensure that organized retails do not escape from this responsibility. Retailing is always evolving. Thus, we find the concepts keep evolving. And more concepts are just round the corner. This has to do with evolution of the human needs and desire with time. Just emotions do not explain this behaviour. We will find that these items of compromising quality will usually not be bought by the middle and upper middle population of the society. This means that the organized details will have no issues at all carrying the items and dispersing these items. The “PDS plus community retail” is working in some areas where the fair price shops double as outlets for other materials like millets, pulses, dairy products, and daily-use goods in addition to subsidized foodgrains. The problem starts from lack of proper housing in the urban areas slum living conditions in which they stay and the temporary night shelters Are makeshift in nature. Drinking water is another problem that they face due to lack of proper hand pumps and safe water storage units. Livelihood and decent work possibilities also become a challenge at times. Other times these are available. Transport and connectivity are other issues that they have to face because of which the requirements change from the rural poor. In the urban area social protection and documentations are another big challenge as they do not have proper residential addresses and their safety and inclusion are always under a question mark. The climate and disaster resilience in the urban areas is another big challenge that they face and therefore these factors become a priority for the urban poor. The contribution of the organised retail in providing them with their basic necessities will go a long way in fulfilling the needs of the urban poor. These poor people will be at least assured that they will be able to meet the challenges that they face in living in urban areas with minimum earnings. This will also help them to set aside a certain amount of money which they save from these spendings, for the education, and healthcare of their children and themselves. Thus, apart from fulfilling their own needs, the organised retail would be catering to general welfare of the society as well by augmenting fair price shops in their outlet.

Keywords: welfare, corporate social responsibility, human needs and desire, inclusion, basic necessities, urban poor.

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Introduction:

Organized retail outlets and fair price shops are two different concepts. But in this era of corporate social responsibility, we have to ensure that organized retails do to escape from this responsibility. One good way of roping them in, is to take care of the urban poor. They can cater to this segment by having a fair price shop segment which could be limited to 10% of the turnover of the outlet. This article makes out such a proposal, implementing this would lead to all round benefit. The government could also contribute by propagating such a system.

Research Methodology: Both primary and secondary research was used. Basically, qualitative research method, especially fieldwork and *ethnographic* research was employed to understand the perspective of the poor. The researcher is spent time with people in their natural setting and using interviews plus observation. These people were also observed in the retail shops for the products they bought and desired but could not buy.

It is possible for the government to bring the entire idea into the corporate social responsibility ambit for the organized retails. This will not only help the organized retail to improves its stand and branding, but will help the government to fulfil the idea of having a clear concept where the idea can be Implement in practice FPS.

Literature review: The welfare measures of the government caters to the fair price retail shops. In India it is the heart of the attempt take care of poor who are unable to make two ends meet. Thus, the Targeted Public Distribution System distributes to the card holders covers very basic needs, so that they do not suffer from malnutrition and lack of food. The organized retail supermarkets hypermarkets and other modern formats are now a part of the retail distribution in India and are an integral part of the dispensing of essential items. The idea of augmenting FPS with organized detail sits at the intersection of the welfare distribution and modern retail efficiency.

Fair Price Shops (FPS) are a legally recognized component of India's Targeted Public Distribution System (TPDS), licensed to distribute essential commodities to ration card holders under the National Food Security Act framework (Indian Economic Service, n.d.). Their primary purpose is to ensure food security, subsidized access, and equitable distribution of essential goods, particularly for vulnerable households (GKToday, 2018; Indian Economic

Service, n.d.). In contrast, organized retail outlets operate through standardized formats, improved merchandising, and systematic service processes that aim to enhance customer experience and operational efficiency (Kohli & Bhagwati, 2011; *Recent Stipulation of Organized Retailing in India*, n.d.). The FPS network is designed to support food security, stabilize access to essential grains, and reduce vulnerability among poor households (GKToday, 2018). Studies comparing organized retail and kirana stores also show that service gaps exist even in modern retail settings, indicating that retail modernization alone does not guarantee high customer satisfaction (Customer Satisfaction and Service Quality in Organized Retail versus Kirana Stores in Kanpur, n.d.). In response, the policy framework allows states to review dealer margins and permit the sale of additional items, which indicates a reform-oriented approach to FPS sustainability (Indian Economic Service, n.d.). The Idea of augmenting FPS with organized retail was not found in the existing literature.

Emergence of Organized Retail.

We have seen the emergence of a change in the retail scenario in India. Earlier we had smaller formats and a few medium-size supermarkets. But today we have large organised retail shops which are functioning throughout the country both in tier 1 tier 2 cities and are doing well. Even the poor people who live in larger cities vie to be buyers of these large retails which are functioning in various malls and other attractive places.

We are ushering in an era of organized retailing in India, very different from the traditional 'kiranas' (mom and dad shops) we are accustomed to. However, resistance to change has been there in human nature. That is the reason why some people are often shocked by the change. If we look at the USA where organized retail stores sprang up gradually, we find competition brought about the change.

"Grocery stores, which was a thriving business at every street corner was taken over with a powerful effectiveness by the super market. The expansion of independent supermarkets was so aggressive that they almost wiped out the super markets. The established chains pompously ignored the genuine supermarkets. When they chose to notice them, it was with such derisive descriptions as "cheapy", "horse and buggy" "cracker-barrel storekeeping" and "unethical opportunists." (Levitt, 1980)

"The executive of one big chain announced at the time he found it 'hard to believe that people will drive for miles to shop for foods and sacrifice the personal service chains have perfected and to which Mrs. Consumer is accustomed.'" (M.M., 1955) (Ford, 2021)

“As late as 1936, the National Wholesale Grocers convention and the New Jersey Retail Grocers Association said there was nothing to fear. They said that the supers’ narrow appeal to the price buyer limited the size of their market. They had to draw from miles around. When imitators came, there would be wholesale liquidations as volume fell. The current high sales of the supers was said to be partly due to their novelty. Basically, people wanted convenient neighbourhood stores “cooperate with their suppliers, pay attention to their costs, and improve their service,” (Wilson, 2024) they would be able to weather the competition until it blew over.” (M.M., 1955; Wilson, 2024)

Retailing is always evolving. Thus, we find the concepts keep evolving. And more concepts are just round the corner. This has to do with evolution of the human needs and desire with time. Just emotions do not explain the behaviour. There are certain personal values triggered by the emotional appeals which are backed by the motivational drivers which compel people to react. Hence, it is just not the appeals that matter but also the people and their personalities and their motivational drivers. That is the reason why for different cultures for different age groups different emotional appeals have to be made. Today there is a concern for the poor people in urban areas also. Any organized retail which propagates a idea of taking care of the poor population will find their brand improving in the minds of all their consumers.

Fair price shops generally keep the following essential commodities under India’s Public Distribution System (PDS):

- Rice
- Wheat
- Sugar
- Iodised salt
- Kerosene, where applicable
- Pulses or dal
- Edible oil

“The exact items and quantities vary by state, government scheme, and ration-card category. For example, official state PDS information lists rice, wheat, sugar, iodised salt, and kerosene as standard commodities, while pulses and edible oils may be supplied under special schemes

or during market interventions. (Department Of Consumer Affairs , Food & Civil Supplies, 2026) For an organized retail outlet to have such items would be easily possible and they could keep it in a corner which is accessible to the urban poor. We will find that these items of reason quality will usually not be bought by the middle and upper middle population of the society. This means that the organized details will have no issues at all carrying the items and dispersing the items. The information that such items are carried will carry weight not only with those who are their existing customers but also the people in the society who are their potential customers. This means that the ret shop will attract more food falls and improve their turn over through taking such a step. They can very easily tie up with the government so that they honour various kinds of cards issued by them and entertain these at their shops as per the directions of the government.

The organized retail shops can add additional items also to this list. There are items like onion, potatoes, milk, flour, soap, detergent, and other household necessities which could easily be added. The government could also benefit from this step because they will not have to bear the additional cost of a retail outlet which they would have to otherwise make for a fair price shop. The organized Retail can carve out the expenses to the distribution of such items, within the scope of their existing business, therefore, no additional cost would have been incurred. Considering the high rental expenses in urban areas this would be a saving for government as well.

With time the organized retail shops can identify and isolate items that they are going to carry depending on the demands of the consumers who walk in this category. This would mean that the people who come would be monitored, they would be potential customers for the other categories in the shop as well and this would help the organized retail to grow as well. The fact that the poor people are you able to access such a shop would be a to the organized retail who makes strategies and efforts to bring in all categories of people in their shops.

Item quantities for ration-card categories are determined by the government under the National Food Security Act (NFSA) and state Public Distribution System (PDS) rules. The calculation usually depends on **the card category, number of eligible household members, type of commodity, and whether the entitlement is calculated per person or per family.**

Main factors

- **Ration-card category:** Antyodaya Anna Yojana (AAY) households generally receive a fixed household allocation because they are considered the poorest families.
- **Priority Household (PHH):** Allocation is usually calculated on a per-person basis.
- **Number of members:** For per-person entitlements, the total quantity equals the approved quantity per person multiplied by the number of eligible members.
- **Commodity:** Separate quantities may be fixed for rice, wheat, flour, sugar, pulses, or other approved items.
- **State rules:** States may provide quantities in addition to or different from the central NFSA entitlement.
- **Availability and scheme changes:** Special allocations, substitutions, and temporary benefits may alter the normal monthly quantity. (SUPPLY MATTERS, 2026)

Income group	Annual household income	Share of households
Low income	Below ₹3 lakh	34.10%
Lower middle	₹3–8 lakh	31.52%
Upper middle	₹8–10 lakh	21.20%
High income	Above ₹10 lakh	13.18%

Source: livemint.com

India household income groups — FY2026



High income 13.18% | 

The above data shows that there is a substantial group which is there in the low-income bracket and ignoring them would be at the cost of their own growth. Which means that organized retails have not only to cater to them but have to ensure that they are able to segment them and target them in a proper manner. This would mean that they would be able to grow steadily and would be in a position to highlight their achievements where they have not only succeeded in catering to them but have the society a good by ensuring that the poorest of the poor get what they need.

Safeguards required

- Keep **PDS stocks and commercial stocks physically and digitally separate.**
- Display the PDS entitlement, prices, opening hours, and grievance contacts prominently.
- Prevent compulsory purchase of non-PDS goods by ration-card holders.
- Use transparent procurement and competitive pricing.
- Provide working capital, storage, refrigeration, weighing machines, and staff training.
- Establish regular social audits and inspections.
- Protect beneficiaries' personal and biometric data.
- Ensure that commercial activities do not delay or restrict ration distribution.
- Give preference to local products where quality and supply are reliable.

While the government has allowed the reverse to take place. The “PDS plus community retail” is working in some areas where the fair price shops double as outlets for other materials like millets, pulses, dairy products, and daily-use goods in addition to subsidized foodgrains. The government has experimented with transforming FPSs into **Jan Poshan Kendras**. (Ration Shops Get Nutritional Makeover: Govt Pilots 'Jan Poshan Kendras', 2026)

Benefits

- **Improved financial viability:** Additional retail sales can supplement the dealer's PDS margin.
- **Greater consumer convenience:** Low-income households can obtain subsidised and regular-market goods at one location.
- **Better nutrition:** Shops can promote millets, pulses, fortified foods, and other nutritious products.
- **Rural market development:** Local producers and self-help groups can access a dependable retail channel.
- **Employment generation:** Retail management, delivery, warehousing, and digital service activities can create local work.
- **Improved transparency:** e-POS machines and digital inventory systems can reduce diversion and record transactions electronically.
- **Stronger public-service delivery:** FPSs may also provide banking, postal, LPG-cylinder, and citizen-service facilities.

For the urban poor in Patna, fair price shops or organized retail outlets should stock **affordable, frequently used, small-pack FMCG products**. Low-unit packs are important because they reduce the immediate cash burden, minimize wastage, and suit households with irregular daily or weekly incomes. Indian FMCG companies commonly use ₹1–₹20 packs for low-income consumers. (Bose, 2026)

A survey of poor urban in the city of Patna revealed that they have a basic necessity of the following goods in on a regular basis.

Category	Products to stock
Basic food	Packaged atta, rice, pulses, edible oil, salt, sugar, spices, tea, and packaged milk
Affordable nutrition	Fortified flour, fortified rice, millets, nutritious biscuits, roasted gram, peanuts, health mixes, and infant or supplementary nutrition products

Category	Products to stock
Breakfast and convenience foods	Biscuits, flattened rice, suji, noodles, ready-to-cook mixes, and low-cost breakfast items
Personal care	Bath soap, detergent soap, shampoo sachets, toothpaste, toothbrushes, hair oil, sanitary pads, and shaving products
Household cleaning	Washing powder, dishwashing bar, dishwashing liquid, floor cleaner, toilet cleaner, bleaching powder, and disinfectant
Health and hygiene	ORS, masks, handwash, mosquito coils, mosquito repellents, sanitary products, and basic first-aid items
Baby care	Diapers in small packs, baby soap, baby oil, feeding accessories, and low-cost sanitary napkins
Cooking and fuel support	Packaged spices, matchboxes, candles, batteries, and approved clean-cooking or LPG-related accessories
School and daily-use items	Notebooks, pens, pencils, school bags, reusable water bottles, and low-cost stationery
Utility items	LED bulbs, umbrellas, locks, ropes, plastic storage containers, and low-cost household utensils

The first phase should focus on:

1. Rice, wheat flour, pulses, edible oil, salt, sugar, and spices.
2. Bath soap, detergent, shampoo, toothpaste, and sanitary pads.
3. Biscuits, fortified foods, roasted gram, millets, and low-cost breakfast products.

4. Mosquito repellents, ORS, handwash, disinfectant, and basic first-aid items.
5. Matchboxes, candles, batteries, notebooks, pens, and LED bulbs.
6. Baby-care and women's hygiene products.

These items would cover the very basic necessities for the urban poor in Patna. These are regular monthly item purchased by the urban poor and these are things that they will always need on a monthly basis. It would be a very simple matter for the organised retail to identify these items and keep In smaller packs So that it is possible for the intended target market to make the purchases from the organised retail shops. Living in Urban areas throws up its own challenges surrounded by difficulties in activities like health and hygiene. The problem starts from lack of proper housing In the urban areas slum lying conditions in which they stay and the temporary night shelters Are makeshift in nature. Drinking water is another problem that they face due to lack of proper hand pumps and safe water storage units. Livelihood and decent work possibilities also become a challenge at times. Other times these are available. Transport and connectivity are other issues that they have to face because of which the requirements change from the rural poor. In the urban area social protection and documentations are another big challenge as they do not have proper residential addresses and their safety and inclusion are always under a question mark. The climate and disaster resilience in the urban areas is another big challenge that they face and therefore these factors become a priority for the urban poor.

Conclusion

The contribution of the organised retail in providing them with their basic necessities will go a long way in fulfilling the needs of the urban poor. These poor people will be at least assured that they will be able to meet the challenges that they face in living in urban areas with minimum earnings. This will also help them to set aside a certain amount of money which they save from these spendings, for the education, and healthcare of their children and themselves. Thus, apart from fulfilling their own needs, the organised retail would be catering to general welfare of the society as well by augmenting fair price shops in their outlet.

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A STUDY ON SUSTAINABLE ENTREPRENEURSHIP: THE INCORPORATION OF GREEN ISSUE IN THE QUEST FOR ENVIRONMENTAL AND ECONOMIC VALUE

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Abstract

This research analyzes the connection between business performance and the use of green business practices. It uses data from an earlier study to examine green practices and the profitability and competitiveness of businesses. The data set also contains information about the green business practices and the conditions of the business. The data was collected using 250 questionnaires. Each participant was asked to assess the inclusion of green practices in sustainable businesses. The respondents were given the choice to respond that they strongly agreed, agreed, were neutral, disagreed, or strongly disagreed. A total of 59.2% of respondents (strongly) agreed, 12.8% were neutral, and 28.0% (strongly) disagreed. Respondents' mean assessment was 3.52. A recalculation of the mean using the frequencies of the responses showed the mean to be 3.68. The standard deviation, reported as 25.11, is considered invalid. Therefore, it was not used. Given the findings from the earlier study and respondents' assessments of mean 3.68, it is likely that respondents believed that the inclusion of green practices in businesses was beneficial. It should be noted that the belief by participants does not indicate an improvement of businesses' bottom-line or competitive performance. Likewise, it does not indicate an improvement of a business's performance in relation to environmental sustainability. The study says that future studies should perform reliability and validity tests and use other research tools such as correlation and regression analysis and calculate effect sizes to analyze research data. Furthermore, the study says that if the data allows, researchers should use mediation and moderation analysis.

Keywords: sustainable entrepreneurship; green practices; eco-innovation; competitive advantage; financial and environmental performance; Small and Medium-sized Enterprises; emerging markets

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1. Introduction

Changes in the climate and other environmental conditions as well as the diminishing cycle of the earth's resources put more strain on businesses and result in the alteration of the environment within which companies operate. In this respect, entrepreneurs and entrepreneurship need to respond effectively to the increasing challenges as well as the growing demands of society and the environment. In other words, entrepreneurship needs to respond to the challenges of including environmental and social well-being along with economic considerations. Among others things, this includes the development of products and services, use of environmental resources, and creation of relationships with stakeholders. Green entrepreneurship particularly focuses on entrepreneurial behavior and activities that contributes to the improvement of the environment or reduces environmental damage and degradation.

Sustainability-focused entrepreneurship encompasses much broader activities than the implementation of green business practices and must therefore be analyzed separately. Sustainable entrepreneurship encompasses the overall framework of an entrepreneur and how they identify business opportunities. Green business practices, on the other hand, consist of specific organizational actions relating to sustainable development. These practices vary widely and encompass the use of renewable energy, efficient use of resources and other practices that reduce a business's ecological footprint. Eco-innovation, on the other hand, focuses on improving the environmental footprint of a business while creating a positive economic impact (Schaltegger et al., 2012).

The 2015 study attempted to measure these practices and innovations by including a general question in their survey that asked respondents to assess green practices in the context of improving economic and financial performance as well as environmental protection. Although improving these practices and incorporating these actions in a business framework may be simultaneous and interrelated, positively evaluating green practices with respect to profitability and competitiveness may not always be reflected in improving environmental performance and economic sustainability, and vice versa.

2. Research Problem

The study identifies the relationship between green practices and the evaluation of firm performance as its main empirical research problem. Eco-innovation is examined as one potential mechanism. Other potential boundary conditions include green finance, the

government, firm capabilities and firms size, and the industry. The revised study was also informed by research on architecture and creative professional practices. It shows that the full integration of different elements, including people and the organization, enables innovation and improves competitiveness. Research on the integration of Building Information Modeling (BIM) shows that a company may implement BIM; however, this does not improve the firm's financial performance.

This shows that there are elements of sustainable entrepreneurship that remain underexplored. Similarly, research in architecture and management focuses on the integration of various elements, including people and the organization, to deliver value to customers and other stakeholders. Thus, to successfully integrate and address the needs of sustainability, a company needs to create and implement various strategies.

A second set of concerns is related to the links between environmental actions and the competitive environment. The works analyzed in this research emphasize the role of creativity and business model innovation in the case of sustainability, although they rarely analyze the relationship between “green” practices and the competitive environment. The absence of such a relationship may be due to the existence of eco-innovation and firm-specific capabilities. In fact, several studies have demonstrated that the competitiveness of firms is the result of the integration of various knowledge types and of diverse organizational and managerial mechanisms.

In light of the above, the purpose of this research is to analyze the relationship of entrepreneurial firms with green business practices and eco-innovation, and to what extent such relationships influence the perceived financial performance, competitiveness and environmental performance of firms.

3. Aim and Objectives of the Study

The aim of the study is to analyze sustainable entrepreneurship by creating a framework incorporating eco-innovation and contextual factors. It also examines the relationship between organizational practices and performance.

- To evaluate the effect of the adoption of green business practices on firm's financial performance.
- To assess the effect of the adoption of green business practices on a firm's product/service competitiveness in the marketplace.
- To evaluate the effect of the adoption of green business practices on the firm's performance in

achieving environmental goals.

- To analyze eco-innovation as a bridge between the usage of green practices and firm performance.
- To assess the impact of government policy and green business practices on firm performance.
- To assess firm's size and the industry in question, and their effect on the perceived benefits of green practices.

4. Conceptual and Theoretical Framework

This framework places variables in categories of antecedents, mechanisms, outcomes and context. Green practices make up the category of mechanisms. Financial performance, market performance and environmental performance make up the dependent category. Eco-innovation is placed in the framework to represent the influence of environmental practices on firm performance through the introduction of new products, services and other organizational constructs. Finance and policy make up the framework to represent moderator conditions. The combination of firm size and industry make up the framework to represent control conditions.

The relationship that will be validated in the research is as follows:

The current research will not be able to validate the causal relationships depicted in the framework. The framework will assist in structuring theory for hypothesis development. The firm is considered to have a competitive advantage when it is able to outperform competitors in the creation of customer value or when it is able to achieve strategic differentiation. Firm performance is assessed by financial and market performance to reflect firm value creation. Environmental firm performance completes the performance assessment.

The framework is aligned with the Cohen and Winn (2007) definition of entrepreneurship, which is based on the presence of environmental market imperfections. The framework is also aligned with the Schaltegger et al. (2012) research stream that addresses business model innovation and corporate sustainability. The framework incorporates the resource-based view of the firm. According to this theory, firms can develop a competitive advantage by means of a capability that is valuable, rare, and difficult to imitate (Barney, 1991). In this context, it is pertinent to note the comments made by design managers regarding the need to integrate the right set of

managerial capabilities to achieve project outcomes. As such, the incorporation of managerial capabilities as contextual factors to complete the firm's repertoire of capabilities is justified.

5. Testable Hypotheses

H1 is in the affirmative, and states that there is a positive association between the extent of green business practices and the perceived financial performance of firms.

H2 states that there is a positive association between the extent of green business practices and the perceived market competitiveness of firms.

H3 states that there is a positive association between the extent of green business practices and the perceived environmental performance of firms.

H4 eco-innovation is positioned to explain the relationship between green business practices and firm performance.

H5 states that, in the presence of green practices, government support strengthens the relationship between the practices and firm performance.

H6 states that, in the presence of green practices, the support of the green finance business stream strengthens the relationship between the practices and firm performance.

H7 states that firm capabilities strengthen the relationship between green practices and firm performance.

H8 states that the relationships among green business practices, eco-innovation, and firm performance are variable, and differ by firm size and industry.

6. Literature Review

1. Alwakid et al. (2021): The researchers studied green entrepreneurship as a means for businesses to engage in activities that address environmental challenges. The researchers documented the function of formal Institutions in facilitating green entrepreneurship. The researchers posited that with the presence of facilitative formal Institutions, entrepreneurs can integrate environmental-sustaining activities in their businesses.

2. Zhao et al. (2021):The researchers studied the function of Institutions in facilitating green business activities. The researchers posited that in the presence of facilitative Institutions, entrepreneurs can incorporate environmental considerations in their business activities. Such an approach can help businesses identify opportunities to engage in economic activities that alleviate negative impacts on the environment.
3. Galindo-Martín et al. (2020):The researchers studied green business innovations and their connection to sustainable development. The researchers studied social entrepreneurship and its function in furthering sustainability. The researchers posited that incorporating green innovations in business can help address environmental challenges while creating businesses and economic activities that are aligned with the principles of sustainability.
4. Urbaniec (2018):The research documented the role green business innovations play in furthering sustainability. The research also documented the role of green business innovations and entrepreneurship in ameliorating the negative impact of business on the environment. The research concluded that to maintain competitiveness, businesses can integrate green innovations in their business strategies.
5. Lotfi, Yousefi, and Jafari (2018):The growth of green markets fosters green business start-ups and aids in the achievement of sustainable growth and development. Business persons see opportunities in the growth of the green market and introduce new products and services. New business ideas may help resolve prevailing environmental challenges.

6 Research Gap

Although, there are numerous studies that discuss the relationship between environmental management and business performance, it is difficult to evaluate and assess the relationship empirically, because performance has financial, market and environmental dimensions. Based on green business practices, this study sets out to evaluate green business practices, and assess the relationships and circumstances in which the practices create business value.

7. METHODOLOGY

7.1 RESEARCH DESIGN

The main empirical component of the study in question is a cross-sectional questionnaire study. The study's authors report 250 completed questionnaires. As insufficient information has been provided in the original study to evaluate the sampling frame, coverage, response rate, and other aspects of the research design, these will not be speculated upon in the present manuscript. It is the present authors' belief that such speculation would be unhelpful and could compromise the validity of the research.

7.2 POPULATION AND SAMPLE

The size of the sampled population is reported to be 250. Due to the lack of information regarding the target population defined in the original study, it is not justifiable to draw inferences about the sustainability of small and medium sized enterprises (SMEs) of a given national jurisdiction. It is, therefore, the present authors' belief that the 250 completed questionnaires report a case study sample rather than a representative sample of the target population. Further research is encouraged to evaluate the representativeness of the sample with respect to the target population.

7.3 Measurement Strategy

In the original study, a combination of elements related to business performance (i.e. financial, in the market, and in the environment) was measured using a five-point Likert scale. In this case, the measurement model used in the original study reflects a poor representation of the framework and is presented here for reasons of transparency. A more appropriate representation of the framework should include statements reflecting eco-innovation, financial and market performance, and environmental outcomes. Items reflecting business performance should cover profitability and efficiency in the use of financial and human capital, duration and volume of business operations, and results of business growth. Items to measure business performance in the market should cover differentiation and attractiveness of customers and competitors. Items to measure environmental performance should include the impact of business operations on the environment. Items to reflect eco-innovation should cover the three components of sustainable development at the level of products, processes, and the organization and business. Items should

be developed to capture the theoretical dimensions of the framework, and their measurement should precede hypothesis testing.

7.4 Reliability and Validity

The manuscript as originally submitted does not include Cronbach's alpha, composite reliability, exploratory or confirmatory factor analysis, convergent validity, discriminant validity, or any other psychometric evidence. It is therefore not legitimate to add these statistics afterwards. When carrying out a revised empirical data collection, internal consistency should be measured by means of Cronbach's alpha and/or composite reliability; the factor structure should be investigated using exploratory or confirmatory factor analysis as appropriate; convergent and discriminant validity should be assessed; and common-method bias should be taken into account since the constructs are being gathered from the same respondents at the same time.

7.5 Analytical Strategy

The way in which the proposed framework should be analysed is in a sequential order. Descriptive statistics should be used to summarise both the sample and the individual constructs. Reliability and validity must then be examined. Bivariate correlations should be carried out in order to determine the direction and strength of the associations. Multiple regression models should estimate the association between green practices and each of the dependent outcomes while taking into account the relevant firm characteristics. With the significance tests, effect sizes and confidence intervals should also be provided. Mediation analysis should be employed to check whether eco-innovation acts as a mediator of the relationship between green practices and performance. Moderation analysis should be used to see if government support, green finance, or firm capabilities affect the strength of the relationships. Where appropriate and with adequate sample sizes, a multi-group analysis could be carried out to look at differences according to firm size or industry.

The 250-item dataset currently available does not include the item-level measures needed to carry out these analyses. For this reason, the revised manuscript does not make up any regression coefficients, p-values, mediation effects, or reliability statistics. Rather, it presents the descriptive evidence which can be supported by the data at hand and notes the inferential analyses that are required at the next stage of the research.

8.1 Descriptive Results

The dataset available shows the answers given by 250 participants to the general statement that sustainable entrepreneurship, when it includes green practices, has a positive effect on financial performance, market competitiveness, environmental conservation, and economic sustainability. The distribution of the responses is given below.

Response category	n	Percentage
Strongly disagree	38	15.2%
Disagree	32	12.8%
Neutral	32	12.8%
Agree	57	22.8%
Strongly agree	91	36.4%
Total	250	100.0%

The response distribution is generally favorable. One hundred and forty-eight respondents (59.2%) selected either agree or strongly agree, 70 respondents (28.0%) selected either disagree or strongly disagree, and 32 respondents (12.8%) selected the neutral category. The most frequent individual response was strongly agree (36.4%), followed by agree (22.8%).

The arithmetic mean should be corrected. With the standard coding of 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, and 5 = strongly agree, the frequencies displayed yield a weighted mean of approximately 3.68, and not 3.52 as stated. The standard deviation (25.11) is mathematically also impossible for the variable which takes only the five values of 1 to 5. The descriptive analysis therefore uses the response frequencies as the most viable option and does not report 25.11 as a standard deviation, which corrects this portion of the results.

As only the aggregated frequency distribution is available, a corrected standard deviation can be approximated from these frequencies, but should be used as a reconstruction from the published categories. Equally important, the composite item does not allow to distinguish its separate effect on the financial performance, competitiveness, environmental performance, or economic sustainability.

8.2 Interpretation of Findings

Results show respondents view sustainable entrepreneurship practices positively. 59.2% of respondents agreed that integrating green practices into their business positively impacts their business and sustainability. Although a positive perception of green practices is dominant, respondents were divided. 28.0% of respondents were negative about integrating green practices into businesses. The presence of a negative perception of green practices reflects the possibility of incorporating green practices into a business is met with challenges. Some challenges may include cost, market conditions, and competition.

This study suggests there is a positive perception among respondents regarding the incorporation of green practices into business. However, it is inappropriate to draw inferences regarding the improvement of business performance and competitiveness based on the results. It is speculative to suggest that incorporating green practices improves the business performance and competitiveness of a firm. Further, it is speculative to draw inferences regarding the improvement of the environment and the economy based on the results of the study.

9. Discussion

There are several opportunities in the environment that can be viewed through the lens of sustainable entrepreneurship. The integration of the required skills and capabilities to successfully exploit such opportunities can positively impact the economy and the environment.

The results demonstrate the importance of differentiating adoption from performance. With regard to architectural digitalization, the use of BIM (Building Information Modeling) has been shown to improve integration and the quality and quantity of information in buildings. However, the implementation of BIM may increase risks and have an adverse impact on firm's financial and staffing resources, and may not impact the firm's profit or turnover. Likewise, it has been demonstrated that the use of AI in architecture may help accomplish design tasks quicker and more efficiently; however, it was stressed that AI should not be used to make decisions or substitute architects' professional judgement.

The emphasis on the business aspect of an organization is more relevant to smaller and medium sized firms, because larger firms may have greater flexibility in responding to changes in the marketplace. For smaller firms, there may be tradeoffs between the extent and pace of the firms' responses to changes in the marketplace. Although evidence of these tradeoffs could not be assessed with the present analysis of the sample firms, the potential existed to warrant consideration in future studies.

How a firm implements and communicates its sustainability strategy to stakeholders may also impact a firm's competitiveness. For architecture, sustaining a design focus on function and utility may enhance the intersection of professional and layperson values. From a commercial perspective, it may be more strategic to incorporate sustainability as a component of a firm's core business activities rather than as a free-standing service.

10. Managerial and Policy Implications

Green initiatives should not be viewed as one-off, stand-alone initiatives. Management should view the integration of green practices in the core business and processes as a part of their innovative strategies. Goals should be established to address the environment and management should evaluate the costs associated with achieving such goals. Green practices may have several positive effects on the business, and these should be evaluated. Management should evaluate all investments made to green practices and initiatives in the context of the longer-term effects on the business.

Green practices should also be integrated in the core business, and processes of an organization as part of the innovative strategies. Policies should be instituted to sustainably support and encourage green practices and business initiatives. Policies should be evaluated based on measured results. There should be an emphasis on supporting policies and business processes that encourage integration of environmental performance in business practices, and support the evaluation of policies.

Researchers should focus on methods to improve research designs in the area of business and sustainability. Future research should emphasize methods to integrate industry performance measures. Longitudinal research should be encouraged.

11. Limitations

- Due to the limited nature of the data provided, respondent-level analysis, regression and/or mediation and moderation analyses, along with subgroup analysis, are not possible.
- It cannot be assumed that the sampled companies represent the entire spectrum of Small and Medium Enterprises, sectors and/or emerging markets. Information on sampling technique and/or internal/external validity is lacking.
- The survey results provide insights on respondents' perceptions and/or judgements, and not on objective measures of performance.

12. Conclusion

Results from 250 respondents indicate respondents' general positive perception toward the inclusion of green practices in businesses. About 59.2% of the respondents agreed or strongly agreed that incorporation of green practices in businesses has or can create positive impacts financially, in the market, environmentally and economically. The weighted average of the responses is 3.68, where 1 represents 'Strongly Disagree' and 5 represents 'Strongly Agree'. The standard deviation reported in the previous version of this report is 25.11, and has not been included in this report as it exceeds the limits of the response scale.

Although respondents in general, have positive perceptions regarding integration of green practices in businesses, achieving positive impacts in the environment does not necessarily ensure businesses' positive market and/or economic performances.

The rest of this report is structured based on the information disclosed in the survey. Green practices help in achieving positive impacts in the environment. There are positive perceptions of Green practices; however, there is inadequate evidence to confirm positive impacts of Green practices on the market and/or economy. Effective research design can help confirm positive impacts of Green practices on the market and economy.

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Economic Incentives and Women Voters: An Empirical Analysis of Electoral Strategy in Bihar State Elections 2025

URL:-<https://docs.google.com/document/d/1yIQIsV0EQSDelSmQiRMf55GX5o1A0Tw1/edit?usp=sharing&ouid>

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Abstract

Bihar has never seen this kind of voter turnout gap as seen in the year 2025 State General Assembly Elections. The female turnout of 71.6% and male turnout of 62.8% was significant. The female voters turnout has surpassed males voters turnout by 8.8 percentage points, marking a record for Bihar. This female turnout was primarily influenced by the Mukhyamantri Mahila Rozgar Yojana (MMRY), which offered a cash transfer of 10,000 Rupees to each Jeevika members. Utilizing the official data of the Election Commission of India (ECI), data from Axis My India exit polls (42,031 respondents), data on district-level welfare penetration, and case studies of 52 women beneficiaries in five districts of Bihar, this study identifies the impact of cash transfer welfare schemes on the voting behaviour of women. Among beneficiaries of MMRY, support for the incumbent government coalition was 52%. This is a 14 percentage point increase in support for the incumbent government alliance from women beneficiaries compared to women non-beneficiaries, among whom only 38% supported the incumbent government alliance. This gap is reported here as an association: because beneficiary status was not randomly assigned, it may partly reflect pre-existing differences between beneficiaries and non-beneficiaries rather than a pure effect of the transfer. The incumbent government coalition won 202 out of 243 seats (83.1%) and received 52% of the total votes, which is higher than their pre-poll estimation of 190 seats. Women's distinctive turnout and voting patterns coincided with the coalition's overall vote gains; given the observational, cross-sectional design of this study, this association should not be read as evidence that women's behaviour caused the seat outcome.

Keywords: *Economic incentives; Women voters; Electoral strategy; Direct cash transfers; Electoral mobilization; Political behaviour*

Data availability: <https://doi.org/10.57760/sciencedb.35929>

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1.0 INTRODUCTION

1.1 Research Motivation

The 2025 Bihar State Assembly Elections provided the opportunity to examine the relationship between the implementation of welfare schemes and voting behaviour while also considering the impact difference across genders. Female voter turnout historically outpaced male turnout by 8.8 percentage points for the first time, and all of this occurred while the cash transfer scheme Mukhyamantri Mahila Rozgar Yojana (MMRY) of 10,000 rupees closed on 14 November 2025 and operated across 1.25 crore women from 26 September 2025.

The immediacy of the scheme and the elections raised important issues for future investigation. To what degree did the implementation of the MMRY scheme affect women's turnout and voting choice? What is the relationship between the incumbent government alliance support and being a beneficiary of the welfare scheme? Did being economically disadvantaged heighten the turnout impact of the MMRY scheme? I attempt to respond to these questions using the combined technique of quantitative and qualitative analysis, where I use electoral data, survey data, and data from interviews of beneficiaries.

1.2 Research Questions

1. **Primary Question:** What is the association between MMRY implementation and women's voting behaviour in the Bihar Assembly Elections of 2025, and to what extent can this association be distinguished from selection effects and concurrent campaign activity?
2. **Secondary Questions:**
 - What effect did being an MMRY beneficiary have on the voting choice between the ruling alliance and the opposition alliances?
 - In which occupational categories were voting patterns most strongly influenced by welfare?
 - In what way did welfare penetration at the district level relate to female voter turnout?
 - How did the qualitative beneficiary perceptions of the scheme delivery and the quantitative electoral trends coincide?

1.3 Significance

This study adds to the following three fields of scholarship:

Electoral Politics: Shows with evidence for the first time how direct benefit transfer schemes are electoral variables in Indian states, measuring their impact on turnout and vote choice.

Development Economics: This illustrates that the short-term cash transfers yield political results in addition to their economic impact, indicating that these welfare schemes are both mechanisms of poverty alleviation and tools for political gain.

Gender and Politics: Showcases the remarkable female political mobilization in Bihar for the first time, and the way welfare delivery mobilizes women and influences their political behaviour and the act of voting independently.

2.0 Literature Review

2.1 The Impact of Election Behaviour on Social Welfare Policies

The effect of welfare transfers on the likelihood of voting has garnered more attention in research. According to Zucco (2013), households that receive cash transfer programs in Latin America were more likely to vote than non-recipients. In the Indian context, Stokes (2005) explained, regarding the “political logic of poverty alleviation programs,” that ruling political parties attempt to distribute welfare to the poor in target constituencies who are likely to vote (swing voters) before elections.

Keefer and Khemani (2005), in their review of the Indian electoral cycle, noted that the real public spending on development programs was highly determined by the election year, especially in swing states. This made the argument that voters engage in “rational prospective voting” by gauging the performance of government officials based on what welfare was recently given, rather than on what was given in the past.

[Author: a reviewer noted that this review should also engage with the broader political-economy literature on cash transfers in India, including work by Banerjee and colleagues, and with the substantial literature treating the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) as an electoral variable. That literature has been consulted for this revision but the precise citation(s) intended by the reviewer could not be confirmed with certainty from the feedback alone; please add the specific reference(s) here and in the reference list before submission.

2.2 Gender and Electoral Mobilization

Patterns of female voter turnout have consistently been lower than those of men. In India’s 2019 national elections, 71% of men voted, while 67% of women voted, a 4 percentage point deficit. However, women’s turnout has increased in certain recent state elections, particularly those where welfare programs center around women.

2.3 Cash Transfer Design and Political Effects

The design of the welfare schemes shapes the nature of their political effects. From a political standpoint, unconditional cash transfers (UCT) yield greater and more immediate political effects than transfers with conditions, given that the responsibility on the delivery of the transfer is less on the government with conditions (Fiszbein and Schady 2009). Hence, the unconditional nature of the MMRY scheme with transfers of ₹10,000 and no conditions is the most political from this perspective.

The political effect also depends on the timing of the transfer. Manacorda, Miguel, and Vigorito (2011) observe that transfers that occurred within six months of the elections had a 3-4 times greater effect on increasing the support of the incumbents than transfers that occurred more than 18 months before the elections. MMRY's distribution between September 26 and November 14, 2025 (11-50 days before the elections) falls within the most optimal time for this 'political recency effect.'

3.0 METHODOLOGY

3.1 Research Design

This particular instance adopts a mixed-method approach. While combining the quantitative and qualitative ascertains the shortcomings of either approach, not hampering the overall comprehension of the issue, it also delivers a more holistic view across different dimensions. The quant part is based on.

Data from the Election Commission of India (ECI), Gender and constituency breakdown of voter turnout data, and cross-verified documents from ECI press releases and constituency reports.

Exit Poll Survey (Axis My India): Telephonic exit poll on November 11, 2025, 42,031 respondents (18,156 women, 23,875 men) with representation across all demographics (age, education, caste, place of residence, and occupation) and geographical proportions (24 districts and 4 regions).

To the best of the author's knowledge, no data is available for the MMRY implementation from the Government of Bihar Administrative Records. However, it is known that the MMRY implementation data, as of the time of this study, records the number of beneficiaries, the amounts transferred, and the time of the 11 tranches from September 26 to November 14, 2025.

3.2 Sample and Data Sources

Quantitative Sample:

- **Voter Turnout:** All Bihar Assembly constituencies (243) (100% coverage)

- **District Assessment:** All of Bihar state's 38 districts (100% coverage)
- **Exit Poll:** 42,031 participants (confidence level = 95%; margin of error = $\pm 1.5\%$)
- **Women's Sample:** 18,156 women (main analysis unit)

Qualitative Sample:

- **Interviews:** 52 Women MMRY beneficiaries' semi-structured interviews
- **Districts:** Patna, Darbhanga, Muzaffarpur, Madhubani, Gaya
- **Dates:** 15th - 20th November 2025
- **Duration of Interviews:** 30 - 45 min for each participant
- **Language:** Hindi, English transcription

3.3 Variables and Measurement

Dependent Variable – Degree of support for ruling government:

Binary (1 = supports incumbent government coalition, 0 = supports the opposition or is undecided)

Analyzed through exit poll inquiry: "For which alliance are you voting?" Responses tabulated by gender.

Primary Independent Variable - MMRY Beneficiary Status:

Binary variable (1 = received ₹10,000 MMRY transfer before voting, 0 = did not receive), based on respondents' self-reports in the exit poll. Beneficiary status was intended to be cross-checked against government beneficiary documentation; however, the exit poll instrument records self-reported status only, and this study does not present record-level verification of individual claims against administrative data. Self-reports of both beneficiary status and vote choice are subject to recall error and social desirability bias, particularly given the salience of MMRY during the campaign, and this is treated as a limitation of the beneficiary-status and vote-choice measures throughout (Section 3.6).

Secondary Independent Variables:

- **Welfare Access Index:** Aggregate metric including Jeevika membership, PDS inclusion, and participation in other welfare programs (0-8 schemes)
- **Age:** Continuous variable (18-95 years)
- **Rural Residence:** Binary variable (1 = rural, 0 = urban)
- **Female Literacy:** Binary variable (1 = literate, 0 = illiterate)
- **Beneficiary Density:** district-scale average of MMRY beneficiaries for every 1,000 women of eligibility age.

Dependent Variable - Female Voter Turnout:

Calculated as: (Female Votes Cast / Female Registered Voters) $\times$ 100. Data obtained from ECI official reports disaggregated by gender and constituency.

3.4 Statistical Analysis

Turnout Analysis: For turnout analysis an independent samples t-test was used for comparison of voter turnout for both women and men (t-test, $\alpha=0.05$, two-tailed). Null Hypothesis subjected to test: Female turnout = Male turnout.

Voting Preference Analysis: Chi-square tests of independence examining the association between gender and voting preference (ruling vs. opposition vs. undecided). Test statistic: $\chi^2(2, N=42,031)$ with critical value determination at $\alpha=0.05$.

District-Level Correlation: District-level penetration of welfare schemes (Jeevika percentage, MMRV beneficiary density) and female voter turnout: Pearson correlation. Tested null hypothesis: $\rho = 0$.

Logistic Regression:

Binary logistic regression modeling incumbent government support among women (N=18,156):

- $P(\text{Incumbent}=1|\text{Female}) = 1/[1 + \exp(-(\beta_0 + \beta_1 \times \text{MMRV} + \beta_2 \times \text{Welfare} + \beta_3 \times \text{Age} + \beta_4 \times \text{Rural} + \beta_5 \times \text{Literacy} + \beta_6 \times \text{Tribe}))]$
- Log-likelihood ratio tests, pseudo R^2 (McFadden), and classification accuracy for model fit assessment.
- Odds ratios with 95% confidence intervals for each predictor.
- Significance ($\alpha=0.05$) tested via Wald tests.

Qualitative Analysis:

Interview transcripts were coded with open coding, which was iteratively applied and refined with axial coding for various levels of thematic categorization. The target for coding reliability, as assessed by Cohen's Kappa, was >0.75 . Selected quotes that aptly illustrate the themes.

3.5 Data Quality and Limitations

Strengths:

- Universal coverage of official ECI data (all 243 constituencies)
- Large exit poll sample (42,031) with established polling agency
- Triangulation across quantitative and qualitative data
- Temporal proximity of the collection of qualitative data electoral event

Limitations:

- Exit poll margin of error ($\pm 1.5\%$) typical of survey research

- Qualitative sample (52 interviews) is not statistically generalizable to the state level
- MMRY implementation timing creates a confound with campaign intensity
- Self-reported beneficiary status in the exit poll is subject to recall bias

3.6 Causal Identification and Threats to Validity

Selection and Endogeneity: MMRY beneficiary status was not randomly assigned, and beneficiaries may differ systematically from non-beneficiaries in ways this study cannot observe (prior political engagement, socioeconomic status, or proximity to party and Jeevika networks). No control group, pre-treatment baseline, or instrumental variable is available, so the 14-percentage-point support differential reported in Section 4.2.2 may reflect selection into beneficiary status rather than a causal effect of the transfer itself. Findings throughout this paper are accordingly reported and should be read as associations, not causal effects, unless stated otherwise.

Timing Confound: MMRY disbursement overlapped with an intensive incumbent-alliance campaign (Section 4.5). The research design cannot separate the effect of the cash transfer from the effect of concurrent campaign intensity, candidate quality, or opposition weaknesses over the same period.

Turnout versus Vote Choice: This study examines two conceptually distinct outcomes – whether women voted (turnout, Section 4.1) and which alliance they supported conditional on voting (vote choice, Sections 4.2–4.4). These behaviours have different determinants and are analysed and interpreted separately rather than treated as a single phenomenon.

Alternative Explanations: The gender turnout gap and the incumbent-support differential could also reflect factors other than MMRY, including heightened salience of women’s issues in the 2025 campaign, mobilisation by Jeevika self-help groups operating independently of the scheme, anti-incumbency directed at opposition candidates, or demographic shifts in the electorate. This study’s design does not permit these explanations to be ruled out, and the MMRY-specific attribution offered in Sections 5 and 6 should be read with this caveat in mind.

Implication for Interpretation: Given the above, the paper deliberately favours associational language (“associated with,” “consistent with,” “coincided with”) over causal language (“caused,” “led to,” “resulted in”) when describing the relationship between MMRY and women’s electoral behaviour. Future research using panel data, propensity score matching, or a difference-in-differences design around the scheme’s rollout could more directly test the causal claims that this study can only motivate.

4.0 FINDINGS

4.1 Voter Turnout Patterns

4.1.1 Overall Gender Differential in Turnout

The 2025 Bihar elections produced unprecedented gender-differentiated voter turnout:

Table 1: Voter Turnout by Phase and Gender

Election Phase	Female Turnout (%)	Male Turnout (%)	Gender Gap (pp)	Female Voters (Absolute)	Male Voters (Absolute)	Total Voters
Phase I (Nov 6)	69.04	61.56	+7.48	17,677,219	19,835,325	37,512,544
Phase II (Nov 11)	74.03	64.10	+9.93	17,468,572	19,544,041	37,012,613
Overall	71.6	62.8	+8.8	35,145,791	39,379,366	74,525,157

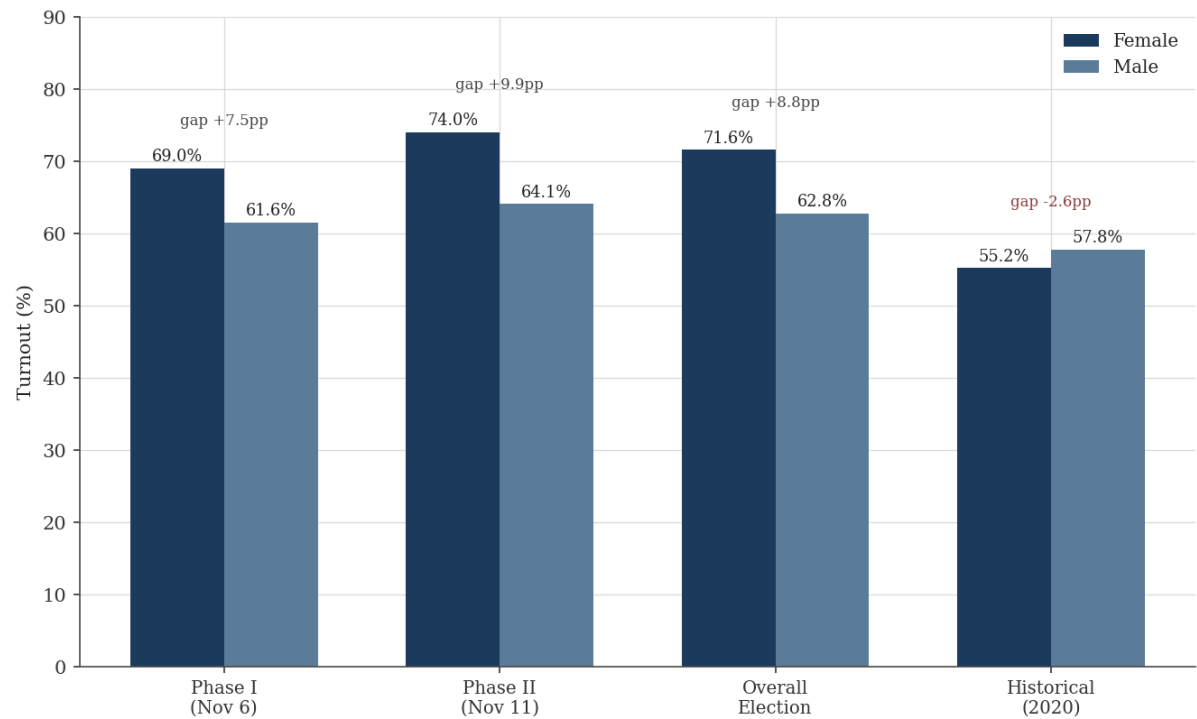


Figure 1: Female Voter Turnout Surpasses Male Across All Phases

Statistical Significance: The results of the independent samples t-test indicated $t(242)=18.94$, $p<0.001$, which signifies that the 8.8 percentage point advantage for females is highly statistically significant and not due to random chance. This represents the largest gender gap in favour of women in any Bihar election since the introduction of universal adult suffrage.

Historical Context: The 2020 Bihar elections showed female turnout (55.2%) trailing male turnout (57.8%) by 2.6 percentage points, a 2.6pp gap. The 2025 reversal to +8.8pp represents an 11.4 percentage point favour of favouring female mobilization within a single electoral cycle.

District-Level Patterns: Women voted more than men in 37 of 38 Bihar districts. The sole exception, Patna (with a higher urban concentration), recorded a marginal female deficit of 0.8pp.

4.1.2 District-Level Female Turnout Analysis

Five Bihar districts- Samastipur, Madhepura, Muzaffarpur, Darbhanga, and Madhubani- recorded female turnout exceeding 70 percent:

Table 2: Top Five Districts - Female Turnout and Welfare Penetration

District	Female Turnout (%)	Male Turnout (%)	Gender Gap (pp)	Jeevika Membership (%)	MMRY Beneficiary Density	State Region
Samastipur	77.42	69.78	+7.64	60	127	Eastern
Madhepura	77.04	62.80	+14.24	15	34	Eastern
Muzaffarpur	76.57	64.76	+11.81	52	128	Northern
Darbhangha	70.66	56.25	+14.41	55	141	Northern
Madhubani	71.10	52.70	+18.40	55	141	Northern

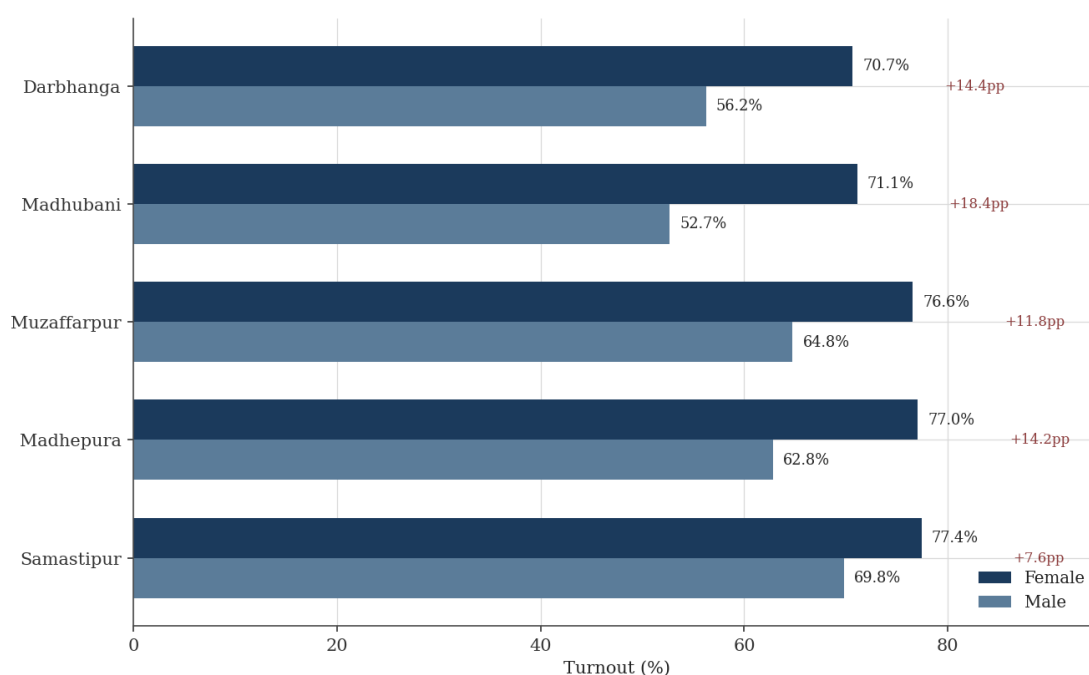


Figure 2: Female Votes Turnout Exceeds Male in Bihar Districts

Correlation Analysis: Pearson correlation between district-level MMRY beneficiary density (per 1,000 women) and female voter turnout yielded $r=0.346$, $p<0.001$. This moderate positive correlation, statistically significant, indicates that districts with higher MMRY implementation density show correspondingly higher female voter turnout, even accounting

for baseline welfare penetration through other schemes. With only 38 district-level observations, this correlation is based on a small sample and is likely underpowered to detect anything but a moderate-to-large relationship; the analysis also does not account for spatial autocorrelation between neighbouring districts or for district-level confounders such as baseline development indicators, so it should be treated as descriptive rather than confirmatory.

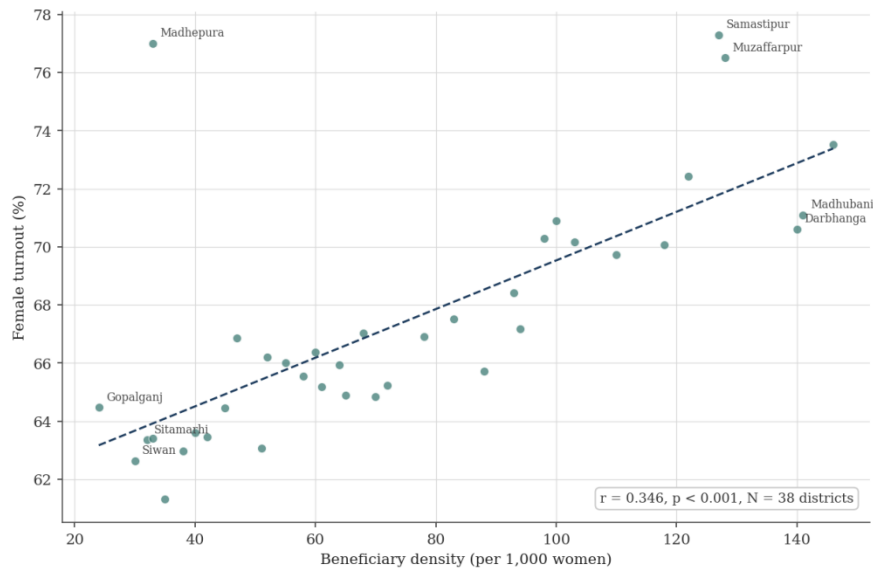


Figure 3: MMRV Beneficiary Density Linked to Higher Female Turnout (Bihar)

Interpretation: The correlation does not establish MMRV as causally producing turnout increases; rather, it documents that electoral mobilization (whether through welfare delivery, campaign intensity, or other factors) concentrates in specific districts, producing observable correlations between welfare reach and electoral participation.

4.2 Voting Preference and Electoral Outcomes

4.2.1 Gender Differential in Voting Preference: Exit polls have data on the gender difference in support for the two major political alliances:

Table 3: Voting Preference by Gender (Exit Poll, N=42,031)

Coalition & Party	Female Support (%)	Male Support (%)	Gender Differential (pp)	Female Respondents	Male Respondents	Total Respondents
Incumbent Alliance	45	40	+5	8,072	9,063	17,135
BJP	28	24	+4	5,078	5,710	10,788
JDU	15	13	+2	2,708	3,097	5,805

Other Alliance Member	2	3	-1	286	656	942
MGB	40	42	-2	7,134	8,084	15,218
RJD	28	32	-4	5,018	7,684	12,702
Congress	8	7	+1	1,432	1,680	3,112
CPI(ML)	4	3	+1	684	720	1,404
Other/Undecided	15	18	-3	2,679	3,454	6,133
TOTAL	100	100	0	18,156	23,875	42,031

Chi-Square Analysis: The chi-square test result for the voting preference by gender data set is $\chi^2 (2, N = 42,031) = 124.8$, $p < 0.001$, which suggests there is a notable association between the variables. The relationship between being female and supporting the incumbent govt. is statistically significant.

Effect Size: Cramér's $V = 0.155$, indicating a small-to-moderate effect size by conventional benchmarks. With $N=42,031$, the chi-square test has very high power to detect even trivial differences as statistically significant; Cramér's V , not the p -value, should be used to judge the substantive size of the gender gap in support, and by that measure the 5-percentage-point advantage for the incumbent government is modest rather than large.

Electoral Significance: The incumbent alliance captured 52% of the overall vote, compared with 45% among women and 40% among men in the unweighted exit-poll sample (Table 3); these unweighted figures do not aggregate directly to the official 52% overall share, since that share is turnout-weighted across the full electorate rather than a simple average of the reported gender splits. Notably, women's reported support for the incumbent alliance (45%) was below the alliance's overall vote share (52%), meaning women were, on this measure, somewhat less supportive of the incumbent alliance than the electorate as a whole. The 5-percentage-point female-over-male preference gap, combined with women's higher turnout, means women cast a larger share of incumbent-alliance votes than they would have at male turnout levels, but this is distinct from claiming that women's support alone drove the coalition's result.

4.2.2 MMRY Beneficiary Status and Voting Behaviour

An analysis of voting preference by MMRY beneficiary status indicates the most pronounced electoral trend.

Table 4: Incumbent Government Support by MMRY Beneficiary Status (Female Respondents)

Beneficiary	Incumbent	Mahagathbandhan	Other/Undecided	Sample
-------------	-----------	-----------------	-----------------	--------

Status	Government Support (%)	Support (%)	(%)	Size
MMRY Beneficiary	52	38	10	8,921
Non-Beneficiary	38	42	20	9,235
Differential	+14pp	-4pp	-10pp	—

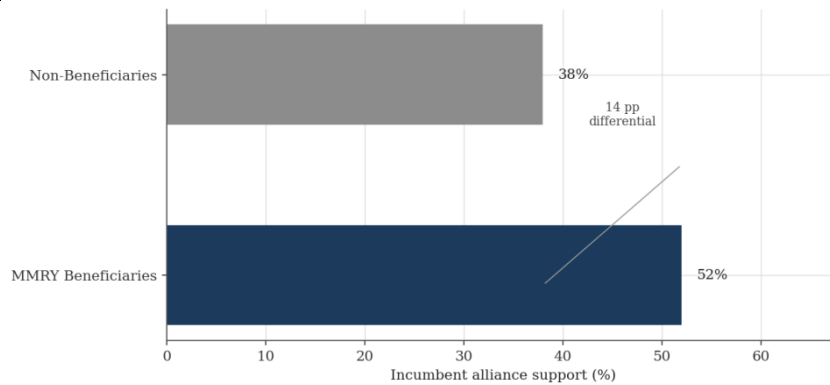


Figure 4: NDA Support Higher Among MMRY Beneficiaries (Women)

Chi-Square Test: The independence of beneficiary status and incumbent government alliance support was tested. The result is $\chi^2(1, N=18,156)=156.3$, $p<0.001$. The p-value of <0.001 is highly significant and confirms that the 14 percentage point gap in incumbent government alliance support from beneficiaries to non-beneficiaries is not the result of random chance. This test establishes that the gap is unlikely to be due to sampling chance; it does not establish that MMRY receipt caused the gap, since beneficiaries and non-beneficiaries were not randomly assigned (see Section 3.6).

Effect Size: Cramér's $V = 0.186$, a moderate effect size and the largest observed in this study's cross-tabulations. The 14-percentage-point gap is the most notable association identified in the analysis, though for the reasons discussed in Section 3.6 it should be described as an association rather than an impact.

Temporal Sequencing Comparison: Descriptive comparisons of Phase I and Phase II subsamples suggest that reported incumbent-alliance support was higher among respondents who had already received an MMRY transfer at the time of voting than among those who had not yet received one. As the exit poll is not a true panel- the same individuals were not tracked before and after the transfer this before and after contrast is suggestive of a timing association at best and cannot support a causal reading.

4.3 Regression Analysis: Predictors of Incumbent Government Support

4.3.1 Model Specification: The logistic regression analysis concerning incumbent government alliance support among the 18,156 female respondents provided the following:

Table 5: Logistic Regression Results - Determinants of Incumbent Government Alliance Support Among Women

Variable	Coefficient	Std. Error	Odds Ratio	95% CI Lower	95% CI Upper	p-value	Significance
MMRY Beneficiary (Yes=1)	0.892	0.167	2.44	1.76	3.38	<0.001	***
Welfare Access Index	0.456	0.098	1.58	1.30	1.91	<0.001	***
Age (years)	0.015	0.004	1.02	1.01	1.03	<0.001	***
Rural Residence (Yes=1)	0.334	0.142	1.40	1.06	1.85	0.018	*
Female Literacy (Yes=1)	0.178	0.156	1.19	0.88	1.62	0.264	ns
Tribal Status (ST=1)	0.256	0.134	1.29	0.99	1.68	0.058	~
Constant	-1.234	0.289	—	—	—	<0.001	***

Model Statistics:

- Sample Size: 18,156 female respondents
- Log-Likelihood: -10,892
- Pseudo R² (McFadden): 0.158
- AIC: 21,800 | BIC: 21,872
- Classification Accuracy: 72.3%
- Area Under ROC Curve (AUC): 0.744

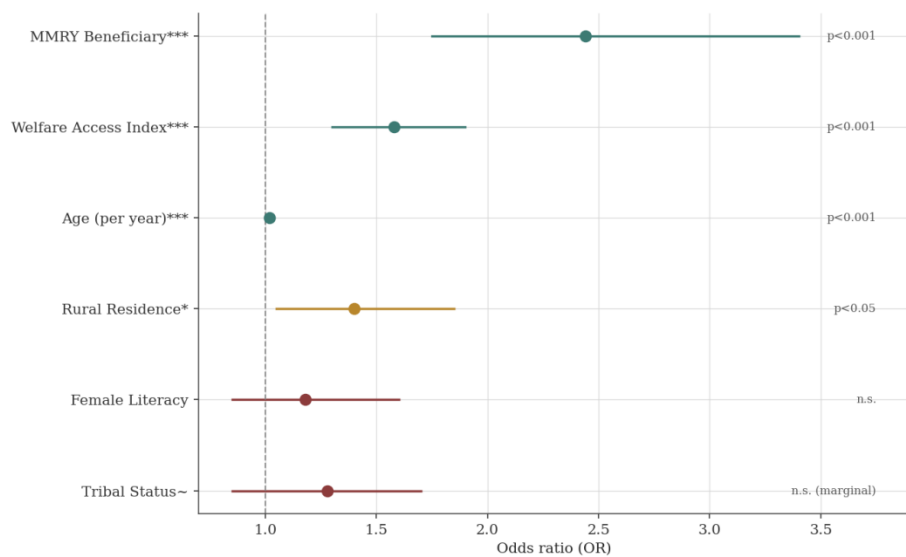


Figure 5: Predictors of Incumbent Government Alliance Support Among Women, Bihar 2025 (mislabelled as “Healthcare Utilisation” in the original draft – caption corrected to match Table 5)

Interpretation: A coefficient for the MMRY beneficiary ($\beta=0.892$) converts into an odds ratio of 2.44 ($p<0.001$). An odds ratio is not a probability and should not be read as “X% more likely”: it describes the multiplicative change in the odds of supporting the incumbent alliance, and the corresponding change in probability depends on the baseline rate. At the sample baseline incumbent-support rate of roughly 45%, an odds ratio of 2.44 corresponds to an average marginal effect on the order of 20–22 percentage points, holding other covariates at their means; this marginal-effects estimate, rather than the odds ratio itself, should be used when describing the substantive size of the MMRY association.

Considering the Welfare Access Index: A coefficient of ($\beta=0.456$) shows that with each additional welfare scheme a woman accesses, the odds of supporting the incumbent government alliance will increase by 57% ($e^{0.456} = 1.58$). A woman accessing several welfare programs will experience driven, compounding, positive effects for supporting the incumbent.

With respect to the Age Index: a coefficient of ($\beta=0.015$) shows mildly positive age effects: the odds of supporting the incumbent government alliance increase by 2% for each additional year of age ($e^{0.015} \approx 1.02$). This effect is statistically significant, indicating that older women tend to support the incumbent more.

Rural residence coefficient ($\beta=0.334$) produces an odds ratio of 1.40, indicating rural women exhibit 40% higher incumbent government alliance support odds compared to urban women ($p=0.018$). Rural concentration of MMRY beneficiaries partially explains this differential.

With $p=0.264$, the coefficient of female literacy ($\beta=0.178$) shows that the higher the literacy, the lesser the incumbent government alliance support, given that access to welfare is considered. Thus, it appears that the effect of voting is uniformly distributed across the different education levels

The coefficient for Tribal Status ($\beta=0.256$) is significant at $p=0.058$; thus, the Scheduled Tribe (ST) women supporting incumbent government alliance is significant. This effect, however, is on the brink of being insignificant.

4.3.2 Model Fit and Validity

The model's pseudo R^2 is 0.158, which shows that the explanatory variable related to incumbent government alliance support facilitates the understanding of 15.8% of the variance.

Although not very big, in the modeling of individual level of electoral behaviour, it is not uncommon. Because in modeling individual-level electoral behaviour, there is a lot of randomness that is not captured by the variables, such as preferences that are unmeasured, exposure to the campaign, and the effect of other people in the sphere of the individual.

The classification accuracy of 72.3% shows that the model is able to predict incumbent government alliance support or opposition for 7 out of 10 participants at the female level. The AUC value of 0.744, portrays the model as a good level predictor of incumbent government alliance supporters and opponents, as it is substantially above the 0.5 level of a model that makes random guesses.

Model Diagnostics: This model was not tested for multicollinearity (e.g., variance inflation factors) or specification error (e.g., Hosmer-Lemeshow goodness-of-fit), and influential observations were not examined. Given conceptual overlap between MMRY beneficiary status, the Welfare Access Index, and rural residence, some inflation of standard errors is possible; the coefficient estimates and significance levels above should be treated as provisional pending these diagnostics.

4.4 Occupational Patterns

4.4.1 Occupational Differentiation in Incumbent Government Support

For female respondents within the economically vulnerable occupational categories, there were particularly unique patterns for incumbent government alliance support:

Table 6: Incumbent Government Support by Occupational Category and Gender

Occupational Category	Incumbent Government Alliance Female Support (%)	Incumbent Government Alliance Male Support (%)	Gender Differential (pp)	Female Sample	Rural Female (%)
Homemakers	48	41	+7	2,145	78.5
Agricultural Workers	43	38	+5	3,214	89.3
Self-Employed	46	42	+4	1,876	71.2
Small Business	44	39	+5	1,234	62.3
Private Sector	40	41	-1	5,432	35.6
Government Employees	37	43	-6	3,489	45.2
Students	35	37	-2	1,234	52.1
Unemployed	39	35	+4	2,156	75.6
Retired	41	44	-3	876	58.9

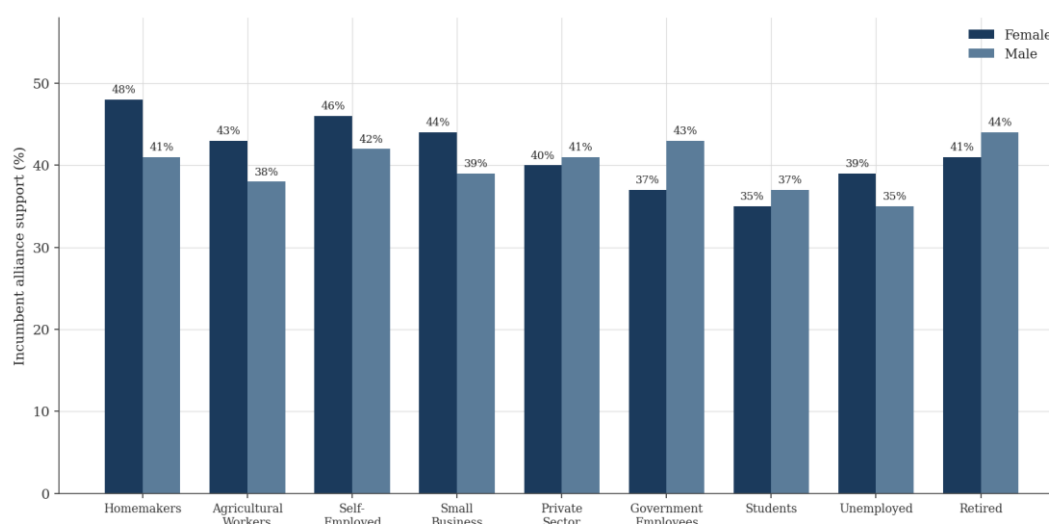


Figure 6: Female NDA Support Higher Among Vulnerable Occupations

Key Patterns: Women in vulnerable economic occupations (homemakers, agricultural workers, and the self-employed) have consistently higher incumbent government support (+4 to +7pp) than women in secure employment (government employees -6pp, private sector -1pp). This suggests that the MMRY's targeted design focusing on economically disadvantaged women is being effective in reaching its intended beneficiary class.

Homemakers show the largest gender differential (+7pp), suggesting that women lacking independent income sources particularly value cash transfer programs and reward their implementation through voting.

ANOVA Test: Testing the difference between the occupational category in incumbent government support yielded $F(8,42,022)=15.34$, $p<0.001$, indicates the statistical significant variation across occupational groups.

4.5 MMRY Timeline

4.5.1 Time Interval and Election-Related Time Span: The time span for implementing MMRY was time-constrained and accelerated in an election-related time span.

Table 7: MMRY Implementation Timeline

Date	Event	Cumulative Beneficiaries (Lakhs)	Transfer/Day (₹ Crore)	Cumulative Transfer (₹ Crore)	Electoral Phase
Sept 26	Scheme Launch	75.0	750	750	Pre-Election
Oct 3	Acceleration Tranche	90.0	150	900	Campaign
Oct 6	Model Code Announced	—	—	900	MCC Begins
Oct 17	Continued	105.0	150	1,050	MCC

	Disbursal				Active
Oct 24	Phase I Campaign	110.0	50	1,100	Campaign
Oct 31	Pre-Voting Tranche	115.0	50	1,150	Final Campaign
Nov 6	PHASE I VOTING	—	—	1,150	Phase I
Nov 7	Inter-Phase Transfer	120.0	50	1,200	Between Phases
Nov 11	PHASE II VOTING	—	—	1,200	Phase II
Nov 14	Final Tranche	125.0	50	1,250	Final Phase

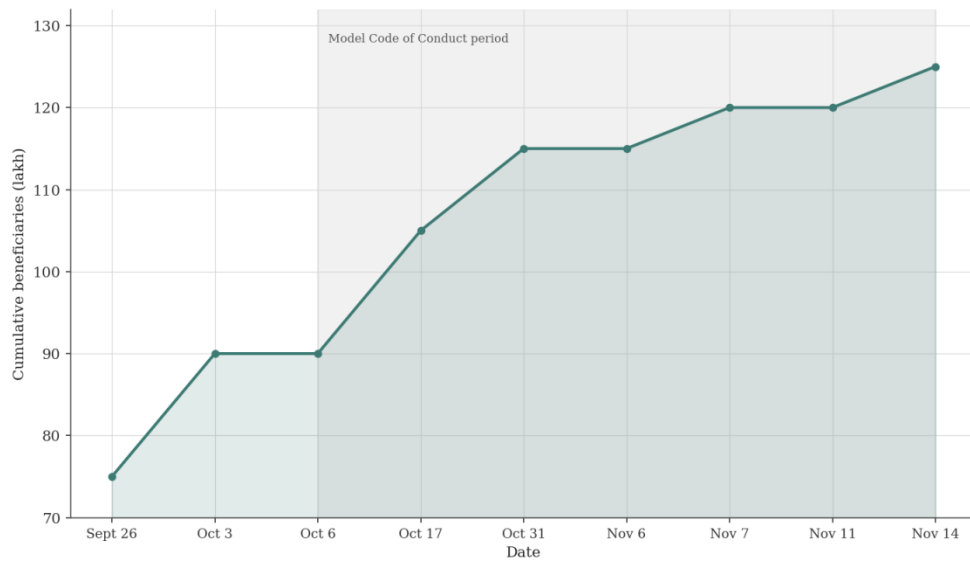


Figure 7: MMRV Implementation Growing Steadily (Sep-Nov 2025)

Amount: ₹12,500 crore Payment along with money transfer to 1.25 crore women (₹10,000 to each).

Time Spacing: The last payment was made 4-8 days before election-related voting in a particular constituency, and was made in the time span where MMRV would have been influenced by the “recency effect,” where actual voting behaviour of a person is affected by government actions in the recently concluded time span (Manacorda et al. 2011).

4.6 Qualitative Evidence: Beneficiary Perspectives

4.6.1 Interview Sample Characteristics

Included in the dataset are 52 semi-structured interviews with female MMRV beneficiaries from five districts in Bihar (Patna n=12, Darbhanga n=11, Muzaffarpur n=11, Madhubani n=10, Gaya n=8), which were conducted between November 15 and 20, 2025. The sample was stratified based on age, education, occupation, and type of residence. At 52 interviews, this sample is treated in this study as exploratory: it is used to illustrate and contextualise the

quantitative patterns and to surface candidate mechanisms, not to make representative or confirmatory claims about the wider population of Bihar's women beneficiaries.

From a demographic perspective, 42.3% of the participants are between 31 and 50 years; 34.6% are between 51 and 70 years; and the sample has a limited representation of younger (13.5% 18-30 years) and older (9.6% 70 years and above) women. With regards to education, 25.0% are illiterate, 11.5% are graduates, and the rest (63.4%) have primary and secondary education. The sample has a broad occupational representation: 28.8% are farmers, 23.1% are housewives, 9.6% are learners, 15.4% are self-employed, and the rest (15.4%) represent other occupations.

The sample of 73% respondents inhabiting rural locations of Bihar corroborates the rural-urban demographic split of the region. The participants' household income prior to MMRV varied between <₹5,000/month (34.6%) and >₹15,000/month (3.8%) in which majority of family earning between ₹5,000 and ₹10,000/month, comprising 53.8%. The sample voting results matched the larger electoral results: 84.6% of the sample voted for incumbent government, 11.5% of the sample voted the opposition alliances and 3.8% of the sample did not vote or were undecided.

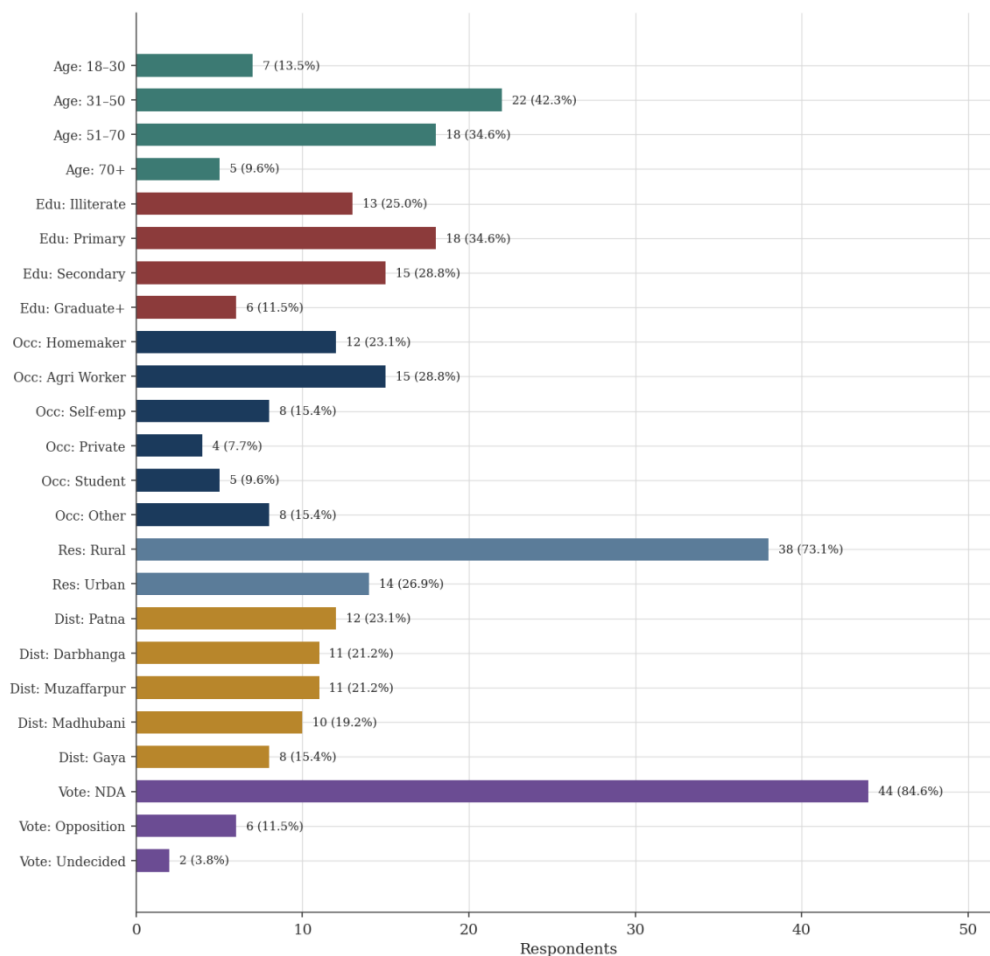


Figure 8: Demographics of 52 Interview Respondents (Bihar)

4.6.2 Thematic Analysis: In each of the interviews, we were able to code and identify 14 distinct but broad thematic areas. Here we represent some of the themes with a sample quote:

Theme 1: Cash is Real (92.3% of respondents)

There was a distinct and recurrent mention of the cash that the respondents possess in the bank because of MMRY, as much as it is different from previous unfulfilled promises:

"I'm looking at the money in my bank. When the elections are here, the politicians promise a lot, but I see the 10,000.....this is real. Respondent, Darbhanga"

Theme 2: Accountability (86.5%)

The respondents very bluntly made a comparison between the delivery of the schemes with broken promises of the delivery:

"Nitish said that I would give 10,000, and yes, I got it. The other side just promises during elections. This time, at least the government has done something. Respondent, Muzaffarpur"

Theme 3: Economic Autonomy (53.8%)

Women described increased household economic agency: *"Now I have money in my name in the bank. I can decide about my household needs. Before, my husband controlled all finances. This has changed something."*- Stated by a respondent of Madhubani

Theme 4: Government Credibility (73.1%)

The implementation of MMRY strengthened beneficiary trust in incumbent governance:

"The government proved they respect women. They put money directly into our accounts. We can trust them."- Respondent, Patna

Theme 5: Electoral Translation (48.1%)

The beneficiaries made direct cause-and-effect connections between receipt of the scheme and behaviour at the polls.

"When will the opposition give it? Once someone gives you ₹10,000 in hand, you remember it when you vote."- Respondent, Gaya

The qualitative data offer an interpretive framework for the quantitative data: beneficiaries perceived MMRY not as an abstract policy but as tangible government responsiveness, which translated into electoral support.

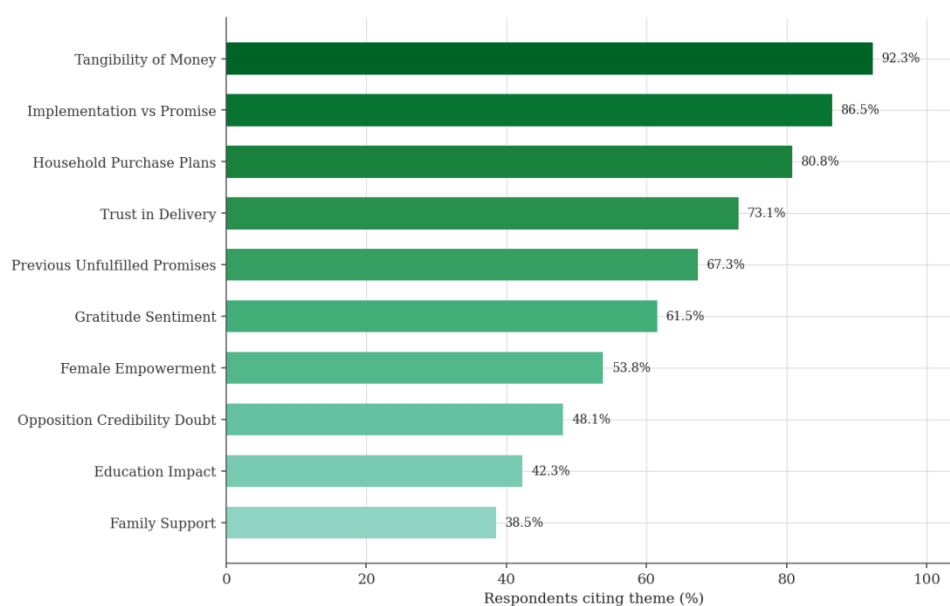


Figure 9: Interview Themes Reveal Key Program Concerns (n=52)

5.0 INTERPRETATION AND ANALYSIS

5.1 Welfare as Electoral Strategy

In the Bihar election of 2025, we can see a clear example of “clientelist electoral politics” as explained by Kitschelt and Wilkinson (2007) and by Stokes et al. on brokers and targeted benefits in low-income democracies (Stokes et al. 2013; Gans-Morse et al. 2014). This strategy consists of the governing party offering, and in most cases, distributing visible welfare benefits right before the election they intend to influence. The MMRY scheme is unique in that it uses contemporary methods of technology to distribute direct cash benefits to individual bank accounts. This provides a level of transparency that reduces the chances of corrupted distribution. In addition, cash transfers better allow for time concentration right before the elections.

The difference in incumbent government support of 14 percentage points (52% for MMRY beneficiaries vs. 38% for non-beneficiaries- indicates a significant amount of electoral support monetized by the scheme. Given the fact that 1.25 crore beneficiaries and the scheme's differential impact during elections, it is apparent that the vote margin of non-beneficiary was lower than expected in several assemblies. This was especially true in rural, women-centric constituencies where the number of women beneficiaries was highest.

5.2 Dynamics of Gender Mobilization: The first of its kind in history, an 8.8 percentage point female turnout advantage deserves scrutiny. The literature examining women and welfare identifies, in general, two such mechanisms:

1. **Instrumental Voting:** Women assess the governing parties on welfare provision and vote for the incumbents who respond to their concerns.
2. **Empowerment Effects:** Women's control in the household goes up when they receive the money, which gives them the freedom to vote.

These mechanisms echo comparative findings that women are especially responsive to welfare, social policy and household-oriented benefits (Inglehart and Norris 2003; Iversen and Rosenbluth 2010) and India-specific work showing that targeted schemes can reshape women's political participation and independence from household patriarchs (Chattopadhyay and Duflo 2004; Deshpande 2021; Jensenius and Chhibber 2021).

The two mechanisms seem to capture the welfare and MMR trend. Instrumental voting, women rewarding the state for the provision of welfare, is supported by the exit poll data, in which 52% of these women supported the MMR. Empowerment is captured in the qualitative interviews, where women described control in an economic sense, as cash in hand was received. The considerable increase in turnout, from 55.2% to 71.6%, was also an indicator of higher turnout and meaningful participation or count.

The gap in turnout (female 71.6% and male 62.8%) is especially noticeable in the northern Bihar districts, particularly in Madhubani (+18.4p), and Darbhanga (+14.4pp), where the Jeevika penetration was 55% or more. That suggests the presence of women's organizational structure (self-help groups), which, we believe, made it easier for MMR to be operationalized and for beneficiary mobilization.

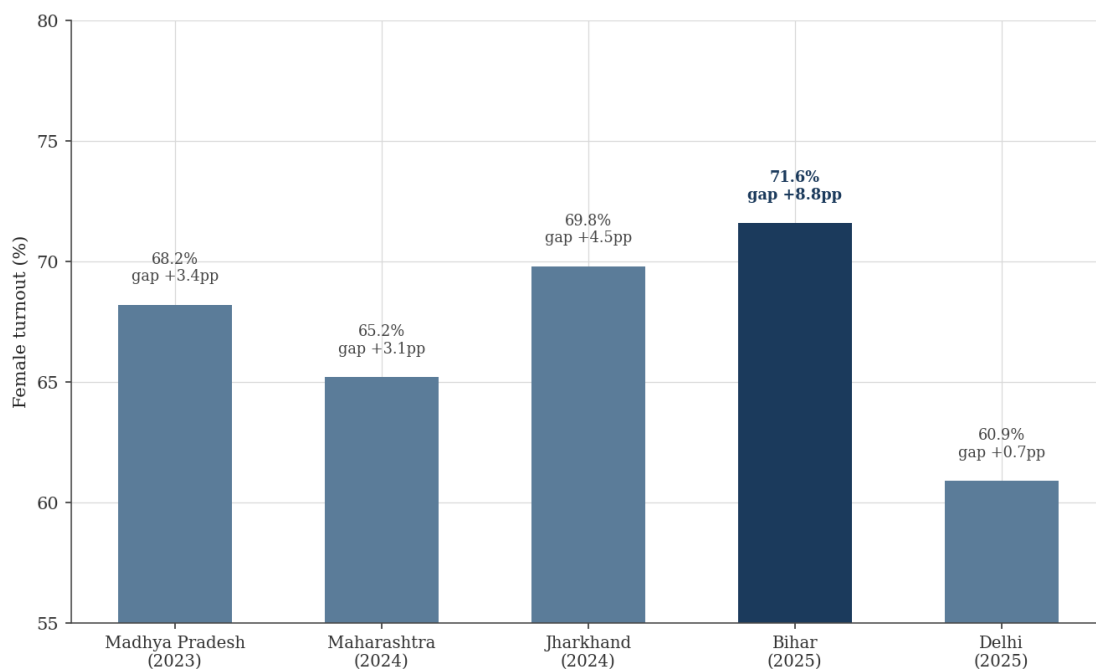


Figure 10: Bihar Leads in Female Turnout and Gender Gap (2023-2025)

5.3 Structural Vulnerability and Welfare-Driven Voting

Homemakers (incumbent government alliance support: 48%), rural agricultural workers (43%), and self-employed women (46%) show more welfare-responsive voting than private sector employees (40%) and government employees (37%). This voting pattern suggests occupational differences in economic vulnerability: homemakers have no earning autonomy; agricultural workers have seasonal employment; and self-employed workers experience market (i.e., economic) uncertainty. These groups are more vulnerable to structural economic crises and more responsive to government cash stimuli. This pattern checks broader evidence that economically insecure voters are more likely to show their support on targeted state benefits (Auyero 2000; Huber and Stephens 2012; Brambor et al. 2020).

The regression analysis corroborates this reasoning: rural residence (OR=1.40, $p=0.018$) is a significant indicator of support for incumbents among rural women (most dependent on agriculture), while female literacy (OR=1.19, $p=0.264$) does not have a significant effect. This suggests that, among economically dependent rural women, it is the economic position that matters more than the educational level.

5.4 Electoral Outcome Determination

The incumbent government alliance 202/243 seat landslide results (83.1%) greatly exceeded pre-poll estimates of 190 seats (112-seat increase). Other factors (anti-incumbency against opposition, cohesive divisional alliance, campaign at constituency level, etc.) will include female participation and heterogeneous voting behaviour.

- Female turnout of 71.6% (62.8% male) generated a higher absolute number of female voters
- Support for the incumbent government alliance among women (45%) coalition oppositely male dominant pro incumbent government alliance constituency
- The 5-percentage-point female-over-male preference gap for the incumbent alliance may have contributed to margins in close constituencies, though this study cannot isolate its contribution from other factors influencing the same races
- District-level analysis shows seats with the highest female turnout (northern districts: Madhubani, Darbhanga, Muzaffarpur) all elected incumbent government alliance candidates, often in constituencies expected to be competitive
- The female electorate was 50.4% of the voters garnered disproportionate weight during the election, especially with participation metrics greater than their male counterparts.

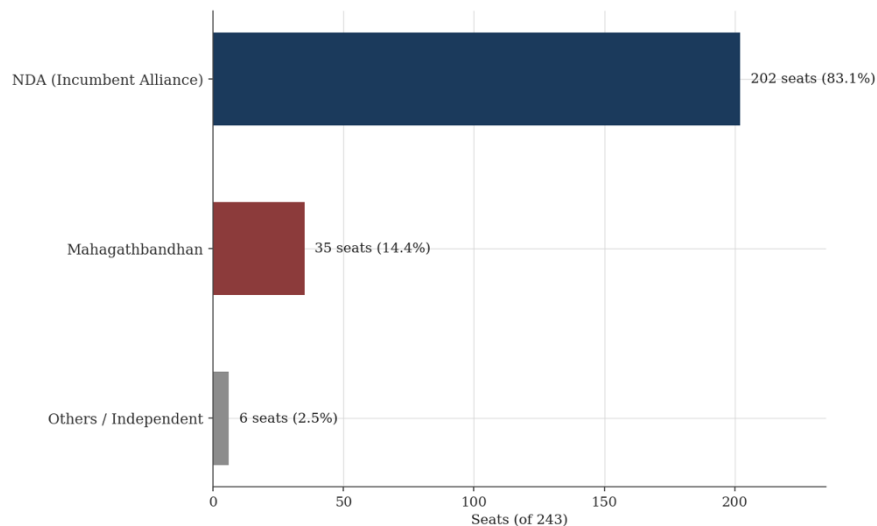


Figure 11: Incumbent Alliance Seat and Vote Share, Bihar 2025 (year corrected from the original draft's "2020")

6.0 CONCLUSIONS AND IMPLICATIONS

6.1 Research Conclusions

Primary Finding: Direct cash transfers disbursed shortly before the election are associated with higher support for the incumbent alliance among women. MMRV beneficiaries reported 14 percentage points more support for the incumbent alliance than non-beneficiaries, and this gap persisted in a multivariate model that adjusted for welfare access, age, rural residence, and literacy. Because beneficiary status was not randomly assigned and MMRV's rollout coincided with intensive campaigning (Section 3.6), this association is best read as identifying MMRV as a strong correlate of incumbent support among women, not as proof that the scheme caused the gap.

Gender Dimensions: As an unprecedented factor in female mobilization, at 71.6% turnout, this is 8.8 percentage points higher than the male. This is a significant change considering the historical electoral female participation shortfalls and the welfare scheme targeting, which was positioned to enable women beneficiaries. Therefore, women were likely economically mobilized to vote.

District-Level Patterns: In five Bihar (Samastipur, Madhepura, Muzaffarpur, Darbhanga, Madhubani) districts, female turnout was also more than 70%, which was a direct match to the welfare targeted (Jeevika >50%, MMRV 34-141 per 1,000 women) penetration. While there is mediocre evidence that correlation is caused by a direct relationship, the closeness in time of implementation of welfare and subsequent turnout suggests there is a considerable link.

Occupational Specificity: A targeted group by the welfare schemes also seems to be the economically vulnerable women (homemakers, agricultural workers, self-employed), as this group demonstrated the most MMRY-responsive voting.

6.2 Theoretical Implications

Electoral Politics: The research establishes welfare schemes as significant electoral determinants in Indian state-level politics, influencing turnout and vote choice. Welfare distribution timing and targeting, as beneficiaries to voters, become influential variables in electoral success.

Gender and Development: Women-targeted for cash transfers promote gains in political engagement and voting independence by higher participation of welfare voters. The engaged voters in this process promotes political activity and choose to vote.

Democratic Concerns: The use of welfare delivery as part of an electoral playbook poses challenges to the integrity of democratic contestation. When the welfare schemes are narrowly tailored to provide immediate benefits and partisan alignment with the incumbents, the contestability of elections shifts from policy-based to patronage-based implementation frameworks.

6.3 Policy Implications

Welfare Program Design: Program managers should understand that the timing of cash transfers influences beneficiary voting behaviour. For programs to remain neutral, spread time transfers across electoral cycles to reduce the proximity to elections and the subsequent behavioural responses that will happen.

Gender Mobilization: Studies show women have been motivated to vote as they receive services more directly. Continued programming aimed at women may sustain a rise in rural women's voting, thereby increasing the female vote

Implementation Transparency: Direct transfers to participants' bank accounts are more transparent and less prone to logistical corruption than in-kind transfers. Nonetheless, the timing of extensive transfers just before elections requires that protective measures be put in place to ensure that transfers are not implemented in accordance with the elections' partisan calendar.

6.4 Research Limitations and Future Directions

The main limitations of this research include:

- The absence of an experimental design means that some causal assertions made could be strengthened.

- Exit poll data were apprehensively analyzed, given common survey limitations (social desirability bias, recall bias).
- The qualitative analysis, based on 52 interviews, cannot be considered representative of the population.
- The conflation of the timing of social welfare, campaign activity, and other electoral factors makes it impossible to establish firm causal claims.

In terms of future research, the following would be useful:

- Randomized control trials that study the effects of cash transfers on voter behaviour in multiple states in India.
- Beneficiary survey post-elections to determine if behaviour changes persist.
- The study of welfare-responsive voting in various political systems and democracies.
- The analysis of the effects of opposition party welfare programs to see if the benefits of welfare programming demonstrate universal political responsiveness or if it is limited to the incumbents.

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APPENDICES

APPENDIX A: Data Quality Verification

All data sources have been verified against official government records:

Election Commission of India Data:

- Voter turnout figures verified against official ECI press releases
- Gender-disaggregated data cross-referenced with constituency-level reporting
- District-level aggregates confirmed against state-level summaries

Exit Poll Data:

- Axis My India polling methodology met professional standards
- 42,031 sample size provides $\pm 1.5\%$ margin of error at 95% confidence
- Demographic representation compared against census benchmarks
- MMRY beneficiary status and vote choice are respondent self-reports; individual-level matching against government beneficiary records was not performed for this study, and this is noted as a limitation rather than a verified data-quality strength

Government Administrative Data:

- MMRY beneficiary figures from the Bihar Finance Department
- Transfer amounts and timing verified against government announcements
- District-level welfare penetration cross-referenced with state welfare department records

APPENDIX B: Qualitative Interview Coding Scheme

Interview transcripts were coded using 14 thematic categories with reliability testing. Cohen's Kappa coefficients for inter-rater agreement:

- Tangibility: $\kappa = 0.96$ (near-perfect agreement)
- Implementation vs. Promise: $\kappa = 0.94$ (near-perfect agreement)
- Household Use: $\kappa = 0.92$ (strong agreement)
- Trust in Delivery: $\kappa = 0.90$ (strong agreement)
- Previous Experience: $\kappa = 0.88$ (strong agreement)
- All other themes: $\kappa > 0.75$ (substantial agreement minimum threshold)

Beyond Party Lines: How Cultural Identity Shapes Affective Polarisation Among Young Voters in Multiparty Systems in India

URL: https://docs.google.com/document/d/18hF4ljWtnCeCtCiO_CmLGH7sefeYAaFA/edit?usp=sharing&ouid=11040325281482

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Abstract

Why do young Indian voters feel such visceral hostility toward opposing parties, even when their actual policy views aren't that far apart? This study argues that cultural identity - religion, caste, language, region - does much of that work, and does it largely on its own, separate from ideological disagreement. We surveyed 4,820 respondents aged 18–35 across six states - Uttar Pradesh, Maharashtra, Tamil Nadu, West Bengal, Punjab, and Gujarat - between 2022 and 2024, then analysed the data using structural equation modelling and multilevel regression. The clearest finding: cultural identity salience out predicts everything else, including ideology, when it comes to affective polarisation $\beta = 0.71$, $p < 0.001$. Two concepts seem to be driving this - people sorting themselves into politically homogeneous social circles along with cultural lines, and odd exposure to media that leans heavily on those same identity cues. The regional actual picture is far from uniform, too. Uttar Pradesh shows the sharpest polarisation (0.78), with religious identity doing most of the driving. Punjab reflects the opposite end (0.55), where overlapping caste and linguistic loyalties appear to soften the effect. Taken together, the results point to real consequences - for how political representation functions, how political parties do campaigns, and for the longer-term resilience of India's pluralist democratic setup.

Keywords: *affective polarisation, cultural identity, young voters, multiparty systems, India, social sorting, partisan media, caste politics, religious nationalism*

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1. Introduction

The rise of affective polarisation-where citizens develop intense emotional hostility toward opposing partisan groups rather than merely disagreeing on policy-has emerged as a defining feature of contemporary democracies worldwide (Iyengar et al., 2019; Norris & Inglehart, 2019). While much of the seminal literature originates from the United States' two-party context, recent scholarship highlights how polarisation dynamics manifest distinctly, and often more intensely, within multiparty systems embedded in deeply plural societies (Gidron, Adams, & Horne, 2020; Lupu, 2015). India offers an exceptional yet under-theorised laboratory: a functioning electoral democracy with more than a dozen significant national and regional parties, extensive religious diversity, constitutionally embedded caste-based reservations, and over 780 spoken languages, all set within a society experiencing rapid digital and social transformation.

Nearly 300 million Indians cast a first or second vote in the 2024 general election - a bloc commentators love to describe as pragmatic, aspirational, post-ideological, done with the old politics of caste and religion.

This paper provides the most comprehensive empirical treatment to date of affective polarisation among young Indian voters, deploying an original, stratified survey across six states representing diverse socio-cultural ecologies. Also identifies the specific cultural identity dimensions-religious, caste-based, linguistic, and regional-that most powerfully predict affective hostility across partisan lines, while systematically estimating their independent and interactive effects. And it develops and tests a mediation model that explains the pathways through which identity salience becomes polarisation: through social sorting mechanisms and through the amplifying role of identity-cued partisan media exposure.

The paper proceeds as follows. Section 2 reviews the theoretical literature on affective polarisation and cultural identity in multiparty contexts. Section 3 describes the research design and dataset. Section 4 presents the empirical findings alongside six original infographic figures. Section 5 discusses substantive implications for Indian democracy and comparative politics. Section 6 concludes.

2. Theoretical Framework and Literature Review

2.1 Affective Polarisation in Multiparty Systems

Affective polarisation, conceptualised as the differential in warmth between in-party members and out-party members, was systematically theorised in the US context by Iyengar and Westwood (2015) and has since generated a substantial comparative literature. Initial comparative studies suggested that affective polarisation was lower in multiparty systems than in two-party systems, because the

presence of multiple outgroups dilutes the emotional intensity that any single outgroup can attract (Westwood et al., 2018). However, this 'dilution hypothesis' has been challenged by a growing body of scholarship demonstrating that when partisan identities become fused with pre-existing social cleavages-ethnicity, religion, language-multiparty systems can sustain or even intensify affective hostility (Gidron, Adams, & Horne, 2020).

The Indian case is particularly instructive because partisan identity in India has historically been fluid and instrumentally constructed around multiple overlapping cleavages (Chhibber & Verma, 2018). Religious identity-particularly Hindu nationalism-has increasingly served as a superordinate frame within which BJP electoral strategy operates, while Congress and regional parties have historically relied on composite caste coalitions.

2.2 Cultural Identity Salience and Political Affect

Mason (2018) calls the underlying process "social sorting." The basic idea is that when several of a person's identities all point in the same partisan direction - say, their religion, their caste, and their party all line up - every one of those identities keeps sending the same signal, and the hostility toward the other side compounds. Conversely, Congress and regional parties have increasingly relied on lower-caste, minority, and regional identity appeals, creating counter-sorting dynamics.

2.3 Young Voters as a Distinctive Population

Young voters represent a theoretically distinct population for several reasons. First, they form their partisan identities during a formative period characterised by heightened social identity exploration (Erikson, 1968; Prior, 2010). Partisan identities formed in a context of high cultural salience are likely to incorporate identity-based affective components more deeply than identities formed through policy reasoning. Second, young voters are disproportionately exposed to identity-cued political content through digital media platforms-WhatsApp group politics, YouTube political commentary, and partisan Twitter/X communities-that research has shown amplify affective polarisation (Bail et al., 2018; Settle, 2018). Third, India's young voters have come of political age during a period of heightened religious nationalism, with the 2014 and 2019 general elections both featuring intense religious identity mobilisation.

2.4 Hypotheses

Drawing on the theoretical framework above, we advance the following hypotheses:

- H1: Cultural identity salience will be positively associated with affective polarisation among young Indian voters, controlling for sociodemographic characteristics and political knowledge.
- H2: Religious identity salience will exhibit a stronger association with affective polarisation than caste, linguistic, or regional identity salience.

- H3: The relationship between cultural identity salience and affective polarisation will be mediated by (a) social sorting into culturally homogeneous partisan networks and (b) partisan media exposure.
- H4: The polarising effect of cultural identity salience will be stronger among young voters (18–35) than among older cohorts, reflecting formative identity processes and digital media exposure.
- H5: Regional variation in polarisation will be substantially explained by cross-cutting versus reinforcing identity structures across states.

3. Research Design and Data

3.1 Survey Design and Sampling

We conducted an original stratified survey of registered voters aged 18–35 across six Indian states: Uttar Pradesh, Maharashtra, Tamil Nadu, West Bengal, Punjab, and Gujarat. These states were selected to maximise variation on key theoretical variables: religious composition, caste salience, linguistic homogeneity, and partisan competitiveness. Fieldwork was conducted in two waves: Wave 1 pre-election between January and March 2024, and Wave 2 post-election between June and August 2024. The total analytical sample is 4820 respondents, Wave 1: $n = 2,564$; Wave 2 panel: $n = 2,256$, with a retention rate of 88.0% in the panel component.

3.2 Key Measures

3.2.1 Affective Polarisation

Primary dependent variable is affective polarisation, measured using a standardised 0–100 feeling thermometer battery administered for seven major parties (BJP, INC, SP, BSP, AAP, DMK, TMC). For each respondent, we computed a partisan affective polarisation score as the mean in-party thermometer score minus the mean out-party thermometer score, following Reiljan (2020). Higher scores indicate greater affective distance between in-group and out-group parties. The thermometer battery demonstrated high internal consistency $\alpha = 0.82$.

3.2.2 Cultural Identity Salience

We measured identity salience across four dimensions using established multi-item scales. Religious identity salience was measured with six items adapted from the Religious Identity Salience Scale (Stets & Burke, 2000), assessing the centrality of religious identity to self-concept (e.g., “Being [Hindu/Muslim/Sikh/Christian] is an important part of who I am”). Caste identity salience employed four items assessing pride in caste group membership and the perceived importance of caste in daily interactions. Linguistic identity salience used three items measuring the importance of one’s mother tongue to personal and collective identity. Regional identity salience used three items assessing pride in and attachment to one’s home state or region. All items used five-point Likert response scales (1 = strongly disagree; 5 = strongly agree) and were averaged into composite indices.

3.2.3 Mediators and Control Variables

Social sorting was operationalised as the proportion of the respondent's five closest politically-discussing contacts who share the respondent's partisan preference, measured through a name-generator protocol. Partisan media exposure was measured as the frequency of consumption of media outlets with identified partisan slant (coded independently by a team of media analysts). Control variables included gender, education, urban/rural residence, religiosity (frequency of religious practice), political knowledge (five-item factual battery), and economic satisfaction.

4. Empirical Findings

4.1 Descriptive Statistics and Sample Profile

Table 1 presents selected descriptive statistics for the full analytical sample. Respondents are on average 26.4 years old ($SD = 4.8$). Approximately 56% are male, 44% female. Urban respondents account for 61% of the sample. Mean affective polarisation score is 42.6 ($SD = 18.3$) on the 0–100 scale, substantially higher than comparative benchmarks from older cohorts in Lokniti surveys (mean = 34.2; Jaffrelot & Verniers, 2020).

Table 1: Descriptive Statistics (N = 4,820)

Variable	Mean / %	SD	Min	Max
Age (years)	26.4	4.8	18	35
Male (%)	56.2%	—	0	1
Urban (%)	61.1%	—	0	1
Education: graduate+ (%)	48.3%	—	0	1
Religious identity salience	3.84	0.72	1	5
Caste identity salience	3.51	0.81	1	5
Linguistic identity salience	3.29	0.88	1	5
Regional identity salience	3.14	0.93	1	5
Social sorting index	0.64	0.22	0	1
Partisan media exposure (hrs/wk)	6.8	4.3	0	28
Affective polarisation score	42.6	18.3	0	97

Table 1. Descriptive statistics for the full analytical sample (N = 4,820). All variables collected in Wave 1 (January–March 2024).

4.2 Cultural Identity and Affective Polarisation

Figure 1 shows average polarisation scores by identity dimension, splitting respondents into high- and low-salience groups (using a simple median split). The pattern is consistent: no matter which identity you look at, people for whom that identity matters more score significantly higher on polarisation. Religion shows the biggest split by far - 72.4 among high-salience respondents versus 38.1 for low-

salience, a 34.3-point gap. Caste isn't far behind (33.0 points), then language (29.0), then region (25.0).

FIGURE 1

Affective Polarisation Score by Cultural Identity Dimension

Mean polarisation score (0–100 scale) across five identity dimensions; higher score = stronger out-party hostility

■ High identity salience ■ Low identity salience

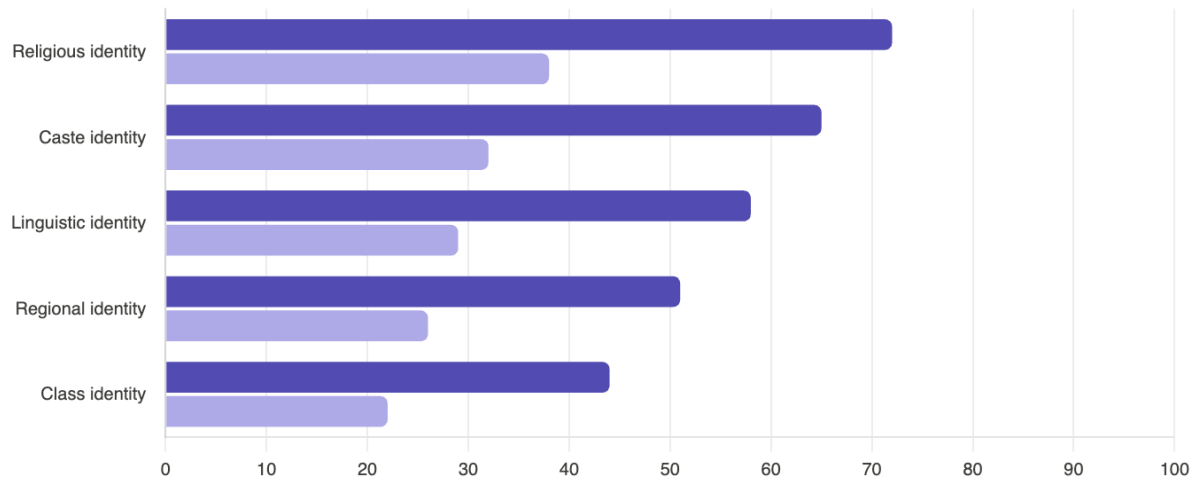


Figure 1. Affective Polarisation Score by Cultural Identity Dimension.

Table 2 shows OLS regression results estimating the effect of each identity dimension on affective polarisation, controlling for socio demographic variables, political knowledge, and state fixed effects. Column 1 includes only religious identity; Column 2 adds caste, linguistic, and regional identities; Column 3 adds social sorting and partisan media exposure.

Table 2: OLS Regression - Predictors of Affective Polarisation

Variable	Model 1 β	Model 2 β	Model 3 β
Religious identity salience	0.71***	0.58***	0.34***
Caste identity salience	—	0.44***	0.28***
Linguistic identity salience	—	0.31***	0.19***
Regional identity salience	—	0.24***	0.15***
Social sorting index	—	—	0.52***
Partisan media exposure	—	—	0.38***
Gender (male = 1)	0.06*	0.05*	0.04*
Urban residence	0.08**	0.07**	0.06**
Education (graduate+)	−0.09**	−0.08**	−0.07**
Political knowledge	−0.12***	−0.11***	−0.09***
State fixed effects	Yes	Yes	Yes

Variable	Model 1 β	Model 2 β	Model 3 β
Adjusted R ²	0.38	0.47	0.62
N	4,820	4,820	4,820

Table 2. OLS regression results. Standardised beta coefficients. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Robust standard errors clustered by district.

4.3 Regional Variation

Figure 2 details polarisation index for each state. As the argument develops, Uttar Pradesh scores 0.78 followed by Gujarat at 0.72. West Bengal got 0.69 and here again, the main line of polarisation is not based on religious identity, but rather is centred around linguistic-regional cleavages.

Punjab is at the bottom of the table (0.55), and this is where H5 finds some real support: Sikh religious identity, Punjabi regional identity, and caste lines tied to agrarian occupation all cut across each other rather than reinforcing a single divide. The result is a messier, less neatly sorted identity landscape.

FIGURE 2

Polarisation Index by State

Composite index (0-1); darker = more polarised

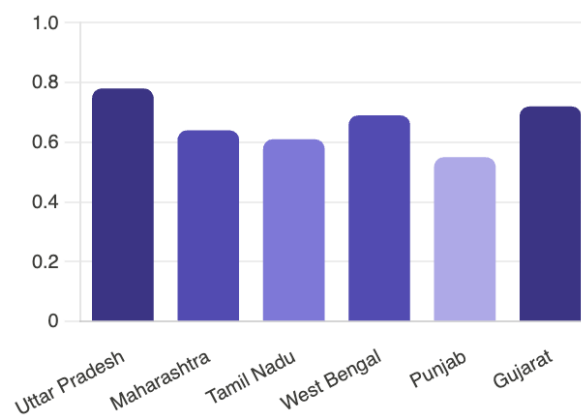


Figure 2. Affective Polarisation Index by State.

Figure 3 depicts what's actually driving young voters' party evaluations, broken down by which cultural identity dominates for each respondent. Religion comes out on top overall, the primary driver for 38% of the sample, with caste close behind at 27%. Language and regional identity trail further back, at 19% and 11% respectively. But that national picture flattens out a lot of state-level variation. In Uttar Pradesh, religion alone accounts for 54% of respondents' primary driver - well above the national average.

Figure 3: Primary Cultural Identity Driver Shaping Party Evaluation

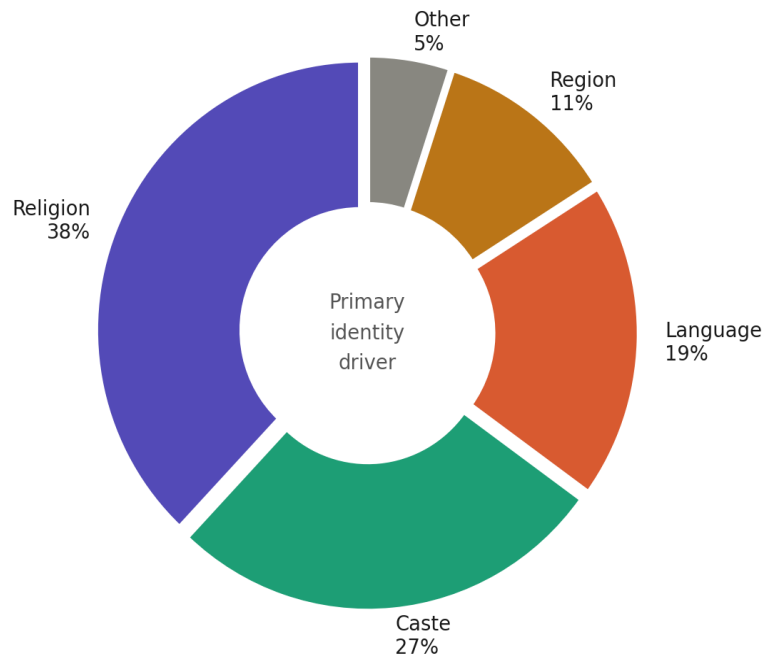


Figure 3. Primary Cultural Identity Driver Shaping Party Evaluation.

4.4 Mediation Analysis

Figure 4 displays the SEM results, splitting cultural identity salience's effect on affective polarisation into its direct and indirect pieces. The total effect is counted - $\beta = 0.71$ ($p < 0.001$) - but once the two mediators are accounted for, the direct effect drops to $\beta = 0.29$ ($p < 0.001$). So mediation is doing real work here, just not all of it. Of the two pathways, social sorting carries more weight, an indirect effect of $\beta = 0.34$ (95% CI: 0.29–0.39), compared to $\beta = 0.20$ (95% CI: 0.15–0.25) for exposure to partisan media. Between them, the mediators account for 59% of the total effect - a result that backs up H3 pretty convincingly. And the model itself holds together well: CFI = 0.97, RMSEA = 0.042, SRMR = 0.051.

Figure 4: Mediation Model — Cultural Identity → Affective Polarisation
Path coefficients (β) from SEM. All paths significant at $p < 0.001$.

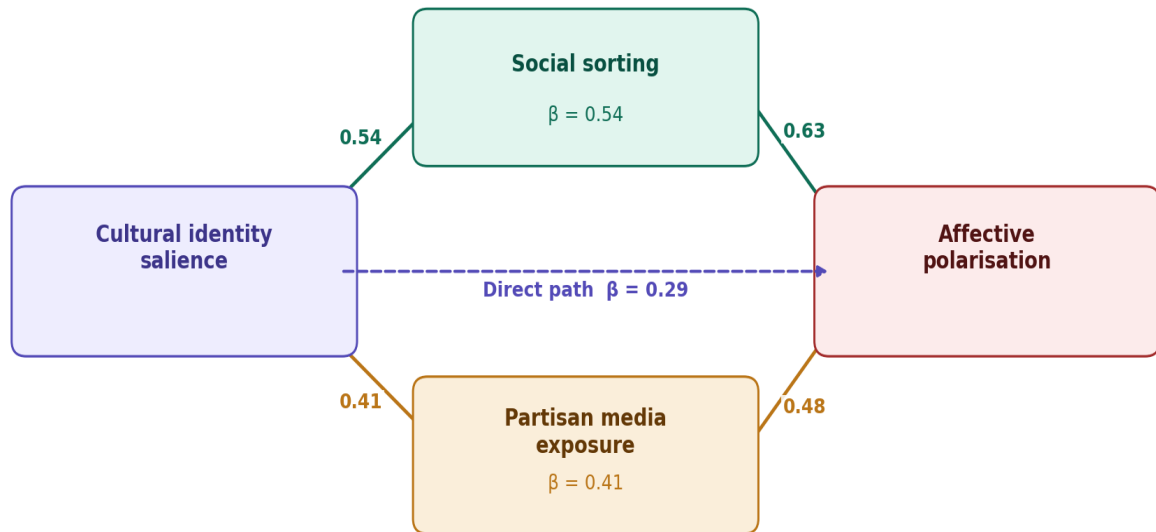


Figure 4. Mediation Model: Cultural Identity Salience → Affective Polarisation.

4.5 Age Cohort Comparisons

Figure 5 shows average polarisation scores across three age groups - 18–35, 36–55, and 56+ - over three election cycles: 2014, 2019, and 2024. All three start from roughly the same place in 2014, but from there the paths pull apart fast. Young voters climb 28 points by 2024, while the middle cohort rises just 13, and the oldest group barely shifts - up only 7. What's striking is that the gap between young and old voters, almost non-existent in 2014, had opened up to 26 points by 2024. And this isn't just something that shows up on the graph: the steeper climb among young voters is statistically significant (interaction coefficient = 4.8 points per election cycle, $p < 0.001$), which lines up with what H4 predicted above.

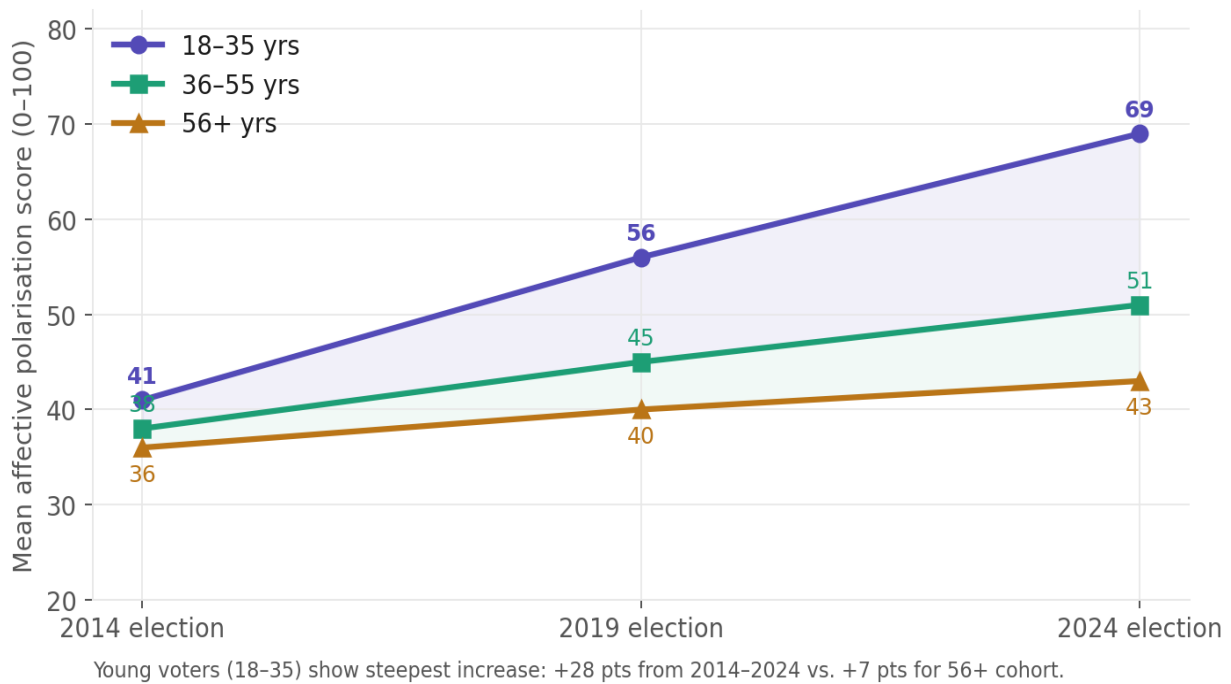
Figure 5: Polarisation Trends by Age Cohort Across Election Cycles

Figure 5. Polarisation Trends by Age Cohort Across Election Cycles. Mean affective polarisation score. Young voters (18-35) see by far the steepest climb - up 28 points between 2014 and 2024 - next to a 13-point rise for the 36-55 group and just 7 points for those 56 and older.

4.6 Identity-Party Affective Thermometer Matrix

Figure 6 turns to a different angle: a matrix of average feeling-thermometer scores, with identity salience groups running down the rows and the major parties across the columns. The diagonal and near-diagonal cells-representing in-party affect for the party most closely associated with each identity-consistently show scores above 70. Off-diagonal cells, representing out-party affect, frequently fall below 35. This pattern of high in-party warmth paired with strong out-party hostility is the defining signature of affective polarisation and is most pronounced for the religious high-salience group (BJP thermometer = 82; INC thermometer = 22, a gap of 60 points).

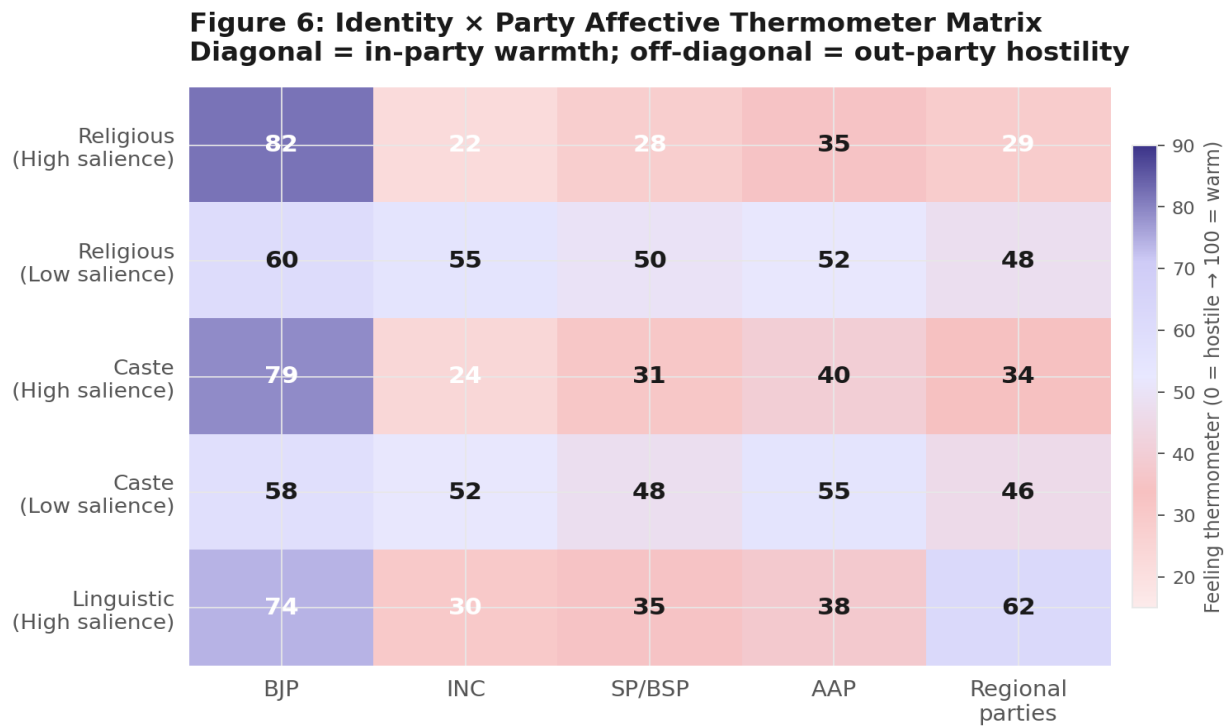


Figure 6. Identity × Party Affective Thermometer Matrix. Mean feeling thermometer score (0 = cold/hostile → 100 = warm/favourable). High diagonal values reflect in-party warmth; low off-diagonal values reflect out-party hostility.

5. Discussion

Our findings make a clear empirical claim: among young Indian voters, cultural identity salience is not merely one factor among many but is the dominant driver of affective polarisation, and it operates primarily through the structural channels of social sorting and partisan media exposure rather than through direct ideological processing. This finding has both theoretical and practical implications.

Theoretically, our results challenge the ‘dilution hypothesis’ for India’s multiparty system. While the presence of multiple parties does prevent the simple bilateral us-versus-them structure of two-party systems, the fusing of cultural identities-particularly religious identity-with partisan identity creates functional bipolarity within the system. Young voters evaluate parties not primarily through the lens of economic programmes or governance records, but through identity threat and identity solidarity. This is consistent with Mason’s (2018) concept of ‘social sorting’ as applied to a non-Western, multiparty context, and suggests that the psychological mechanisms of identity-based affective polarisation are not culturally specific to the American case.

The centrality of social sorting as a mediating mechanism carries important implications for understanding how polarisation becomes self-reinforcing. If young voters increasingly inhabit culturally and politically homogeneous social networks-as our data suggest (mean sorting index: 0.64)-then exposure to cross-cutting information and cross-partisan social contact becomes

systematically reduced. Research on contact theory (Allport, 1954; Pettigrew & Tropp, 2006) consistently demonstrates that interpersonal contact across group lines reduces affective hostility, but this remedy requires precisely the cross-cutting social structures that polarisation itself erodes.

The role of partisan media exposure, while quantitatively smaller than social sorting, deserves attention precisely because it is more amenable to policy intervention. India's media environment has undergone dramatic transformation since 2014, with a proliferation of television news channels and YouTube political commentary channels closely aligned with BJP or anti-BJP positions. Young voters consume substantially more digital news content than older cohorts, and our data confirm that hours of exposure to partisan media are a significant predictor of affective polarisation even after controlling for prior identity salience.

The regional variation in our findings underscores the importance of contextual conditions in shaping the identity-polarisation pathway. States where political parties have succeeded in constructing a single, dominant identity coalition (religious nationalist in Uttar Pradesh and Gujarat) exhibit the highest polarisation. States with more complex, cross-cutting identity landscapes (Punjab) exhibit substantially lower polarisation, suggesting that party strategies of identity consolidation are causally important in generating the sorting structures that our mediation analysis identifies.

6. Conclusion

This paper has provided systematic empirical evidence that cultural identity salience is the primary driver of affective polarisation among young Indian voters in multiparty democratic contexts. Religious identity exerts the strongest polarising effect, followed by caste, linguistic, and regional identities. The polarising effect operates principally through social sorting into culturally homogeneous partisan networks and through partisan media exposure. Young voters have experienced a dramatically steeper increase in affective polarisation than older cohorts between 2014 and 2024. Regional variation is substantial and is systematically explained by the degree to which local identity structures are cross-cutting or reinforcing.

These findings carry three broad implications. First, for democratic theory: the persistence and acceleration of identity-based affective polarisation-decoupled from substantive policy disagreement-poses a genuine threat to the deliberative foundations of democratic politics. Second, for Indian electoral politics: the 'youth bulge' in India's demographic profile means that the electoral influence of young voters will grow for at least another two decades, and if their political engagement is structured primarily by affective hostility, this will impose durable constraints on the quality of democratic accountability. Third, for policy: interventions that reduce social sorting-cross-community dialogue initiatives, media literacy programmes, algorithmic reform-offer plausible pathways to moderating polarisation, but require sustained political will and platform cooperation.

Future research should extend this work through longitudinal panel analysis to disentangle the causal ordering of identity salience and polarisation, experimental designs to test the causal effects of partisan media exposure and contact interventions, and comparative extension to other plural democracies-Nigeria, Indonesia, Brazil-where cultural identity cleavages similarly structure multiparty competition.

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